Modification of the NGA West2 GMMs for Application to Induced Earthquakes

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1 Introduction

Small to moderate magnitude potentially induced earthquakes (PIEs) tend to occur at shallower depths than tectonic earthquakes. The existing ground-motion models (GMMs) generally do not extrapolate appropriately to very shallow depths for small magnitude earthquakes: they significantly underestimate the short-period ground motions at distances less than 10 km for magnitudes less than 5.0. In this study, three of the NGA-W2 GMMs are updated to be applicable to shallow-depth induced earthquakes: Abrahamson et al. (2014), Campbell and Bozorgnia (2014), and Chiou and Youngs (2014).

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2 Ground-Motion Data Base

The data set of ground motions from PIE events was provided by BCHydro and is described by Gregor (2018). This report includes two primary sources of ground motion data from PIE events: Mahani and Kao (2018) contains ground motions from the Montney Play region of north-eastern British Columbia; Rennolet et al. (2018) contains ground motions from PIE events in Oklahoma and Kansas. Additional data has been recorded by Nanometrics in the northeastern British Columbia and western Alberta regions; however, these data were not publically available at the time of this study.

The full data set includes 175,850 recordings from 3990 earthquakes with magnitudes between 0.4 and 5.8 and distances between 2 km and 600 km. The majority of these data are from the central U.S.: there are 174,737 recordings from the Central US and 1063 recordings from Canada. The Central U.S. data includes larger magnitude events related to waste-water injection. The Canadian data includes smaller magnitude earthquakes (mostly $M < 3$) that likely related directly to fracking. The magnitude and distance distribution of the Central U.S. and Canadian data sets are shown in Figure 1.

The three model developers in this study selected their own subsets of the data from this larger data set. A key issue for using this data set to modify the NGA-W2 GMMs to be applicable to small magnitude earthquakes at short distances is the estimation of the moment magnitude from the local magnitude. If there is a systematic difference between the moment magnitude for small earthquakes in the PIE data set compared to small earthquakes in the NGA-W2 data set, then there will be a bias in the event-terms. As discussed in Section 4, there appears to be two distinct populations of small magnitude earthquakes that have large differences in the mean (e.g. shifted by a constant) that may be due to the conversion from local magnitude to moment magnitude. This magnitude conversion issue has not been resolved yet. The three developers used different approaches to address the potential bias in the moment-magnitudes estimates provided in the data set for PIE events.

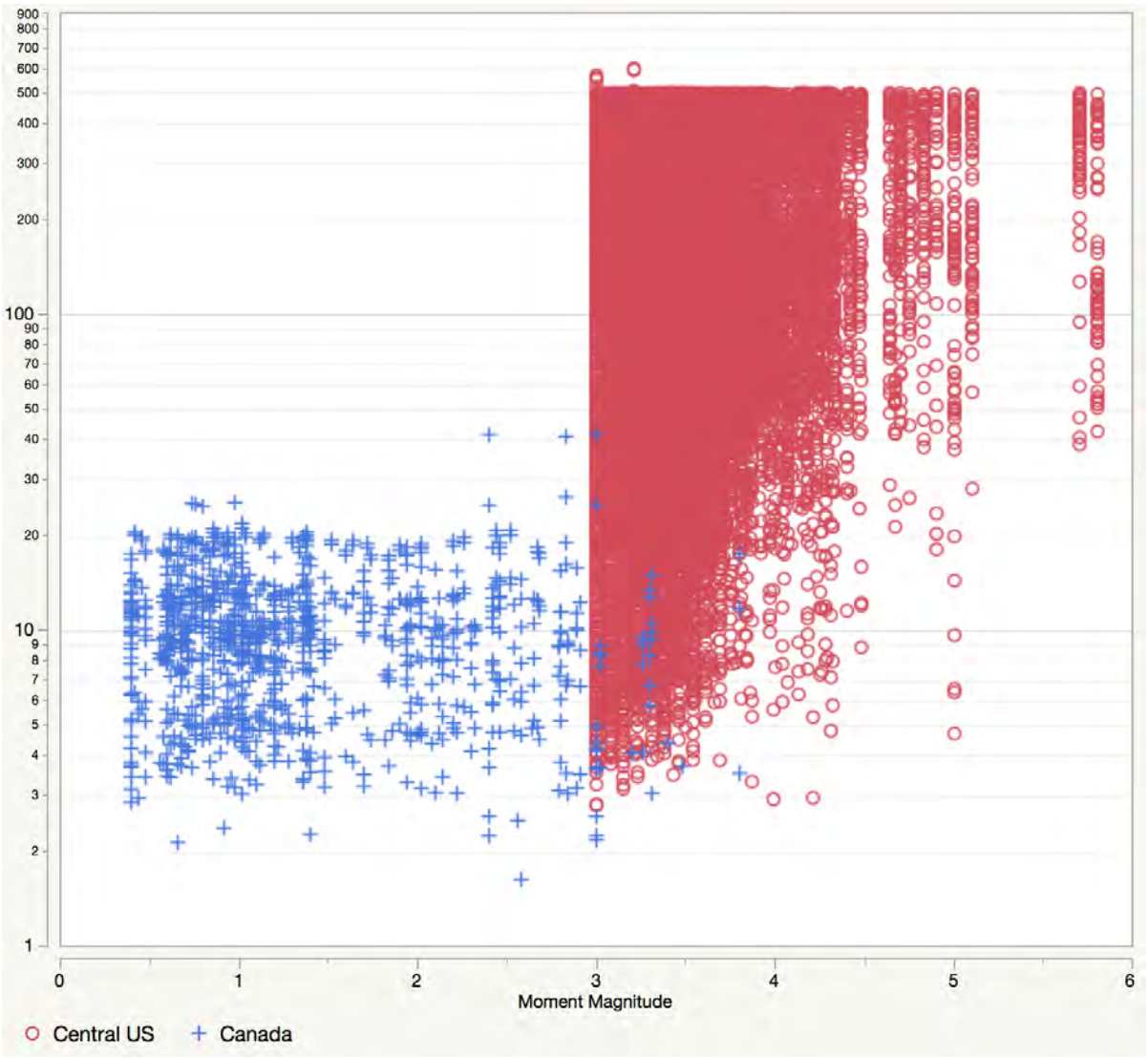


Figure 1: Magnitude-distance distribution of the full data set.

3 Update to ASK14

3.1 Introduction

The Abrahamson et al. (2014) ground-motion prediction equation (ASK14) is modified to better model the ground motion from small to moderate magnitude earthquakes at distances less than 15 km. The objective is to preserve the magnitude and distance scaling for large magnitude earthquakes in the ASK14 model, but adjust the scaling for small and moderate magnitudes to be consistent with the observed ground motions at short distances from shallow earthquakes. In particular, the modification to ASK14 is developed such that the modified model is applicable to small to moderate magnitudes at short distances but with no change in the ASK14 model for $M > 6$.

3.2 Finite-Fault Term

Most modern GMMs use a measure of the closest distance from the rupture to the site as a proxy for the wave propagation effects. For large earthquakes, all of the energy is not released at the closest point on the rupture to the site. The GMMs account for this by adding a finite-fault term to the closest distance to compute the effective point-source distance, R_{ref}, used in the geometrical-spreading term ($ln(R_{ref})$).

In the ASK14 model, the effective point-source distance is given by:

$$R_{ref} = \sqrt{R_{rup}^2 + H^2(M)} \tag{1}$$

in which R_{RUP} is the rupture distance and $H(M)$ is the finite-fault term. It has been common to call the $H(M)$ term the fictitious-depth term, but it is more than a depth term because it includes the distance along strike as well as the distance down dip that is added to the closest distance.

For tectonic events, the chance that there is a seismic station located at a short rupture distance from small magnitude earthquakes is much smaller than for large magnitude earthquakes. There

are two reasons for this. First, the chance that the top of the rupture extends to the surface or near the surface is less for smaller magnitude earthquakes. Second, if the rupture does reach the surface, the chance of a randomly located seismic station being at a short rupture distance is less for small magnitude events because the surface area that is within a small R_{RUP} is smaller due to the the shorted rupture length. As a result, most of the empirical ground-motion data from tectonic events with very short rupture distances are from larger magnitude earthquakes. For this reason, the finite-fault factors in GMMs have been focused on being applicable to larger magnitudes.

As an example, the short-distance scaling in the NGA-W2 GMMs for PGA is shown in Figure 2 for M4 and M6.5 earthquakes. For the M6.5 case, as the rupture distance goes to zero, the finite-fault term reduces the slope and causes the curve to flatten out for all four GMMs. This is consistent with the concept that the seismic slip controlling high-frequency ground motion (e.g. asperity) comes from different parts of the rupture plane and is not concentrated at the closest point on the rupture. Both empirical ground-motion data and numerical simulations using finite-fault rupture show this behavior.

In the development of the NGA-W2 GMMs, the developers spent considerable effort on how the GMM extrapolates to short distances for large magnitudes, but there was little less attention paid to the extrapolation of the GMM to short distances for small magnitudes. For the M4 case shown in Figure 2, three of the NGA-W2 GMMs show a finite-fault effect (flattening of the slope at short distances) that is similar to the finite-fault effect for large magnitudes. This does not make physical sense for small magnitudes: the rupture area for small magnitudes is small, so the seismic slip controlling the ground motion is concentrated in a small region and a strong finite-fault effect is not applicable. As shown in this figure, the ASK14 GMM does not flatten out at short distances as much as the other NGA-W2 GMMs for the M4 case. This behavior in the ASK14 model was intended to capture the reduction in the finite-fault effect for small magnitude earthquakes.

The ASK14 model included a magnitude dependence to the finite-fault term for $M < 5$ as

shown in equation (2).

$$H_{ASK14}(M) = \begin{cases} \text{4.5 km} & \text{for } M \geq 5.0 \\ \text{4.5 km - 3.5 km (5-M)} & \text{for } 4 < M < 5.0 \\ \text{1.0 km} & \text{for } M \leq 4 \end{cases} \quad (2)$$

This magnitude dependence of the H(M) term improved the physical behavior of the finite-fault term in the ASK14 model for small magnitudes ($M < 5$), but it did not capture the reduction in the finite-fault term for M5 earthquakes.

3.2.1 Modification to ASK14 Finite-Fault Term

The modification to the finite-fault terms for ASK14 was developed to reflect the source area scaling for earthquakes with $M < 6$. The magnitude dependence of the finite-fault term was revised based on assuming that the peak ground motion is generated by the slip in the center of the rupture for small and moderate earthquakes. With this assumption, the finite-fault term is given by:

$$H(M) = \sqrt{(0.5L)^2 + (0.5W)^2} \quad (3)$$

Next, assuming that the rupture area (in km^2) can be approximated by 10^{M-4} and that the aspect ratio is 1.0, the H(M) term is equal to the 4.5 km value used in ASK14 for $M = 5.57$. For larger magnitudes, the finite-fault term is held fixed to the 4.5 km in the ASK14 model because the ASK14 adequately captures the finite-fault effects for larger magnitudes through the combination of the magnitude-dependent geometrical spreading term and the H term.

The resulting modified H(M) term is given by:

$$H(M) = \begin{cases} 4.5 & \text{for } M \geq 5.57 \\ \exp(1.50 + 1.15(\text{M-}5.57)) & \text{for } M < 5.57 \end{cases} \quad (4)$$

A comparison of the magnitude dependence of the finite-fault term in the ASK14 model and the modified form given in equation (4) is shown in Figure 3.

3.3 Data Set

For use in developing the modification of the short-distance scaling of the GMM, we limited the selected subset of data to recordings with hypocentral distances less than 50 km. The ASK14 model is not applicable to earthquakes smaller than M3, so the data set is also limited to earthquakes with $M \geq 3$. Seven recordings that appeared as significant outliers were also removed (RSNs 156006, 256249, 468270, 468433, 470699, 498570, 498937). The selected data set includes 21124 recordings from 3469 earthquakes. The magnitude-distance distribution of the selected subset is shown in Figure 4.

3.3.1 Magnitude issues

As noted by Chiou (Section 5), there is a systematic difference between the residuals from earthquakes with direct estimates of the moment magnitude (called Set 1) as compared to the residuals from earthquakes in which the moment magnitude is estimated using a conversion from a different magnitude scale (called Set 2). To evaluate the potential bias due to the magnitude conversion, the magnitude dependence of the residuals for each of the two sets was computed at a spectral period of 2 sec which will be below the corner frequency for the smaller magnitudes but still in the reliable period range for most of the ground-motion data.

Figure 6 shows that the residuals from Set 1 are systematically high for $M < 4$ and are closer to the zero mean for larger magnitudes. The residuals from Set 2 show a large negative bias relative to the ASK14 GMM for small magnitudes, but again are closer to zero for the larger magnitudes. Moment magnitude estimates can become sensitive to the details of the magnitude conversion for small magnitudes. The trends in Figure 6 indicate bias in the conversion from local magnitude to moment magnitude for earthquakes with $M < 4.1$.

For small magnitude earthquakes, the scaling with magnitude is much stronger than for large magnitude earthquakes, so even a small bias in the magnitude can lead to a large effect on ground motion: For example, a change of 0.2 magnitude units in the conversion can lead to a factor of 2 change in the ground motion.

To limit the effect of a possible bias in the residuals caused by the magnitude conversion for $M < 4.1$ on the adjustments to the short-distance scaling for the ASK14 model, a correction to the moment magnitude was applied. The event terms for T=2 sec for the $M < 4.1$ data were modeled as a linear function of magnitude for the Set 2 subset. The slope was not significantly different from zero, so just a constant shift was applied. For Set 2, the corrected moment magnitude is given by:

$$M = M_{set2} - 0.25 \tag{5}$$

The same approach was applied to the Set 1 subset. For Set 1, the mean event term for T=2 is much smaller than for Set 2. For Set 1, the corrected moment magnitude is given by:

$$M = M_{set1} + 0.05 \tag{6}$$

In the following development of the adjustment to the ASK14 short-distance scaling, the corrected moment magnitude given in equations (5) and (6) are used. The key aspect of this magnitude correction is that it allows the data to be used to adjust the short-distance scaling while avoiding the issue of the magnitude conversion bias.

3.4 Exploratory Analysis

In an initial exploratory analysis, the residuals of the ground motion from the induced earthquakes were computed relative to the ASK14 GMM. The residuals from the unmodified ASK14 model for PGA are shown in Figure 8. This figure shows that there is a large under prediction of the PGA values at distance less than 15 km.

Next, the ASK14 GMM was modified to use the updated magnitude-dependence of the finite-fault term shown in equation (4). This new model is denoted ASK14-FF. Figure 8 shows the distance dependence of the residuals of the PGA using the ASK14-FF model. Using the improved finite-fault factor reduces the residuals at short distances for M4.5 to M5.5, but there is still a

significant slope in the residuals at short distances indicating that the reduction in the finite-fault term was not adequate to capture the very short-distance scaling for small magnitudes.

3.5 Functional Form of the Adjustment

Based on the preliminary evaluations of the ground-motion residuals from the PIE events, it is clear that the distance scaling in ASK14 does not capture the ln(dist) slope at short distances (less than 15 km) for small and moderate magnitudes and short spectral periods. Therefore, the distance scaling in the ASK14 model is adjusted to include a change to the short-distance scaling along with a magnitude taper, $T_m(M)$, to limit the adjustment to small and moderate magnitudes. The adjustment function is given by:

$$f_{adjust}(M, R_{RUP}) = (c_1 + f_R(R_{RUP}, M))\, T_m(M) \tag{7}$$

in which the distance adjustment function is given by

$$f_R(R_{RUP}, M) = \begin{cases} c_2 ln\left(\frac{Max(R_{eff},5)+0.1}{R_c+0.1}\right) & \text{for } R_{eff} < R_c \\ 0 & \text{for } R_{eff} \geq R_c \end{cases} \tag{8}$$

The R_c term is a model parameter that sets the distance limit of the adjustment to the geometrical spreading term. Based on visual inspection of the distance trends in the residuals, the R_c ranges from 15 km at short periods to 10 km at long periods. To keep the adjustments to distance scaling smooth over periods, R_c was set at 15 km for all periods. The short-period value was used for all periods because the change in the distance scaling is much stronger on the short-period ground motions than on the long-period ground motions The $MAX(R_{eff}, 5)$ term in equation (8) models the flattening of the trend of the short-distance residuals seen at distances less than 5 km seen in Figure 7.

Most of the data used to develop the adjustment to the short-distance scaling is from M3 to M4 earthquakes. While these events clearly show a change in the geometrical spreading at short

distances, there is a question if this applies to M5 earthquakes as well. The ground-motion data from the 2008 Mogul Nevada earthquake (Anderson et al., 2009) provide insights into this issue. The Mogul earthquake was a magnitude 5.0 earthquake with a hypocentral depth of 3 km. There were seismic stations installed directly over the rupture which provided data at short distances from this event. The PGA is plotted as a function of distance in Figure 9. For this tectonic earthquake, there is a clear increase in the distance slope at distances less than 10 km. The ratio of the median PGA at 3 km to the median PGA at 10 km is about a factor of 14 whereas the ratio expected for this event from the ASK14 GMMs is about a factor of 2.6. The data from this event suggest that change in the short-distance slope should extend up to M5. The empirical data for large magnitudes used in the development of the NGA-W2 GMM shows that the short-distance scaling is appropriate for $M > 6.0$. Therefore, the magnitude taper was set to include the full effect up to M5.0 and reduce to no effect at M5.5. The magnitude taper function, $T_m(M)$, is given by

$$T_m(M) = \begin{cases} 1 & \text{for } M \leq 4.0 \\ \text{(5.5-M)/1.5} & \text{for } 4.0 < M < 5.5 \\ 0 & \text{for } M \geq 5.5 \end{cases} \tag{9}$$

3.6 Regression Results

A random-effects regression, with a random effect for the earthquake, was used to develop the adjustment model. The resulting unsmoothed coefficients from the regression are plotted in Figure 10.

The c_1 term represents the constant shift from the ASK14 model at distances of 15 to 50 km from earthquakes with $M \geq 4.1$. The final smoothing of the C_1 terms at short periods is shown in Figure 10.

At periods greater than 3 sec, there is a strong increase in the c_1 terms; however, this increase is not considered reliable as the data that are usable at this period range is greatly reduced (see Figure 5). Therefore, the smoothed c_1 terms are set to zero for the long periods.

The distance slope adjustment terms shown in Figure 10 are largest at the PGA and reduce as the spectral period becomes larger. To keep the adjusted model reasonably smooth, the c_2 coefficients are heavily smoothed. In particular, the C_1 values at the short periods (less an 0.1 sec) are limited so that they are generally consistent with the slopes at the moderate periods (0.15 to 0.3 sec). As with the c_1 term, the c_2 coefficients at long periods are set to zero. The final smoothed coefficients for are listed in Table 1.

The residuals are recomputed using the final smoothed coefficients. The within-event residuals are shown as a function of distance in Figures 11, 12, and 13 for the PGA, T=0.1 sec, T=0.2 sec, T=0.5 sec, T=1 sec, and T=2 sec. For the PGA, there is still a trend of under prediction at distances of 5-10 km which reflects the smoothing of the c_2 terms which limited the adjustments to the slopes (see Figure 10) At periods of 0.1 to 1 sec, this trend in the short-distance residuals is not seen. At a period of 2 sec, there is an under prediction at short distance of 5-8 km.

The between-event residuals are shown in Figures 16 for T=1 sec. With the approach used to model the small magnitude scaling, the model accounts for a potential bias in the magnitude scaling for $M < 4$ earthquakes. Therefore, the between-event residuals for small magnitudes do not reflect the residual relative to the final modified ASK14 model and do not indicate that the magnitude scaling in the ASK14 model is consistent with the samll magnitude scaling ($M < 4$) in the PIE data set. The small magnitude between-event residuals should only be considered in terms of the standard deviation

3.7 Model Results

The final adjusted ASK14 GMM is given by adding the adjustment term to the median fro ASK14:

$$ASK14_{Adjust}(M, R_{RUP}, V_{S30}, \ldots) = ASK14(M, R_{RUP}, V_{S30}, \ldots) + f_{adjust}(M, R_{RUP}) \qquad (10)$$

Note that the adjustment to the ASK14 model does not include using the default Z_{TOR} value as

suggested by Atkinson XXXX. The depth-dependent stress drop implied by the Z_{TOR} scaling in the ASK14 model is maintained.

The distance scaling of the adjusted ASK14 GMM for PGA, T=0.2 sec and T=1 sec are compared to the distance scaling in the unmodified ASK14 GMM in Figure 17 for M3, M4, and M5 earthquakes. For the M3 case (top left frame of Figure 17), there is a large increase, up to a factor of 5, in the PGA at short distances due to the steeper distance slope. This difference reduces to about a factor of 2 for T=0.2 sec and T=1 sec. For the M4 case (top right frame of Figure 17), the scale factor is same as for the M3 case because the magnitude taper, $T(M)$, is unity for $M < 4$.

For the M5 case (bottom frame of 17), the PGA increase is reduced to a factor of 3 for $R_{RUP} = 3km$ due to the taper of the short-distance slope for magnitudes between 4.0 and 5.5. At long periods, the spectrum from the modified ASK14 model is larger by about a factor of XX than the spectrum from the unmodified ASK14 model at short distance due to the reduction in the finite-fault term at M5.

The median 5%-damped response spectra for the modified ASK14 model and the unmodified ASK14 model with the CEUS site factors are compared in Figures 18 and 19 for rupture distances of 3 and 10 km, respectively. For the 3 km case, the adjustment has a large effect at short periods (factors of 3 to 5) and a small to zero effect at periods greater than 2 sec. Figure 18 shows that at a distance of 3 km, the modified ASK14 M5 spectrum is larger than the modified ASK14 M6 spectrum for periods less than 0.4 sec. This leads to an important issue of oversaturation that is discussed in the next section. For the 10 km case, the modified ASK14 spectra, the increase at short periods is about a factor of 2 and reduces to a factor between 1 and 1.2 for periods greater than 0.2 sec. There is no oversaturation in the magnitude scaling at this distance.

3.8 Oversaturation

Typically, GMM developers do not allow the model to over saturate at short distances. That is, the median ground motion is required to increase monotonically with magnitude at all distances.

One reason for this constraint is from the concepts used in kinematic finite-fault simulations: the ground motion from a large earthquake is the sum of the ground motions from many smaller earthquakes. If we think of the smaller earthquake as just a patch on the rupture plane of the larger earthquake, then it makes sense that the ground motion from the larger earthquake would be at least as large as the ground motion from the small patch.

The problem with this simple concept is that the slip for the large earthquake is not uniform. As a result, the patch of slip that coincides with the a subevent at the closest point on the rupture surface to the site may not have high slip. With the patches of high slip (asperities) being randomly located on the rupture plane for long ruptures, the chance that the patch of slip closest to the site will have higher than average slip is small. That is, the chance of a site being located right next to the asperity is small for large earthquake magnitudes with long ruptures. In contrast, for a small earthquake, the slip must be concentrated in the small rupture area. If the site is located close to a small earthquake, it will be located close to the maximum slip for the source. On average, the slip on the subevents for larger earthquakes will be larger than the slip for a single subevent for smaller earthquakes, but the slip will be variable along the rupture of the larger earthquake, so a site located at a short distance from the large event rupture may be next to the high-slip patch. This explanation provides a physical basis for the oversaturation seen in the short-distance ground-motion data for M4 to M6 earthquakes.

Figure 20 shows the magnitude scaling of the PSA(T) for a $Z_{TOR} = 3km$ and rupture distance of 3 km. At long periods ($T \geq 0.5sec$), the magnitude-dependence of the corner frequency and the weaker adjustment to the distance slope leads to monotonically increasing ground motion with increasing magnitude. At short periods, there is oversaturation for this very short distances. For T=0.2 sec, the earthquakes with magnitudes between 4.5 and 5.8 have ground motions that exceed the ground motion from a M6.0 with the oversaturation up to a factor of 1.4 at M5.

Figure 20 shows the magnitude scaling for a $Z_{TOR} = 1km$ and rupture distance of 1 km. At this short distance, the oversaturation at short periods extends from M3.7 to M5.5. The oversaturation is up to a factor of 2 at M5.

Finally, Figure 20 shows the magnitude scaling for a $Z_{TOR} = 1km$ and rupture distance of 10 km. At 10 km and greater distances, the short-period ground motion increases monotonically with magnitude, and there is no oversaturation.

3.9 Mogul Earthquake

As a check, the PGA distance scaling for the modified ASK14 model for M5 is compared to the PGA data from the Mogul earthquake in Figure 21. The modified ASK14 model is consistent with the observed PGA values at a distance of 3 km, but it over estimates the observed PGA values for distasnce of 10-50 km by a factor of 2. The increase in the distance slope at short distances for the Mogul earthquake is stronger than in the modified ASK14 model. For example, the scaling from 10 km to 3 km from the Mogul data is about a factor of 13 whereas the scaling from 10 km to 3 km in the modified ASK14 for M5 earthquakes is about a factor of 5. For comparison, the scale factor from 10 km to 3 km in the original ASK14 model for M5 is a factor of 2.5. This comparison may suggest that the magnitude taper applied to the distance slope (c_2 term) should be extended to higher magnitudes.

3.10 Conclusions and Limitations

The data from induced earthquakes in the Oklahoma region show a significantly steeper distance slope at distances less than 15 km, but the short-distance scaling is mainly constrained by M3-M4 earthquakes. The extrapolation of this short-distances scaling trend to larger magnitudes ($M > 5$) needs a physical basis to constrain the model. The 2008 Mogul, Nevada earthquake provides evidence for the steeper distance slope for M5 earthquakes, but it is just one event. If the short-distance scaling in the Mogul earthquake is typical for shallow events, then there must be oversaturation in the M4 to M6 range. Finite-fault modeling should be used to develop a stronger technical basis to determine the amount of oversaturation that is appropriate. This issue of oversaturation will likely be a key issue for the potential for shallow induced earthquakes to cause significant structural damage at short rupture distances.

The proposed model represents a significant improvement over the ASK14 model in terms of it applicability to shallow M3 to M5.5 earthquakes at short rupture distances. The revisions capture key physical behaviors, such as the magnitude dependence of the finite-fault term that should be included in GMMs. The revised ASK14 model is considered to be applicable to both induced and tectonic earthquakes.

The amplitude of the long-period ground motions, which scale with the moment magnitude, indicate that the converted magnitudes for small earthquakes ($—M < 4.1$) were systematically too high by 0.25 units for Set 2. A correction was applied to the converted moment magnitudes based on the long-period amplitudes. This correction allowed the estimation the adjustments to the short-distances scaling in the ASK14 model, but going forward, the magnitudes of the induced earthquake data set should have a detailed review and be revised if needed.

A second limitation is the lack of spectral values for periods between 0.01 and 0.1 in the PIE database. In this period range, the c_1 and c_2 terms are based on the judgment and not data. The modifications for periods less than 0.1 sec should be reviewed with data sets that include spectral values at short spectral periods as they become available.

Table 1: Coefficients for the modified ASK14 model

Period (sec)	c_1 (CEUS)	c_1 c_2	R_c
0.01	0.18	-1.20	15
0.02	0.18	-1.20	15
0.03	0.18	-1.20	15
0.05	0.26	-1.07	15
0.06	0.29	-1.03	15
0.075	0.33	-0.98	15
0.10	0.33	-0.90	15
0.12	0.25	-0.86	15
0.15	0.15	-0.81	15
0.20	0.0	-0.73	15
0.25	0.0	-0.68	15
0.30	0.0	-0.64	15
0.40	0.0	-0.56	15
0.50	0.0	-0.51	15
0.75	0.0	-0.41	15
1.0	0.0	-0.34	15
1.5	0.0	-0.24	15
2.0	0.0	-0.17	15
3.0	0.0	-0.07	15
4.0	0.0	0.00	15
5.0	0.0	0.00	15
7.5	0.0	0.00	15
10.0	0.0	0.00	15

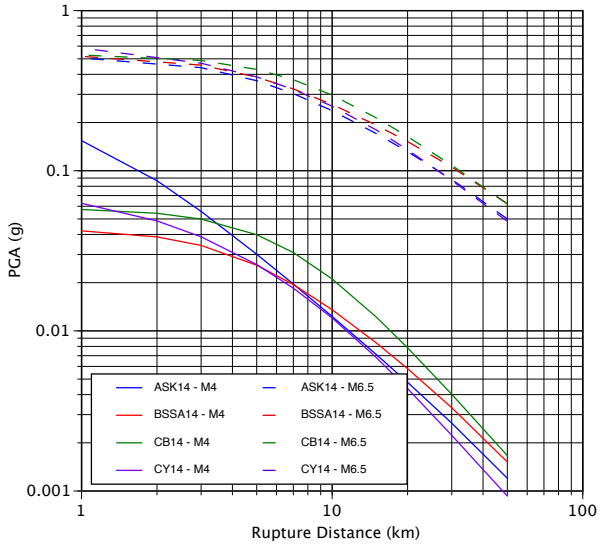


Figure 2: Short-distance scaling in the NGA-W2 GMMs for M6.5 and M4.0 earthquakes.

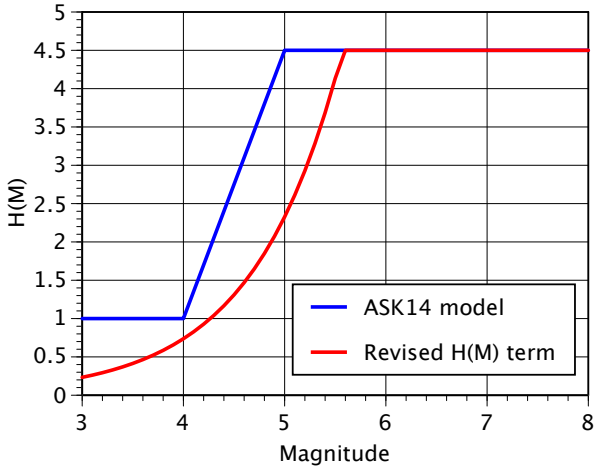


Figure 3: Comparison of the ASK14 finite-fault term and the revised finite-fault term.

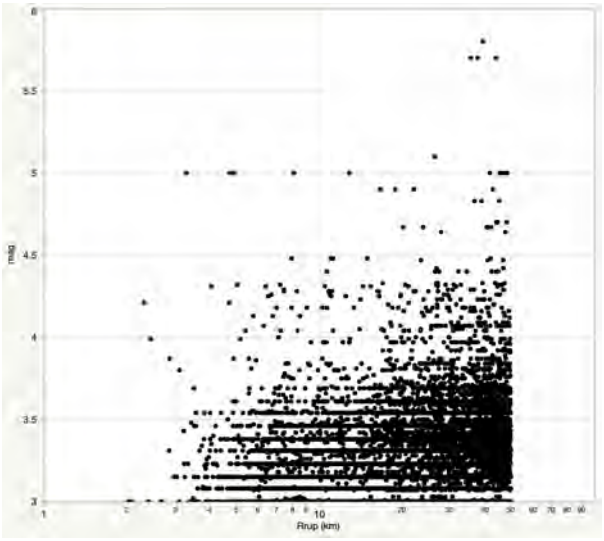


Figure 4: Magnitude-distance distribution of the selected data set.

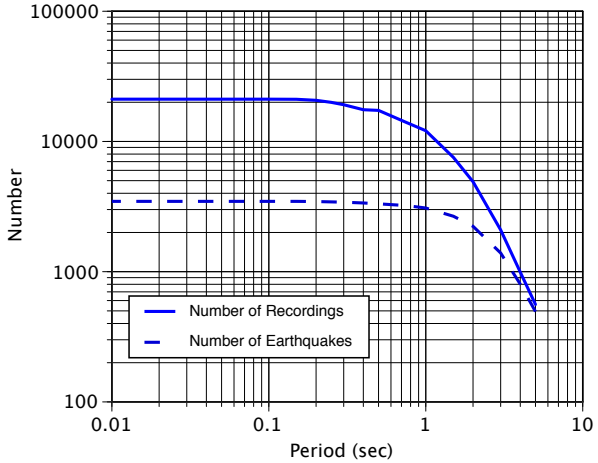


Figure 5: Number of recordings and earthquakes for the selected data set.

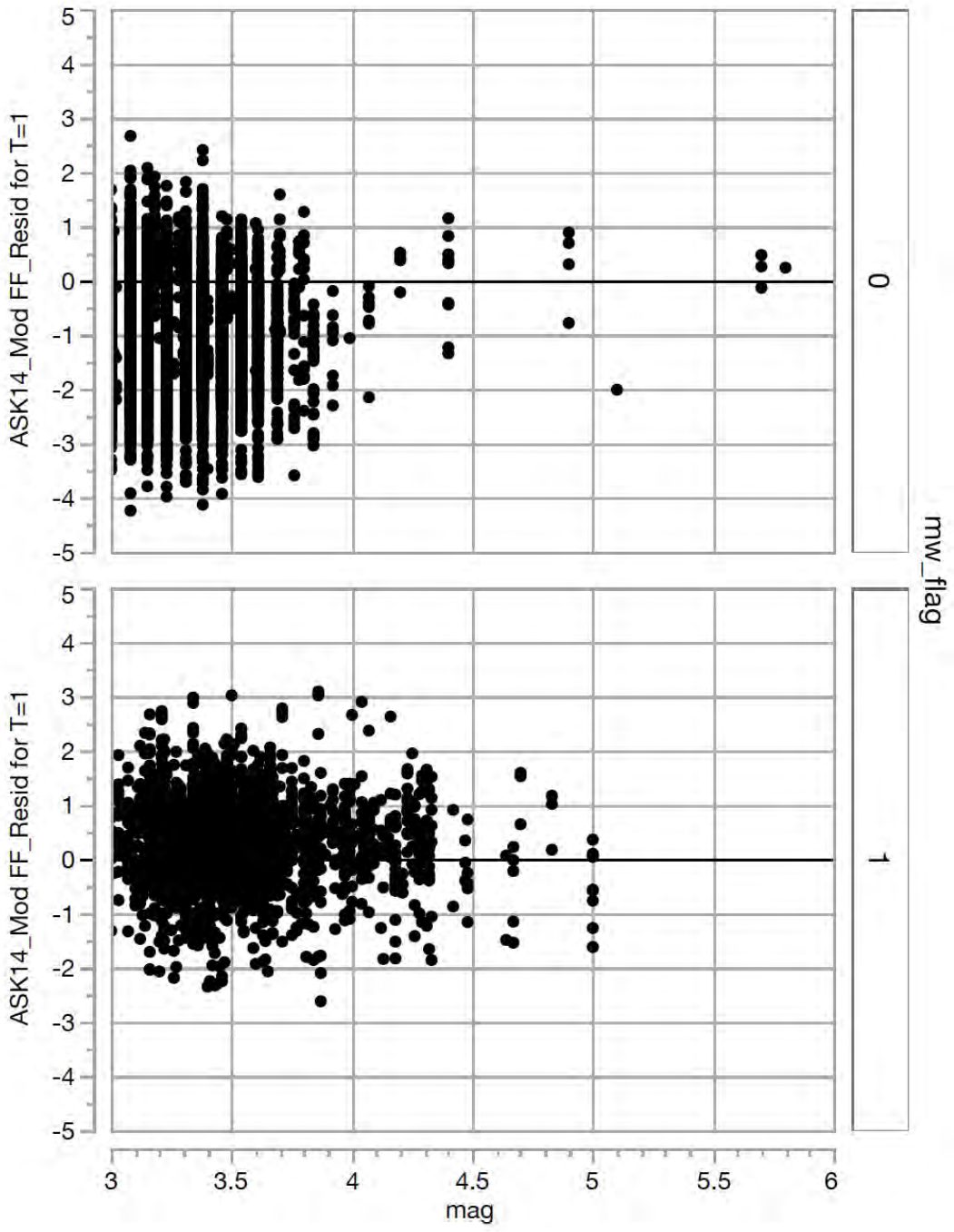


Figure 6: Total residuals from ASK14 GMM with the modified finite-fault term for T=1 sec as a function of magnitude. Top panel is for Set 2 and bottom panel is for Set 1.

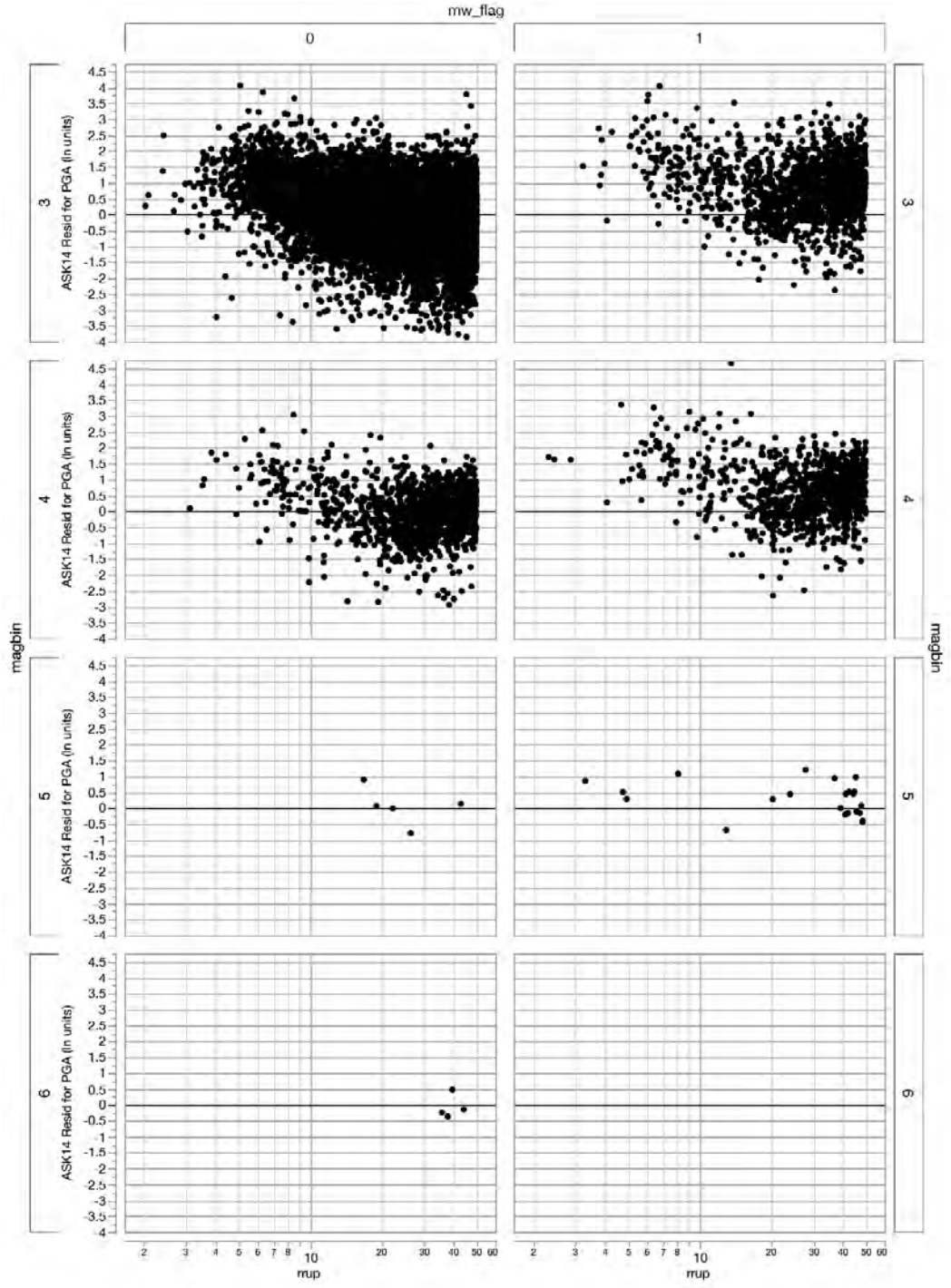


Figure 7: Total residuals from the unmodified ASK14 GMM for PGA by magnitude bin. Left panels are for Set 2 (converted to moment magnitude). Right panels are for Set 1 (magnitudes not converted)

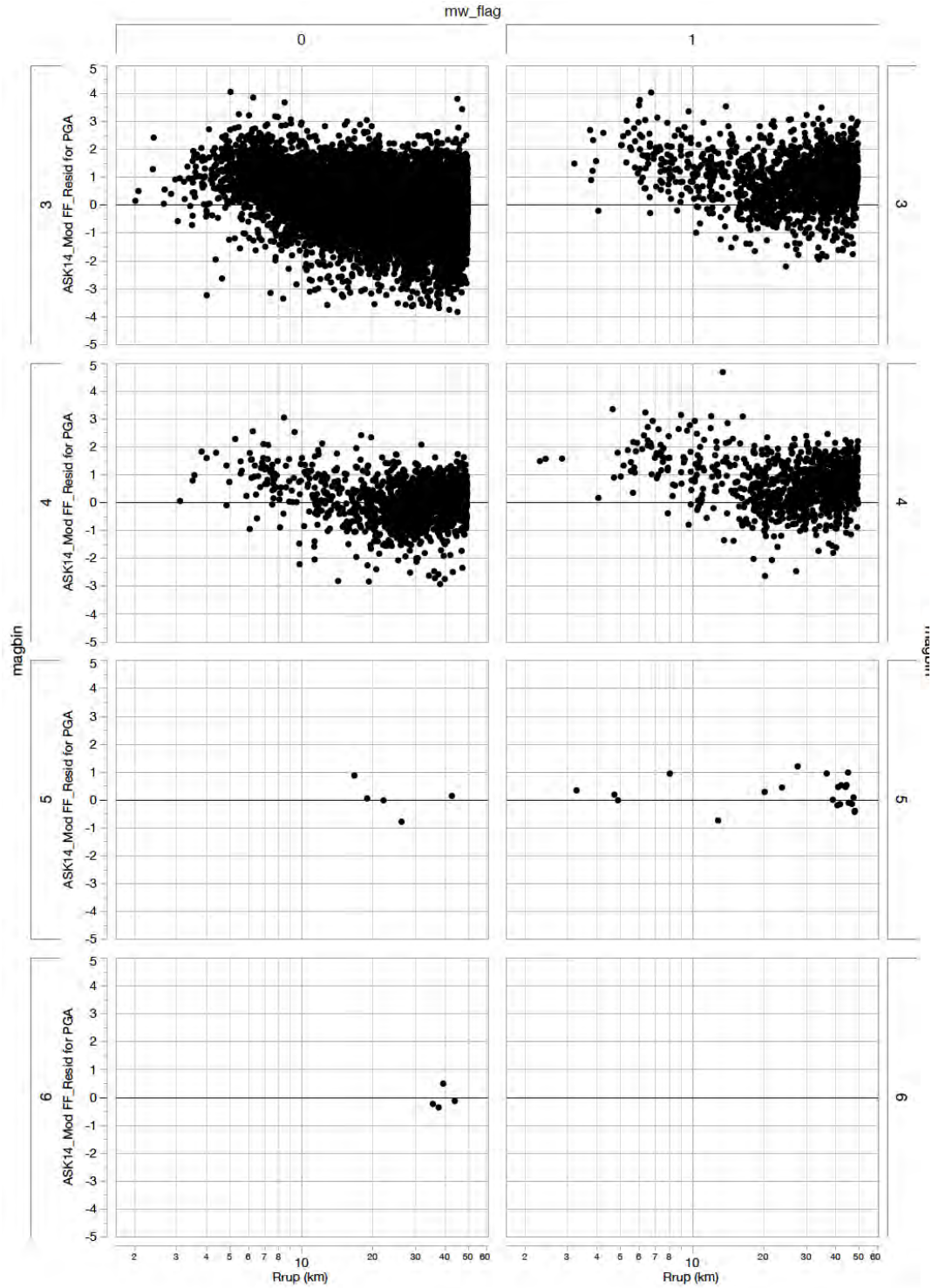


Figure 8: Total residuals from the ASK14 GMM with the modified finite-fault term for PGA by magnitude bin. Left panels are for Set 2 (converted to moment magnitude). Right panels are for Set 1 (magnitudes not converted) *** IS THIS CORRECT FIGURE ****

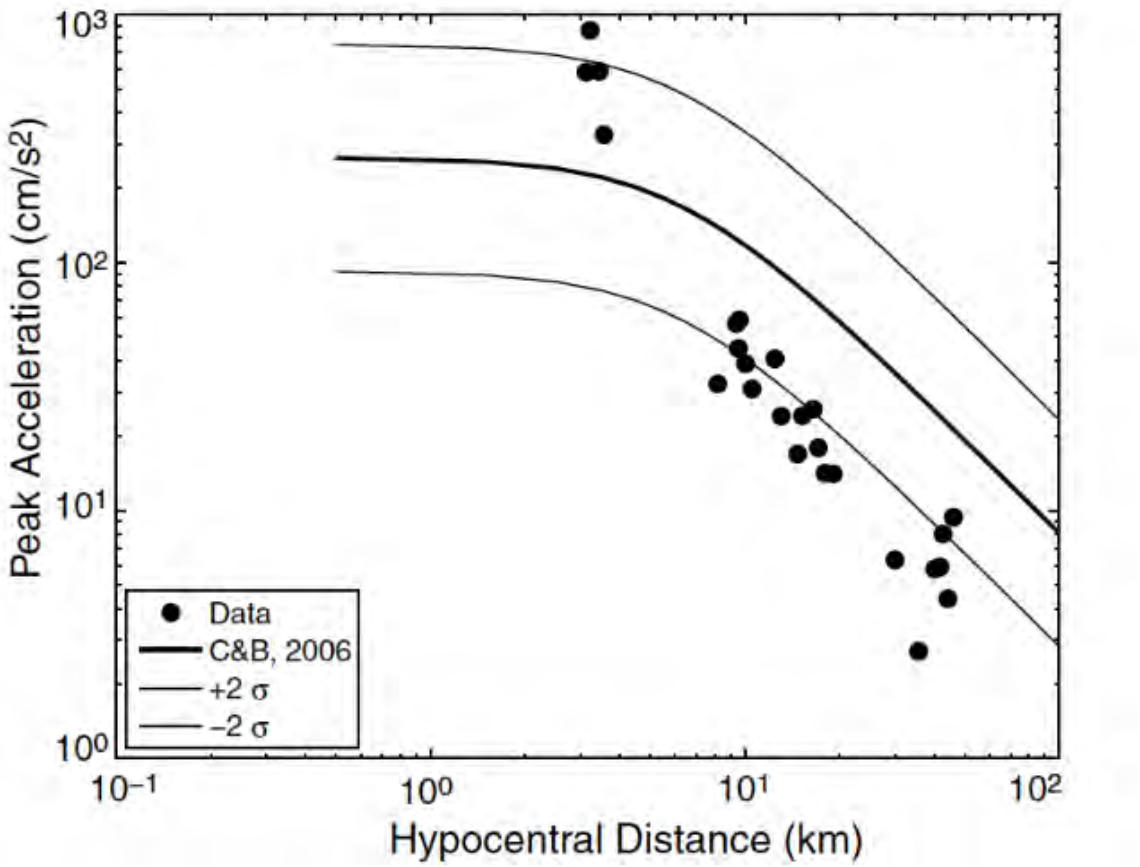


Figure 9: Short-distance scaling in the 2008 Mogul earthquake. (from Anderson et al. 2009)

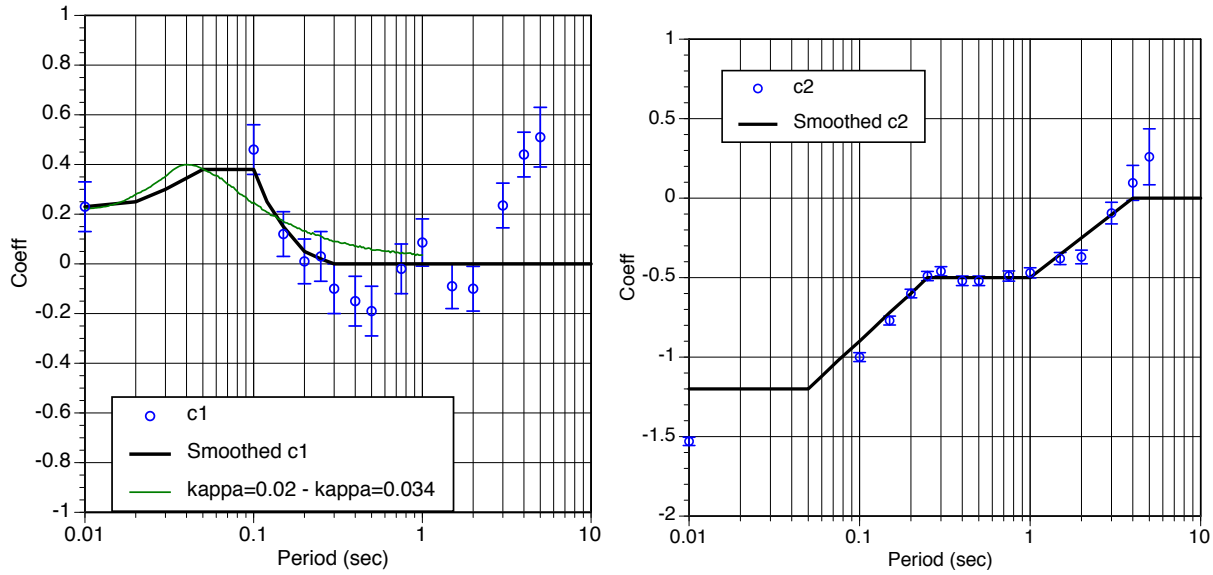


Figure 10: Smoothed coefficients: c_1 for the constant term (left frame) and c_2 for the ln(dist) slope (right frame).

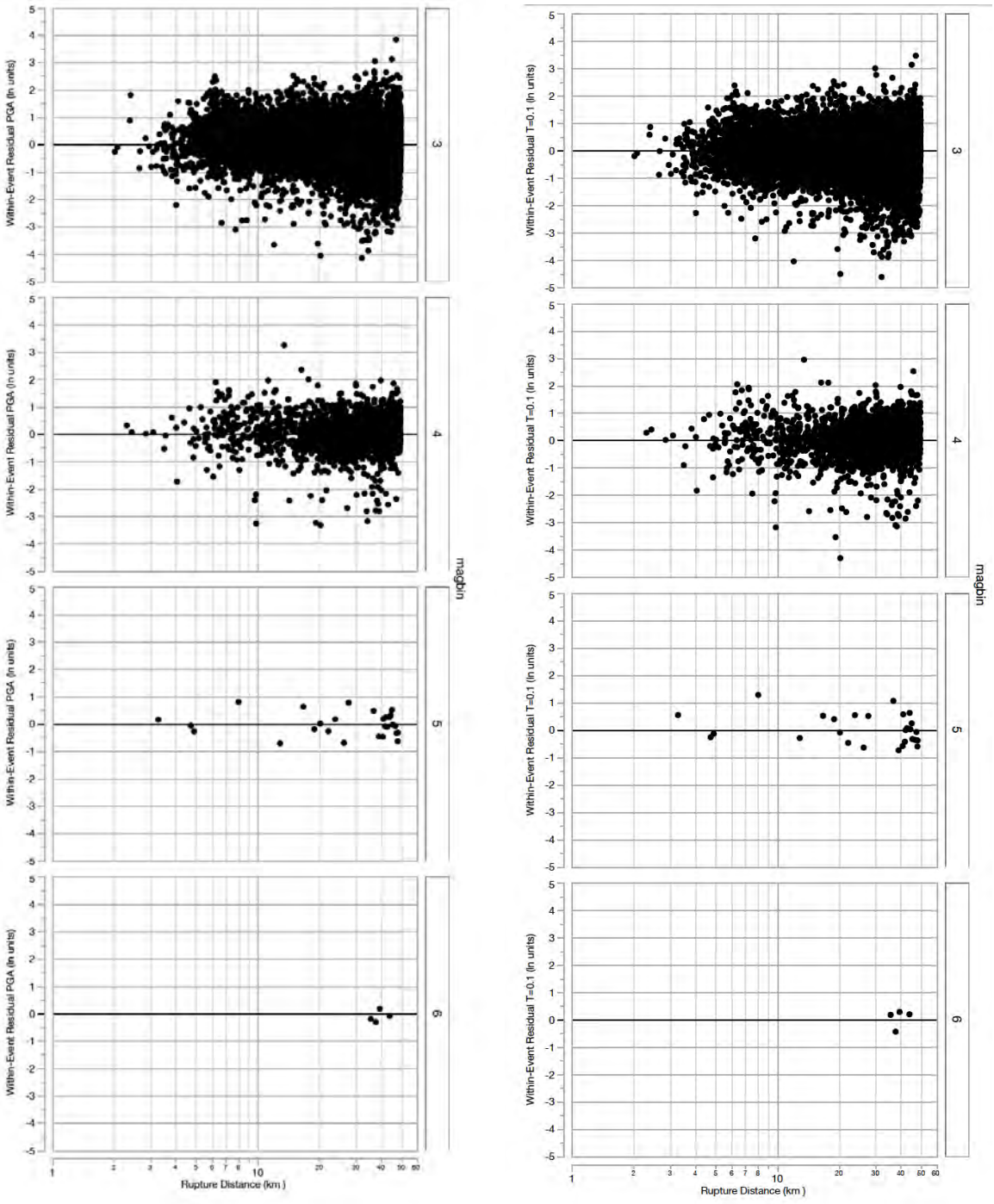


Figure 11: Within-event residuals for PGA (left frame) and T=0.1 sec (right frame) as a function of distance for different magnitude bins: $3.0 \leq M < 3.5$ (top row), $3.5 \leq M < 4.5$ (second row), $4.5 \leq M < 5.5$ (third row), and $5.5 \leq M < 6.0$ (bottom row)

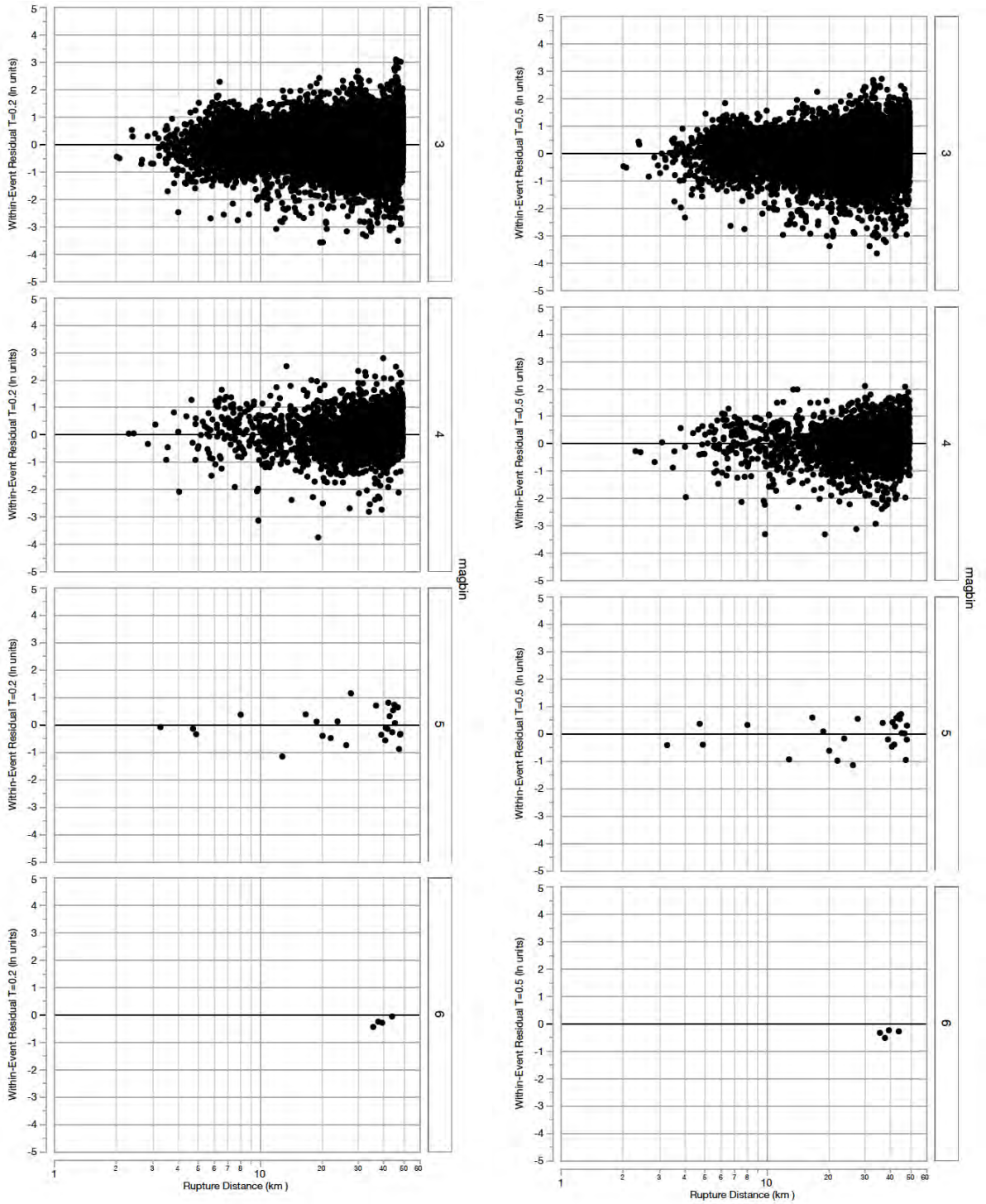


Figure 12: Within-event residuals for T=0.2 sec (left frame) and T=0.5 sec (right frame) as a function of distance for different magnitude bins: $3.0 \leq M < 3.5$ (top row), $3.5 \leq M < 4.5$ (second row), $4.5 \leq M < 5.5$ (third row), and $5.5 \leq M < 6.0$ (bottom row)

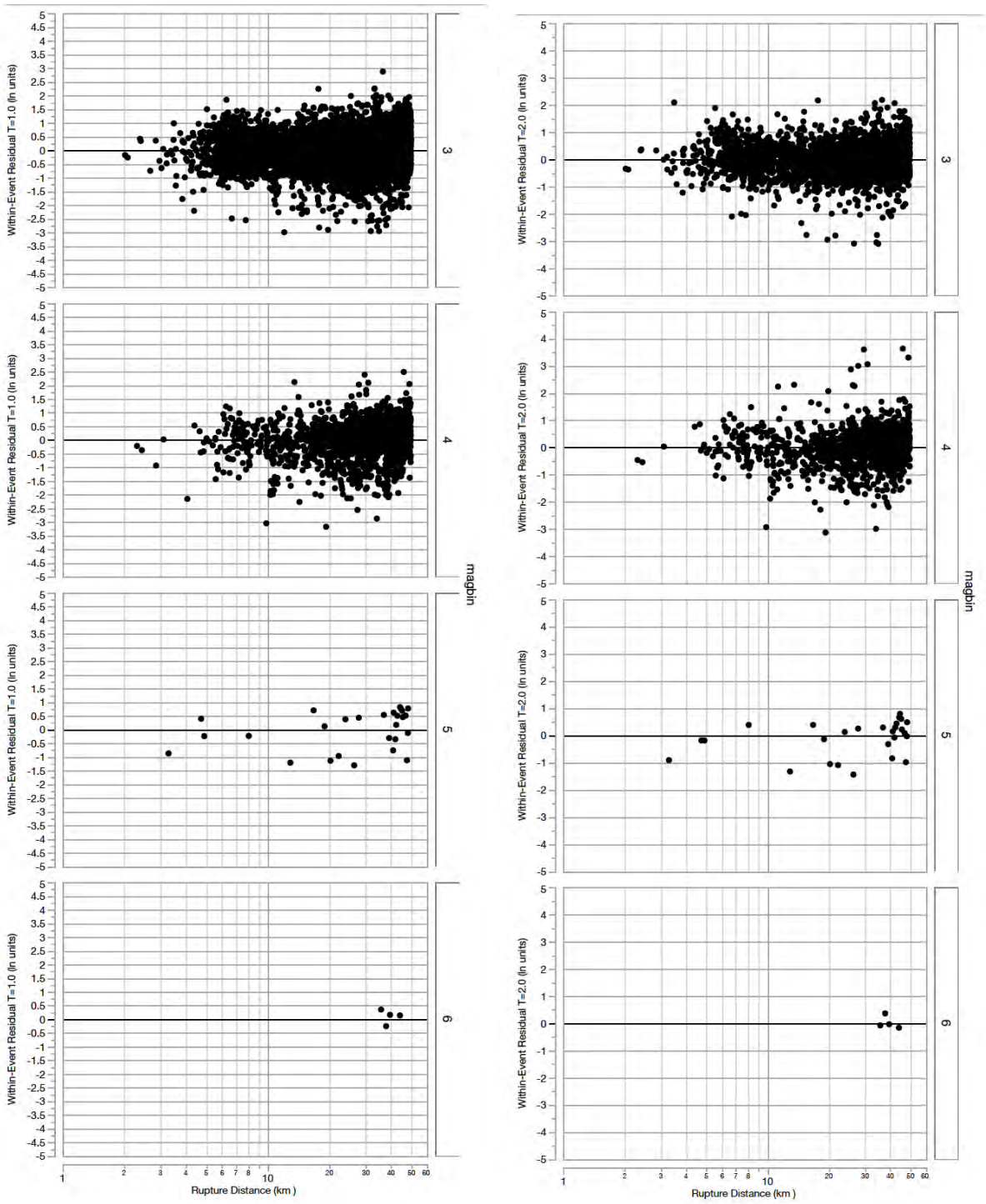


Figure 13: Within-event residuals for T=1 sec (left frame) and T=2 sec (right frame) as a function of distance for different magnitude bins: $3.0 \leq M < 3.5$ (top row), $3.5 \leq M < 4.5$ (second row), $4.5 \leq M < 5.5$ (third row), and $5.5 \leq M < 6.0$ (bottom row)

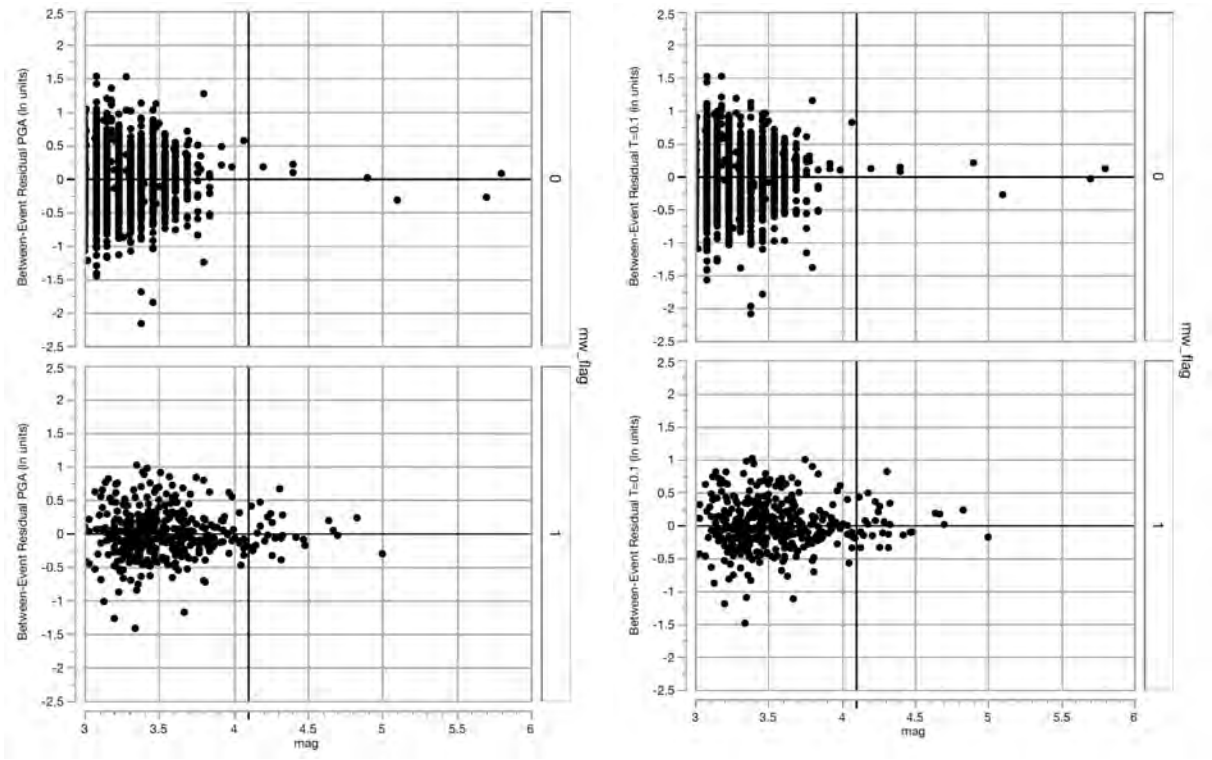


Figure 14: Between-event residuals for PGA (left frames) and T=0.1 sec (right frames) as a function of magnitude. The top row is for earthquakes with magnitudes converted to moment magnitude. The bottom row is for earthquakes with moment magnitude reported. Only events with $M \geq 4.1$ are used for the constant term.

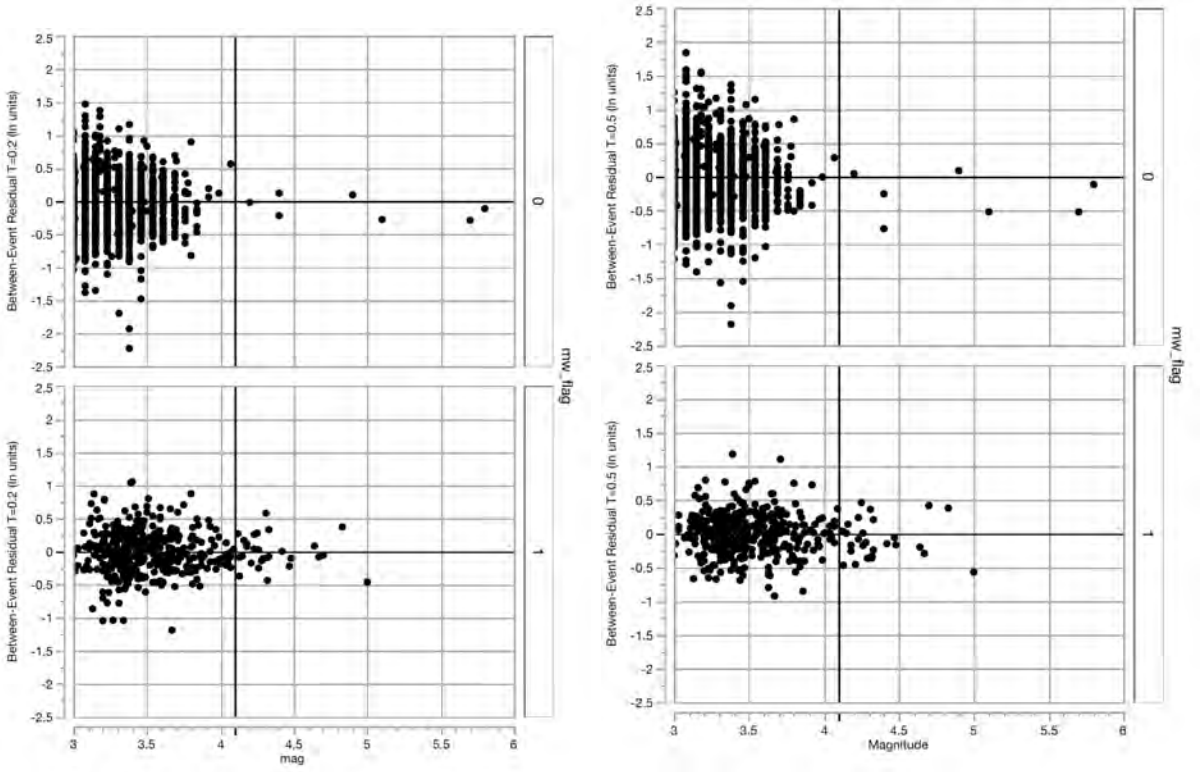


Figure 15: Between-event residuals for T=0.2 sec (left frames) and T=0.5 sec (right frames) as a function of magnitude. The top tow is for earthquakes with magnitudes converted to moment magnitude. The bottom row is for earthquakes with moment magnitude reported. Only events with $M \geq 4.1$ are used for the constant term.

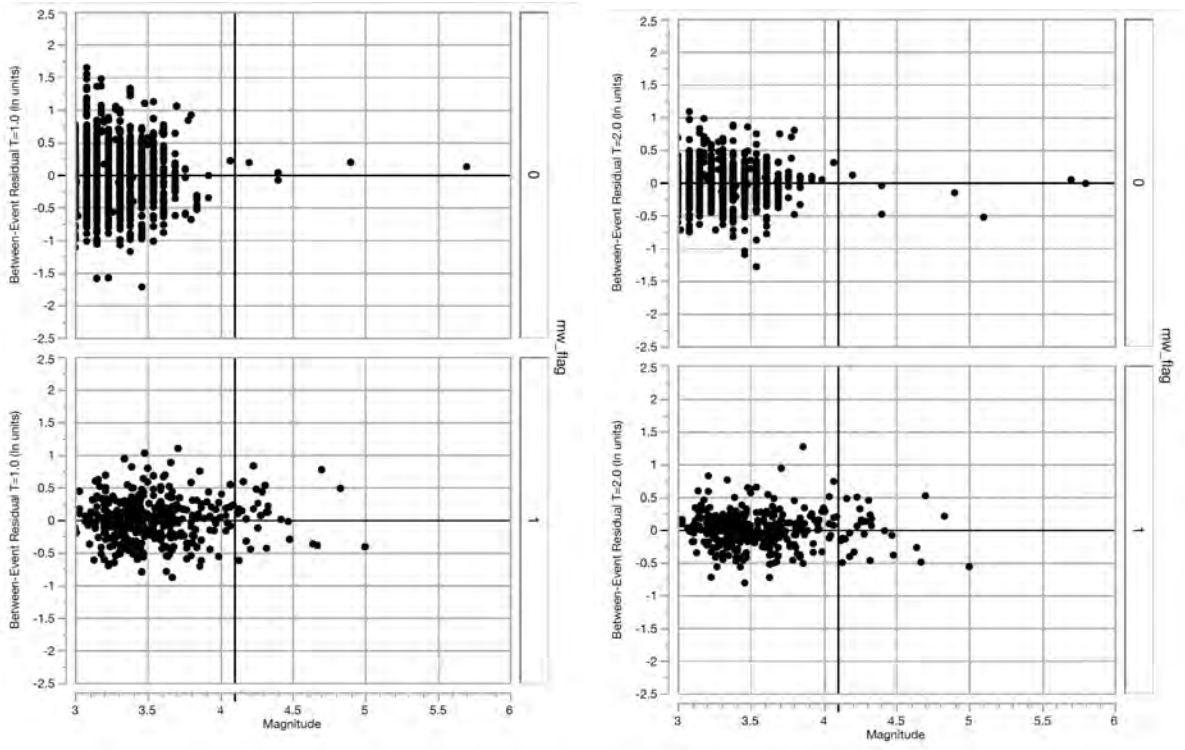


Figure 16: Between-event residuals for T=1 sec (left frames) and T=2 sec (right frames) as a function of magnitude. The top tow is for earthquakes with magnitudes converted to moment magnitude. The bottom row is for earthquakes with moment magnitude reported. Only events with $M \geq 4.1$ are used for the constant term.

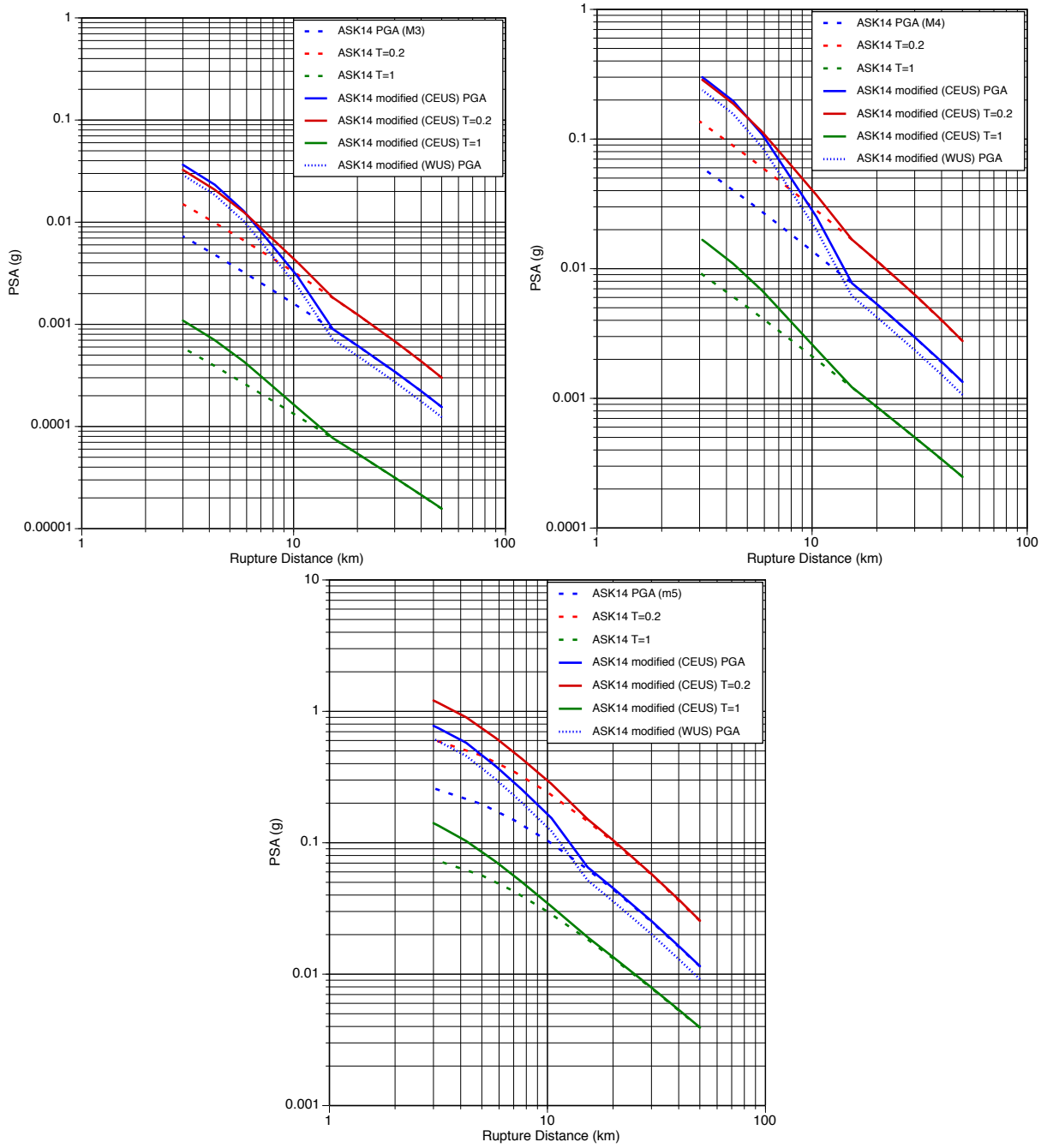


Figure 17: Comparison of the distance scaling for the ASK14 and modified ASK14 models for M3 (upper left), M4 (upper right), and M5 (lower) earthquakes for $Z_{TOR} = 3km$.

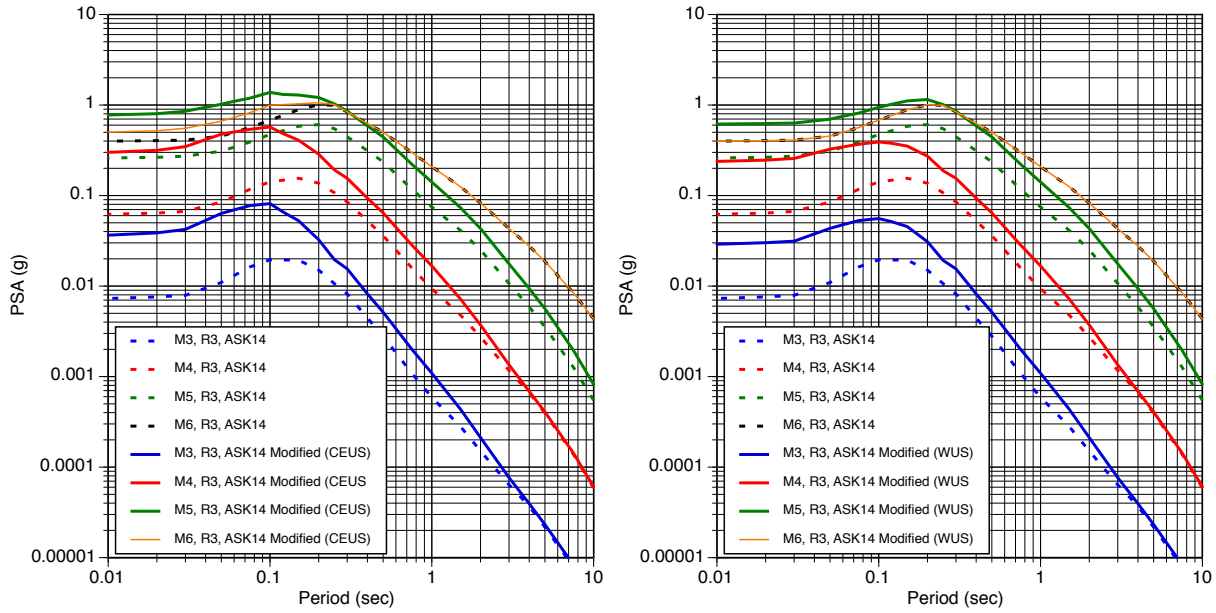


Figure 18: Comparison of the response spectra for the ASK14 and modified ASK14 model for M3, M4, M5, and M6 earthquakes with $Z_{TOR} = 3km$ and a rupture distance of 3 km. The CEUS site term is shown in the left panel and the WUS site term is shown in the right panel.

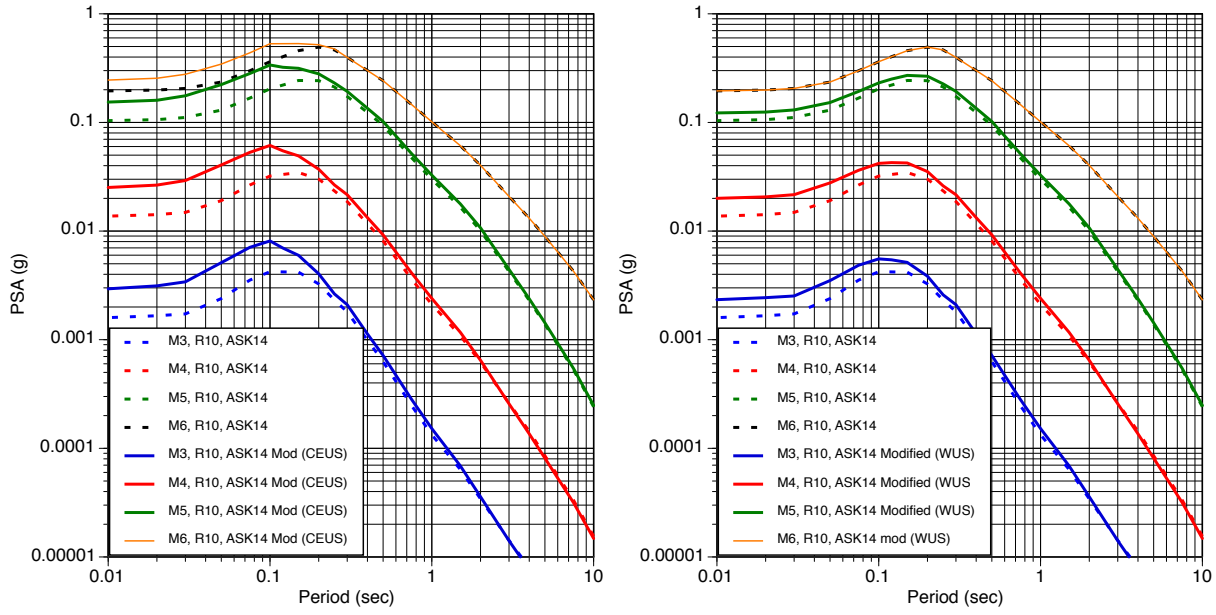


Figure 19: Comparison of the response spectra for the ASK14 and modified ASK14 models for M3, M4, M5, and M6 earthquakes with $Z_{TOR} = 3km$ and a rupture distance of 10 km. The CEUS site term is shown in the left panel and the WUS site term is shown in the right panel.

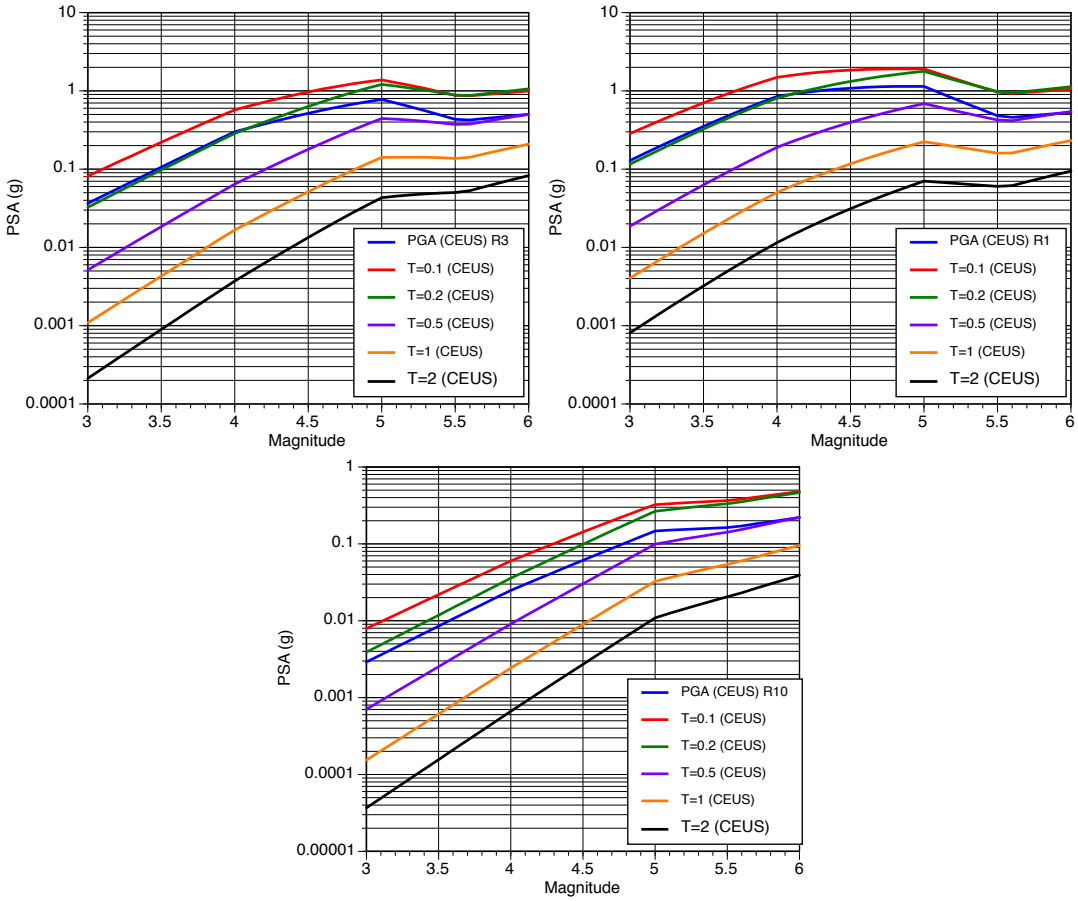


Figure 20: Magnitude scaling of the modified ASK14 GMM. The upper left panel is for $Z_{TOR} = 3km$ and a rupture distance of 3 km. The upper right panel is for $Z_{TOR} = 1km$ and a rupture distance of 1 km. The bottom panel is for $Z_{TOR} = 1km$ and a rupture distance of 10 km. There is oversaturation in the magnitude 4.5 to 6 range at short periods at short distances.

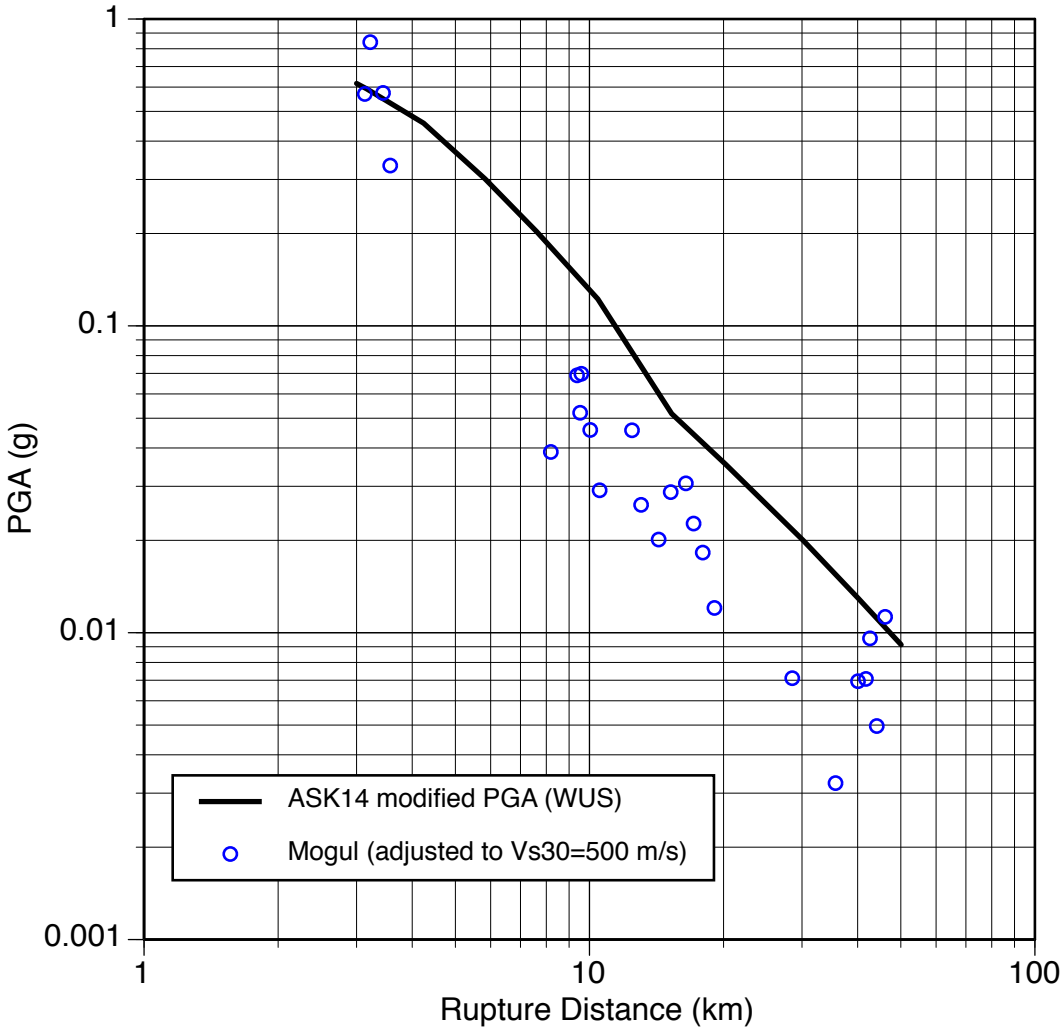


Figure 21: Comparison of the PGA scaling with distance for the modified ASK14 model and the data from the Mogul earthquake.

4 Update to CB14

4.1 Introduction

This section explains the changes made to the GMM of Campbell and Bozorgnia (2014). The changes are minor, and only affect the distance scaling for smaller event ($M < 6.5$) at short distances. There is a clear trend in the scaling of estimated event terms with V_{S30} for the induced data; however, this is an indication that the V_{S30}-scaling of Campbell and Bozorgnia (2014) (estimated from California data) is not applicable to Oklahoma. It is assumed that V_{S30}-scaling is a regional/local effect, and would be different for other regions as well (e.g. British Columbia). Hence, no new V_{S30}-scaling coefficient is estimated.

4.2 Data Selection

The underlying data set is taken from Gregor (2018). The Oklahoma and Kansas data was used to update the model of Campbell and Bozorgnia (2014); this data comes from Rennolet et al. (2018). Only records with distances $R < 50$km are used; all earthquakes with $M \geq 3.0$ are selected. Only records from events and stations with at least 3 recordings are selected. In total, 19,442 records from 2347 events and 159 stations have been selected for further analysis. The magnitude/distance distribution of the selected data is shown in Figure 22.

4.3 Residuals to CB14

In Figure 23, the residuals of the observed data with respect to the median predictions of Campbell and Bozorgnia (2014) are shown, for PGA. The residuals are calculated as

$$\delta R = \ln PGA - \mu_{CB14} \tag{11}$$

Based on Figure 23, there is only a small trend with magnitude and hypocentral depth; hence, no change is made to the scaling of Campbell and Bozorgnia (2014) with respect to these param-

eters. Even though there is a trend with V_{S30}, this is a regional effect, and would change if the model needs to be applied to other regions.

The strongest trend can be observed with the hypocentral distance R_{HYPO}, for shorter distances. Since this trend occurs at distances $R_{HYPO} < 20$km, it is likely due to the near-fault term of Campbell and Bozorgnia (2014), which is magnitude independent in its original form. Hence, in the adjusted model a magnitude dependence is introduced to this term. The full adjustment is described in the next section.

4.4 Adjustment to CB14

The functional form of the model of Campbell and Bozorgnia (2014) is described in their paper, Equations (1) to (25). The general form is

$$\ln Y = c_0 + f_{mag} + f_{dis} + f_{flt} + f_{hng} + f_{site} + f_{sed} + f_{hyp} + f_{atn} \tag{12}$$

where the geometric attenuation term is (Equation (3) in Campbell and Bozorgnia (2014))

$$f_{dis,CB14} = (c_5 + c_6 \boldsymbol{M}) \ln \sqrt{R_{RUP}^2 + c_7^2} \tag{13}$$

For the adjustment based on the induced data, the coefficient c_7 becomes magnitude dependent

$$c_{7,adj} = \begin{cases} MAX(a(M_w - M_b) + c_7, 0.1) & M_w \leq M_b \\ c_7 & m_b < M_w \end{cases} \tag{14}$$

Hence, the near-fault parameter c_7 is magnitude dependent for $M_w < M_b$, and reverts back to the value of Campbell and Bozorgnia (2014) for larger magnitudes. The value of the magnitude break point M_b is not well constrained by data. Hence, the regression is performed for three different values of $M_b = 4.5, 5.5, 6.5$, to include some epistemic uncertainty for this value.

In the regression, the parameter a in Equation 14 is estimated. In addition, the event terms,

δB, and station terms, δS, for each event and station, respectively, are estimated as well. The associated standard deviations τ, ϕ_{SS} and ϕ_{SS} are also estimated.

Hence, the full model is

$$\ln Y \sim N(mu, \phi_{SS}) \tag{15}$$

$$\mu = c_0 + f_{mag} + f_{dis,adj} + f_{flt} + f_{hng} + f_{site} + f_{sed} + f_{hyp} + f_{atn} + \delta B + \delta S \tag{16}$$

$$\delta B \sim N(0, \tau) \tag{17}$$

$$\delta S \sim N(0, \phi_{S2S}) \tag{18}$$

where $f_{dis,adj}$ is the same as Equation (13), but with the updated near-fault term of Equation (14). The other terms of Equation (16) and their coefficients are taken directly from the model of Campbell and Bozorgnia (2014).

The parameters are estimated via Bayesian inference, using the program Stan Carpenter et al. (2017). Stan estimates the posterior distribution of the parameters via Markov Chain Monte Carlo (MCMC) sampling, in particular employing Hamiltonian Monte Carlo Betancourt (2017); Neal (2011). Hence, the outcome are samples from the posterior distribution. For each period, we run 2 chains with 400 iterations each. The first 200 iterations are discarded as burn-in.

Bayesian inference requires to specify prior distributions for the parameters. Here, we use weakly informative prior distributions. The parameters which are estimated and their prior distributions are given in Table 2.

4.5 Results

The estimated coefficients a are shown in Figure 24 as a function of period. The estimated standard deviations ϕ_{SS}, ϕ_{S2S} and τ are shown in Figure 25.

The residuals for the regression for $T = 0$s are shown in Figures 26, 27, and 28, for magnitude break points $M_b = 4.5, 5.5, 6.5$, respectively. Overall, the residuals are very similar, indicating that the three models can explain the data in a similar manner. As stated before, there is a trend in

the station terms, indicating that the V_{S30}-scaling of California is not applicable to Oklahoma and Kansas. There is a trend in the magnitude scaling, but it is not accounted for due to suspected problems with the converted magnitudes. The trend in the within-event/within-site residuals at short distances is reduced.

Figure 29 shows the within-event/within-site residuals for the three magnitude break points, and for different periods. Overall, there is only a minor trend at short distances, and almost no differences between the different models using different magnitude break points M_b.

In Figure 30, the scaling of PSA against distance is shown, for the original CB14 model, and the adjusted models. At longer distance ($R_{RUP} > 20km$), the adjusted models revert back to the original CB14 model. At short periods, the adjusted models predict much larger ground motions at small magnitudes, while the large magnitudes ($M_w > M_b$) are unchanged.

Figure 31 shows spectra for the original CB14 models and the adjusted models for different magnitudes at a distance of $R_{RUP} = 5km$.

Table 2: Parameters that are estimated for the adjusted model of Campbell and Bozorgnia (2014) and their prior distributions.

Parameter	Prior Distribution
a	Normal(0,1)
ϕ_{SS}	Exponential(0.5)
ϕ_{S2S}	Exponential(0.5)
τ	Exponential(0.5)

$T(s)$	$a_{4.5}$	$q_5(a_{4.5})$	$q_{95}(a_{4.5})$	$a_{5.5}$	$q_5(a_{5.5})$	$q_{95}(a_{5.5})$	$a_{6.5}$	$q_5(a_{6.5})$	$q_{95}(a_{6.5})$
0.	5.19377	5.00596	5.37631	2.99101	2.88262	3.09864	2.08045	2.00844	2.15627
-1	4.45522	4.26733	4.64169	2.56422	2.45656	2.66832	1.78807	1.71294	1.86434
0.1	5.95433	5.65384	6.22941	3.43445	3.22978	3.60217	2.37323	2.23646	2.51238
0.132996	6.66617	6.37014	6.95696	3.88381	3.69171	4.07092	2.68498	2.55224	2.82112
0.149993	6.98831	6.70385	7.28332	4.08167	3.89179	4.23118	2.83446	2.70276	2.95708
0.17001	7.42133	7.14707	7.68916	4.35758	4.17772	4.51664	3.03139	2.89443	3.14397
0.2	7.77686	7.46179	8.04166	4.56397	4.38894	4.71842	3.18761	3.07631	3.28271
0.25	7.13028	6.84931	7.40325	4.18316	4.00066	4.32911	2.92343	2.8149	3.03506
0.30303	6.62241	6.32145	6.92644	3.88824	3.70375	4.06063	2.70612	2.56792	2.81236
0.4	6.41555	6.07558	6.7321	3.7756	3.5889	3.9637	2.64088	2.47951	2.77133
0.5	6.02685	5.56654	6.42201	3.61972	3.39175	3.80192	2.5407	2.36829	2.70099
0.59988	5.04455	4.4369	5.5537	3.01186	2.65546	3.28199	2.09418	1.84223	2.30166
0.750188	4.28015	3.8648	4.69173	2.49741	2.22863	2.72702	1.72973	1.56288	1.89674
1.	4.19977	3.80227	4.56624	2.44724	2.20096	2.67529	1.70773	1.53018	1.86209
1.40056	4.36731	3.96002	4.79211	2.56648	2.34737	2.78367	1.77715	1.59879	1.95079
2.	4.98579	4.34251	5.63926	3.057	2.65992	3.41278	2.13323	1.85992	2.38599
2.5	3.94064	3.21146	4.74757	2.78402	2.20484	3.48777	2.02729	1.58706	2.46994
3.33333	3.16542	2.32706	4.08742	2.42296	1.7153	3.40492	1.97394	1.29969	2.88374
4.	1.94542	0.720842	3.16119	1.85874	0.978041	2.74535	1.46698	0.833895	2.13232
5.	1.49876	0.151956	3.03031	1.79463	0.698877	2.88297	1.56038	0.711514	2.47404

Table 3: Coefficents for Modified CB14.

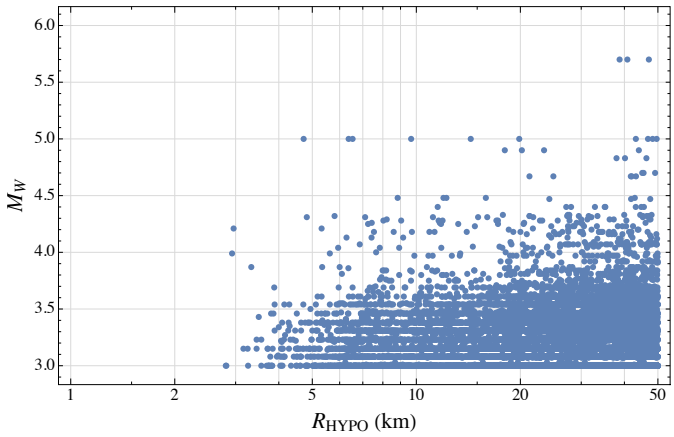


Figure 22: Magnitude/distance scatter plot of the data used to update Campbell and Bozorgnia (2014).

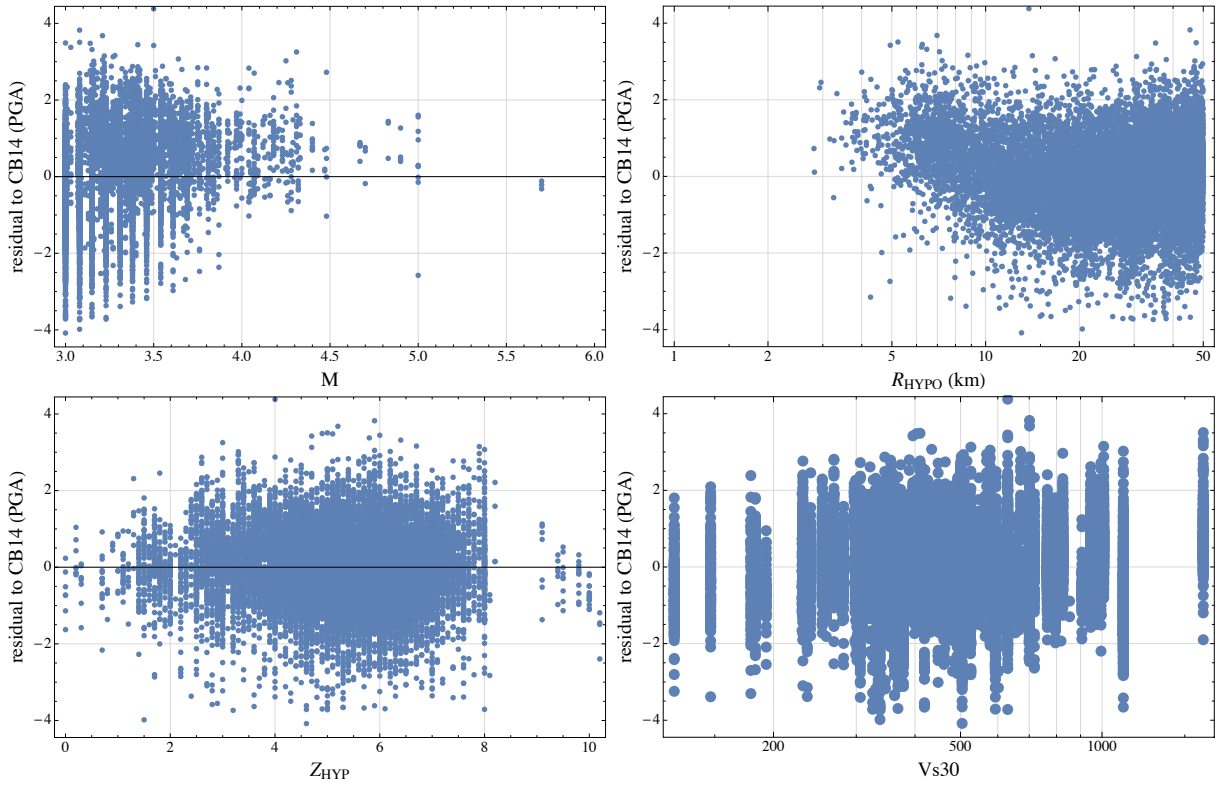


Figure 23: Residuals to Campbell and Bozorgnia (2014).

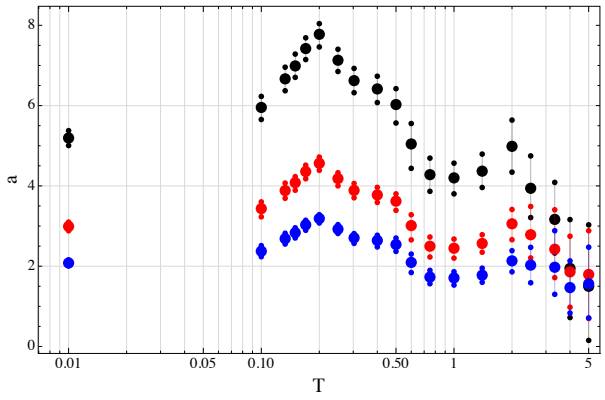


Figure 24: Estimated intercept $c_{0,adj}$ (left) and slope of the near-fault term a (right), for different break points of the magnitude (cf. Equation (14)). Shown are the mean and the 5% and 95% quantiles of the posterior distribution. For the intercept, the uncertainty is small, and barely identifiable in the plot. The intercepts are not very sensitive to the magnitude break point.

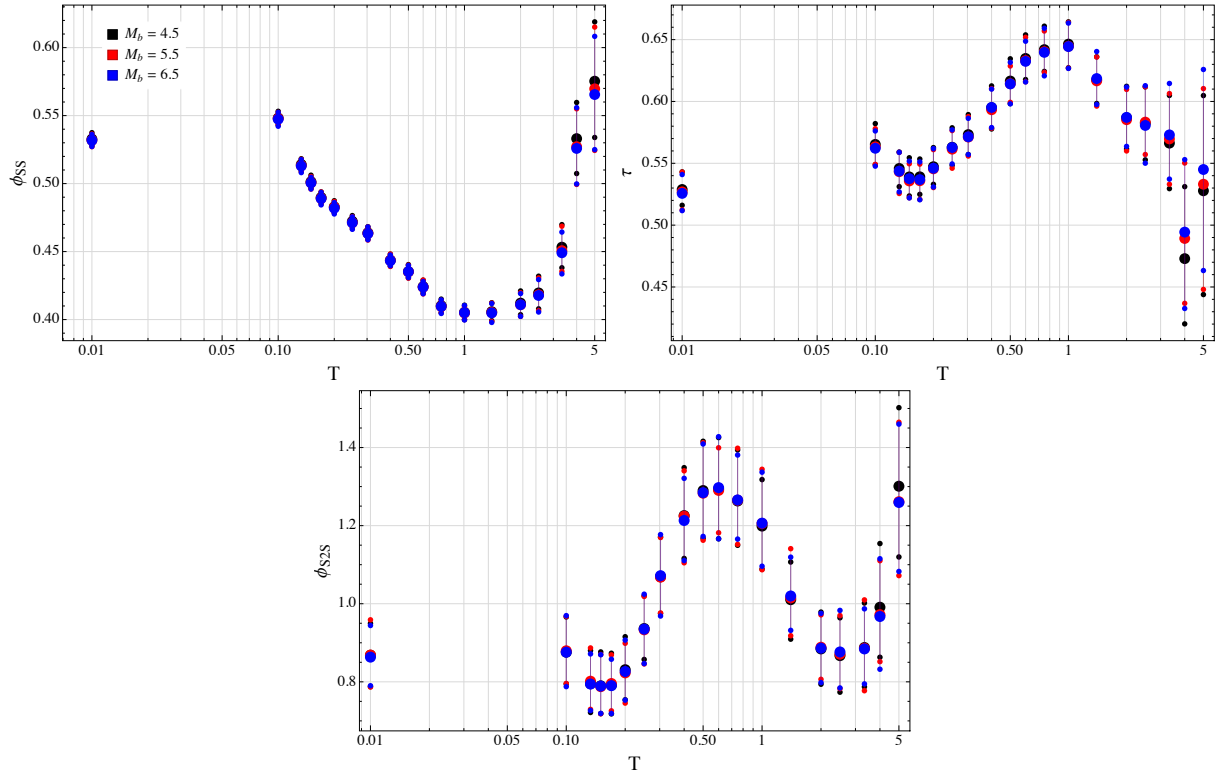


Figure 25: Estimated standard deviations.

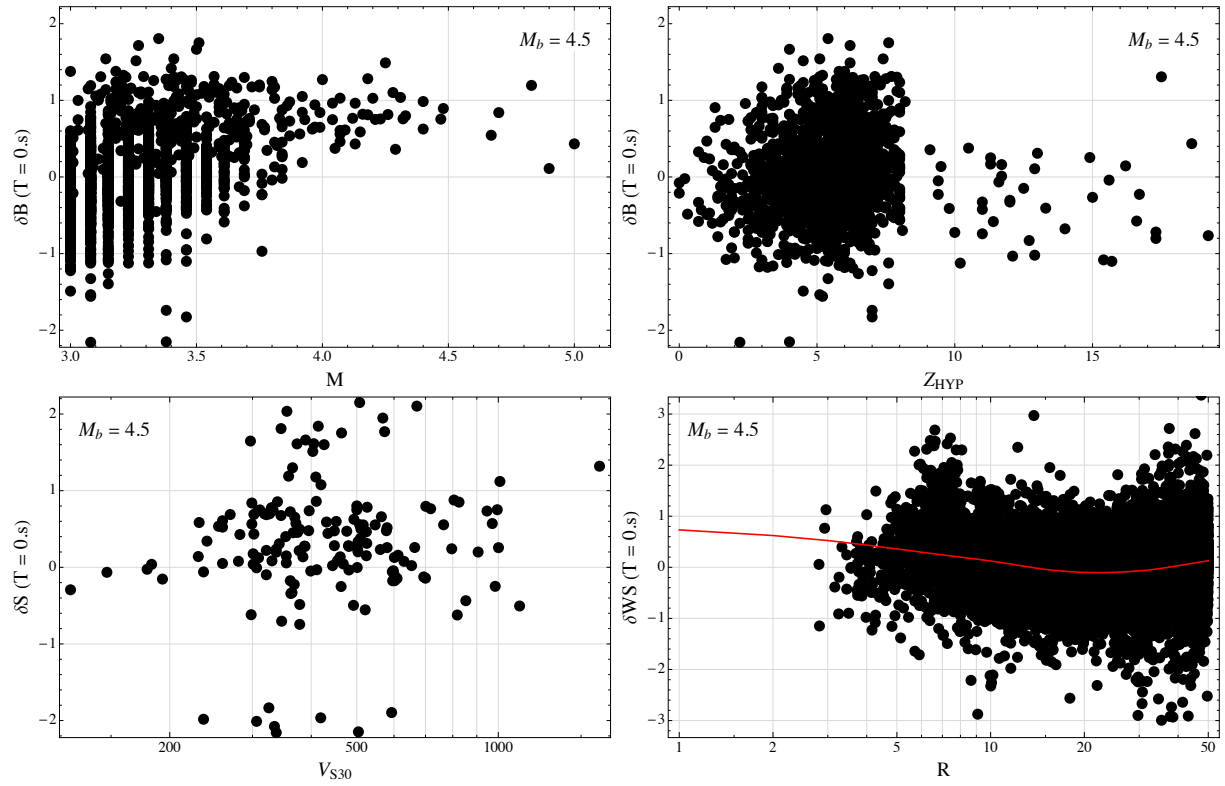


Figure 26: Between-event residuals δB, site-to-site residuals δS, and within-event/within-site residuals δWS, for a regression with magnitude break point $M_b = 4.5$, and period $T = 0$s.

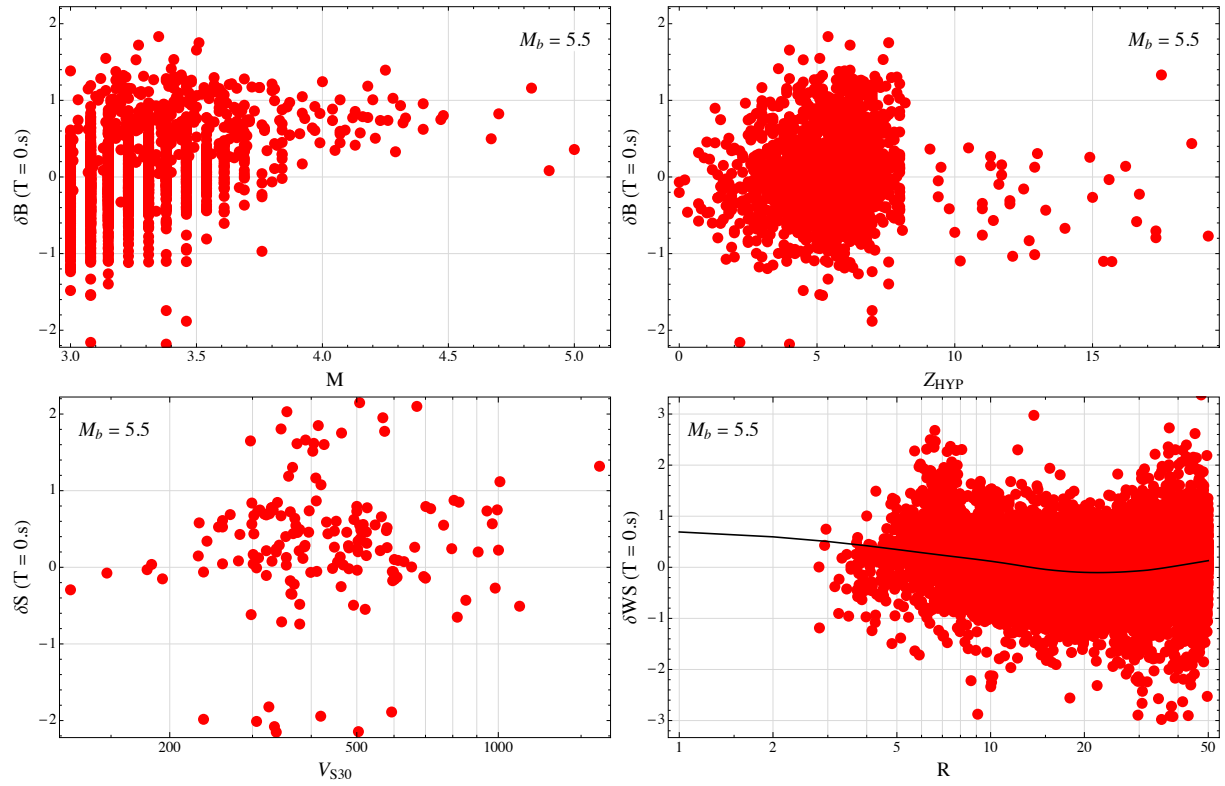


Figure 27: Between-event residuals δB, site-to-site residuals δS, and within-event/within-site residuals δWS, for a regression with magnitude break point $M_b = 5.5$, and period $T = 0$s.

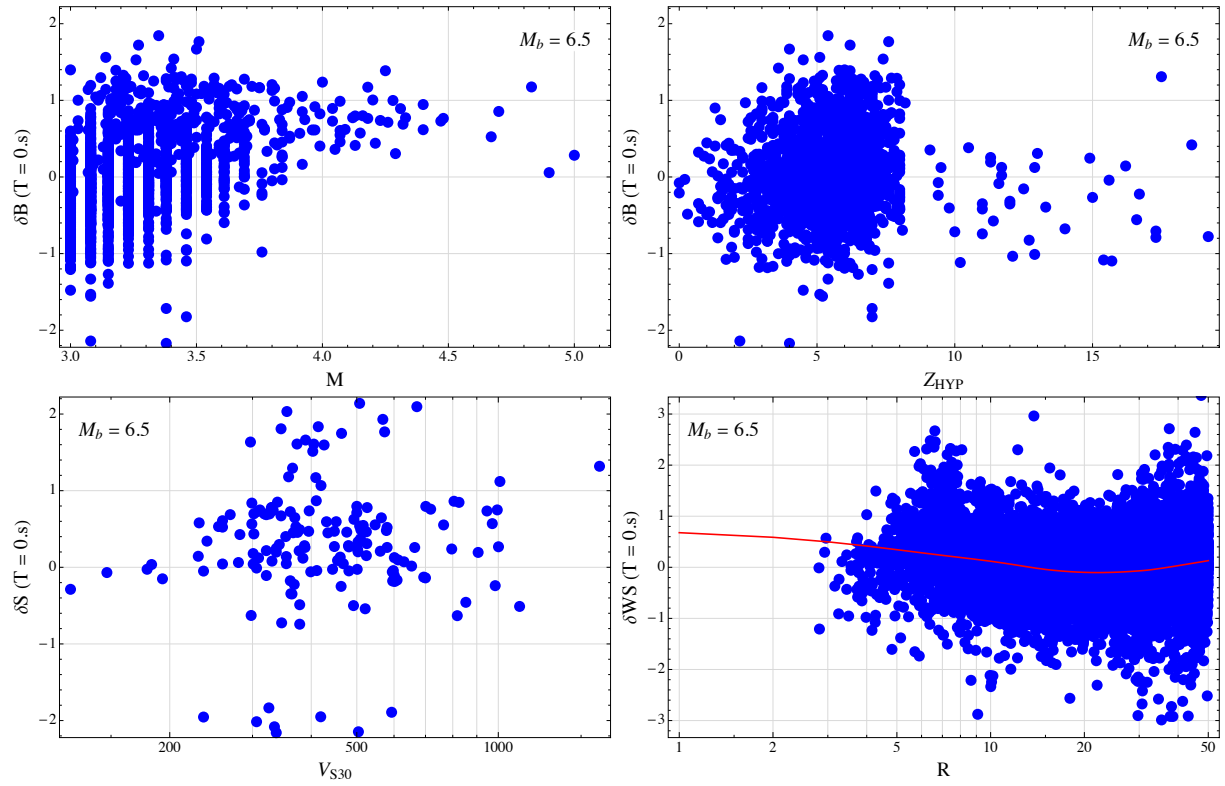


Figure 28: Between-event residuals δB, site-to-site residuals δS, and within-event/within-site residuals δWS, for a regression with magnitude break point $M_b = 6.5$, and period $T = 0$s.

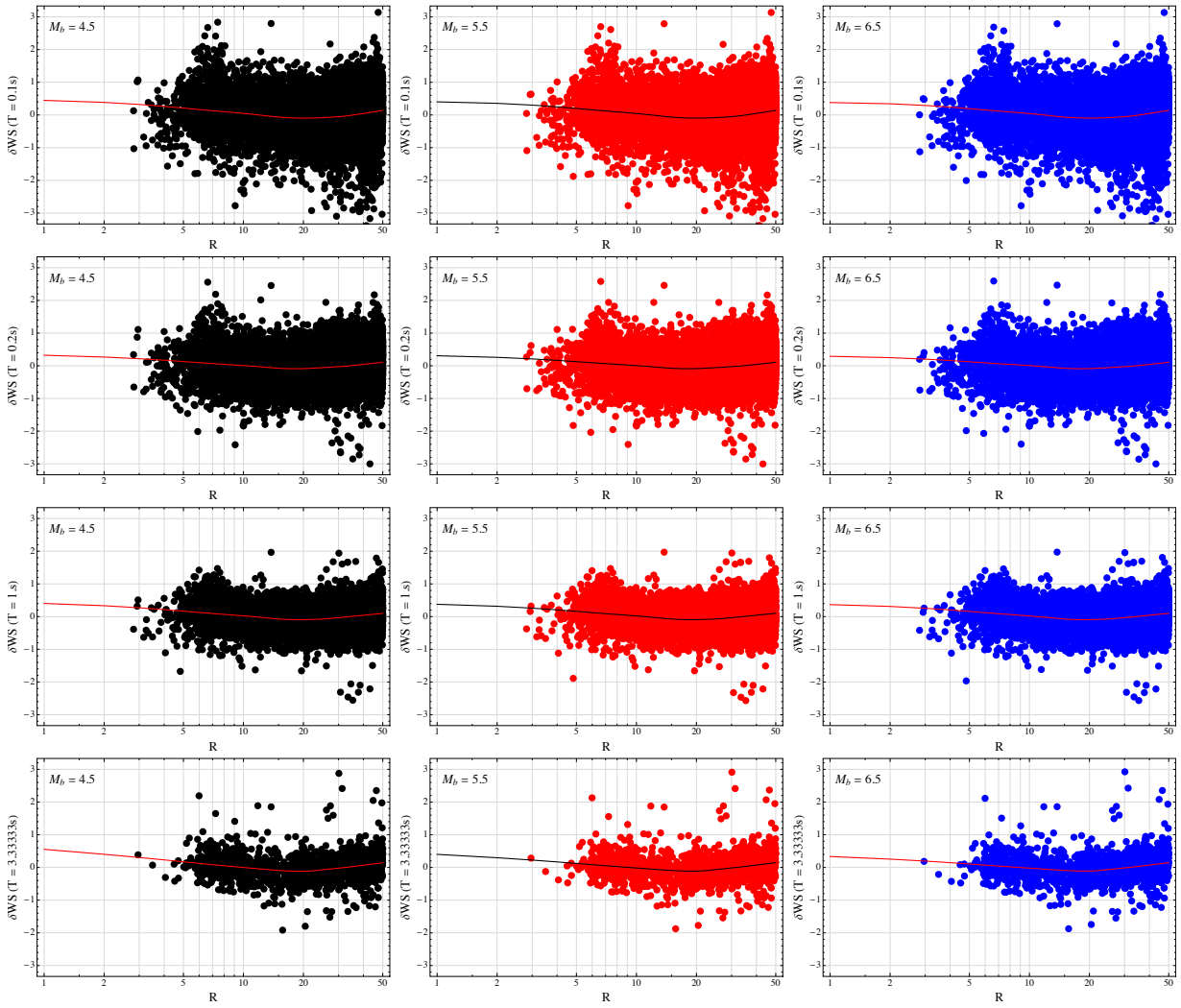


Figure 29: Within-event/within-site residuals, for different periods.

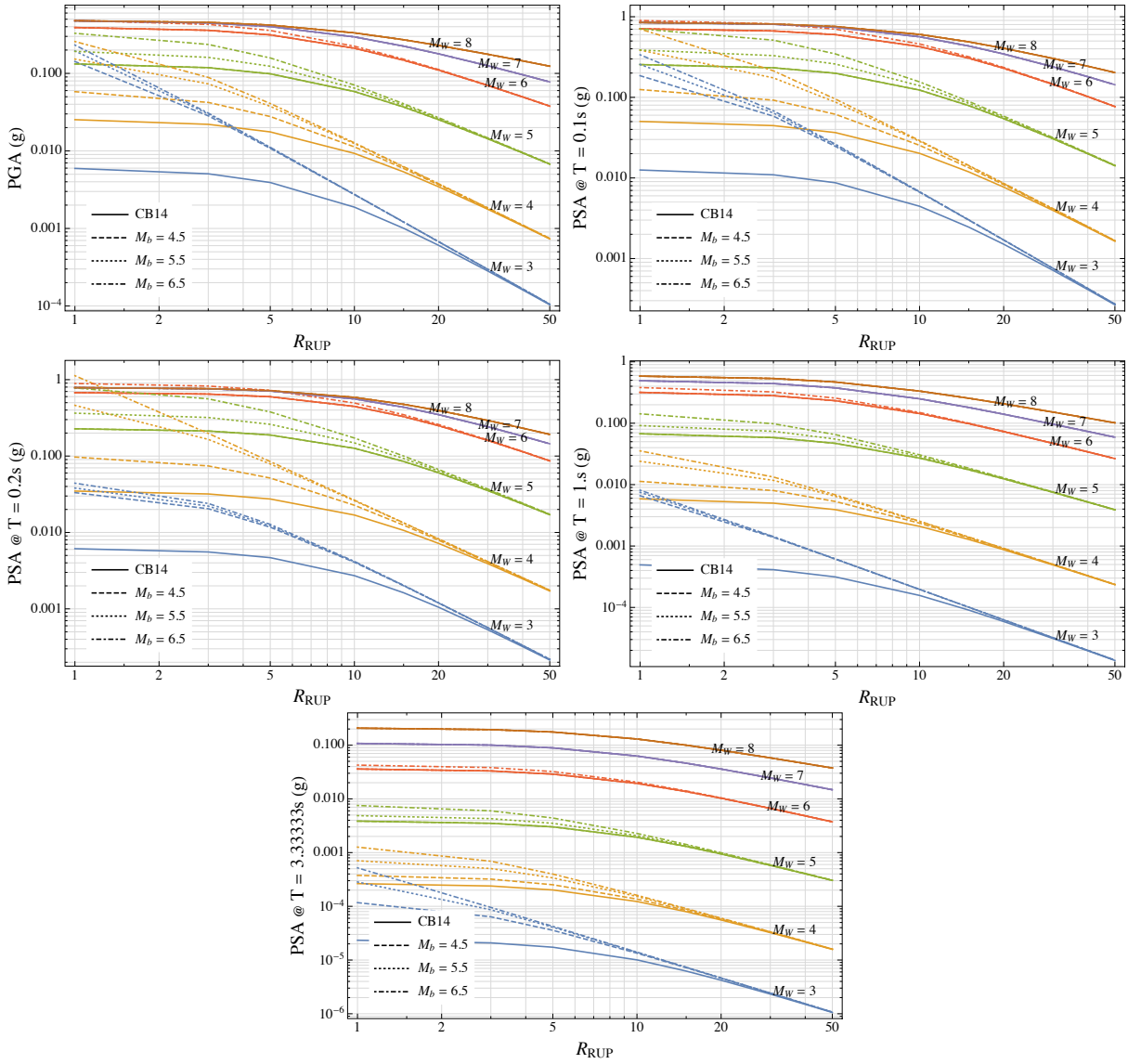


Figure 30: Scaling of PSA with distance, for the original CB14 model, and the adjusted models using different magnitude break points for the near-fault-term.

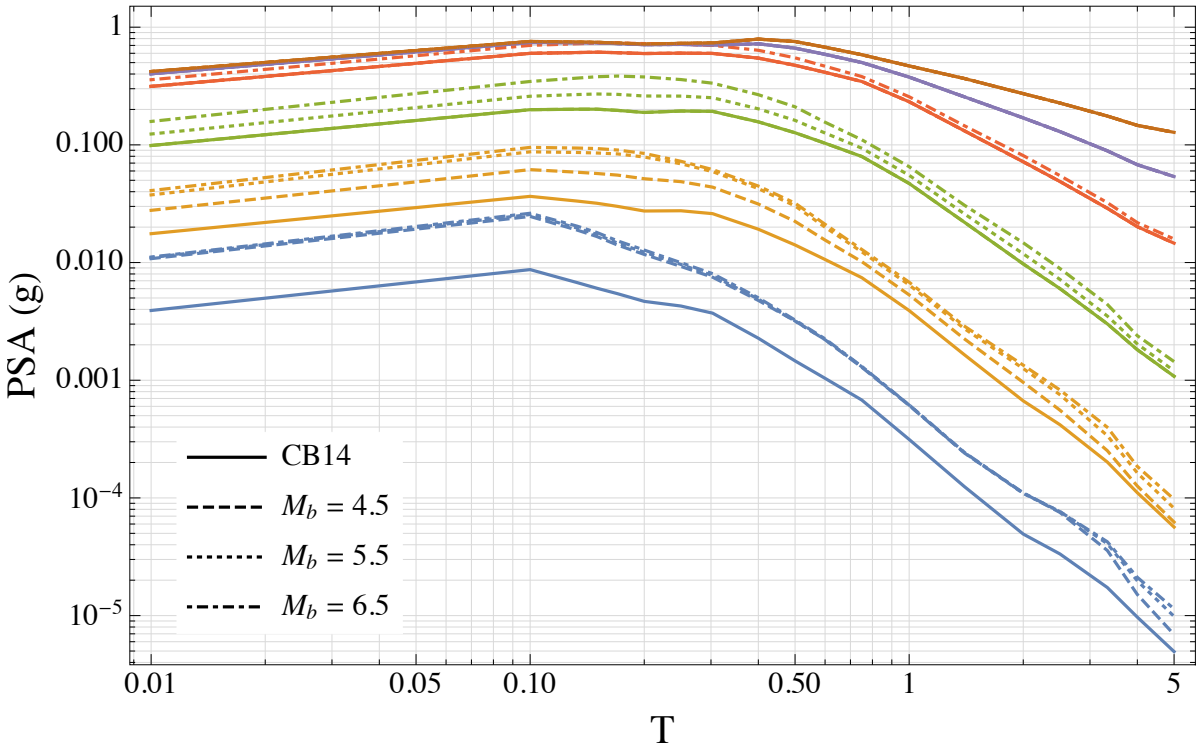


Figure 31: Spectrum at $R_{RUP} = 5km$ for the original CB14 model, and the adjusted models. Spectra are shown at $M_w = 3, 4, 5, 6, 7, 8$, wih the color scale the same as in Figure 30.

5 Update to CY14

5.1 Introduction

This section presents the third ground-motion model for PIE, derived by updating the ground motion model of Chiou and Youngs (2014) (CY14) to be applicable to shallow-depth PIE. The updated CY14 uses a simplified formulation for median amplitude, commensurate with the predictor availability of PIE data set. To achieve a close fit to the unique attenuation characteristics of PIE data, we further modify the formulation to replace the original break-in-slope at $R_{RUP} = 50$ km by a sharp break at $R_{RUP} = 18$ km. Good fits to the PIE data have been achieved by adjusting several coefficients of the modified CY14 functional form.

5.2 Exploratory Analysis of the Full PIE Dataset

The exploratory analysis involves the analysis of data residuals with respect to CY14. Such residual analysis is a useful first step in updating CY14 to the PIE data. It helps identify the key parts of CY14 that need updates. It also helps guide the modification of functional form. Data selection in the initial exploratory analysis is almost non-restrictive: we only require each earthquake to have at least 5 acceptable-quality recordings, regardless of their distances.

For the purpose of computing total residual ϵ, following assumptions are made for the predictors unavailable from the PIE dataset,

- $F_{RV} = F_{NM} = 0$, and dip $= 90°$; that is, all earthquakes are assumed to have strike-slip faulting on a vertical fault plane.
- $R_{RUP} = R_{Hyp}$, a reasonable assumption that has been widely adopted for small earthquakes.
- $Z_{TOR} = Z_{Hyp}$. Note that PIE's Z_{Hyp} is roughly independent of $\mathbf{M}$.
- $\Delta Z_{TOR} = 0$.
- Hanging wall effect is negligible, so we may assume $R_X = 0$.

- Directivity effect is negligible, so we may assume $\delta DPP = 0$.
- $\delta Z_{1.0} = 0$

Data residual ϵ is further partitioned:

$$\epsilon = c_0 + \delta B + \delta S + \delta WS \tag{19}$$

, where δB is the random earthquake-specific residual, δS is the random station-specific residual, and δWS is the record residual after accounting for δB and δS. Residual partitioning is achieved using the *lmer* function in the LME4 package of R.

Earthquake residuals shown in Figure 32 and Figure 33 reveal several important patterns. For $PSA[T = 0.1s]$ (Figure 32), earthquakes having a genuine moment magnitude (reported mainly by Saint Louis University) show a weak **M** trend in δB (the bottom panel of Figure 32), whereas earthquakes having a converted **M** (typically from local magnitude) show an upward trend in the case of Oklahoma-Kansas (OK-Kan) and a significant downward trend in the case of Graham and Septimus, Canada (top panel of Figure 32). The causes of such discrepancy are still not understood. For $PSA[T = 1s]$ (Figure 33), earthquakes having a genuine moment magnitude again show a weak δB trend with **M**. For the updating of CY14, we decide to use only ground-motion data from earthquakes having a genuine moment magnitude. With this decision, updating CY14's magnitude scaling term to OK-Kan data is minor and straightforward.

Station-specific residuals (δS) are shown as a function of V_{S30} in Figure 34. The upward δS trends with $log(V_{S30})$ suggest that PIE's $ln(V_{S30})$ scaling is weaker than that of CY14, hence a need for adjusting CY14's coefficient for $ln(V_{S30})$ scaling.

To examine the appropriateness of CY14's distance attenuation relation, the record residuals (δWS) of PIE data are grouped by magnitude and shown as a function of R_{RUP} in Figure 35 and Figure 36 for PSA of 0.1s and 1s period, respectively. In the range of $R_{RUP} < 20$ km, PSA from the OK-Kan region attenuates at a much steeper rate than CY14's suggested rate, but at a shallower rate in the range of $30 < R_{RUP} < 100$ km. Adjusting the distance attenuation relation

to match this unique attenuation characteristics has been the primary scope for CY14 updating.

5.3 Updating CY14 to OK-Kan Data

5.3.1 Formulation for Median Amplitude

The median amplitude of PIE ground motion is modeled by a simplified CY14 functional form, for which terms of style-of-faulting effect, depth-to-top-of-rupture effect, fault-dip effect, directivity effect, and basin-depth effect are excluded. They are excluded because the current PIE dataset does not contain necessary predictor information to support meaningful update; see also discussions in section 5.2. In addition, the term for break-in-slope is replaced by a new term that yields a sharp break at 18 km distance. Finally, anelastic attenuation is not included in the updated CY14 because we use only data of $R_{RUP} \leq 100$ km; see next section about record selection criteria.

The simplified CY14 formulation is shown in Equations (20) to (25). The coefficients that have been adjusted are shown in red color, while those unchanged from CY14 are shown in black.

$$ln(y_{ref}) = c_1 + c_2(\mathbf{M} - 6) + \frac{c_2 - c_3}{c_n} ln\left(1 + e^{c_n(c_M - \mathbf{M})}\right) + f_{dis} \tag{20}$$

$$f_{dis} = \begin{cases} c_4 \; ln(R) & R_{RUP} \leq r_1 \\ c_{4x} \; ln(R) + (c_4 - c_{4x}) \; ln(R_1) & R_{RUP} > r_1 \end{cases} \tag{21}$$

$$R = \sqrt{R_{RUP}^2 + H^2} \tag{22}$$

$$R_1 = \sqrt{r_1^2 + H^2} \tag{23}$$

$$H = c_5 \; cosh\left(c_6 \; (\mathbf{M} - c_{HM}, \; 0)_{max}\right) \tag{24}$$

$$ln(\mu) = ln(y_{ref}) + b_1 \left(ln(\frac{V_{S30}}{1130}), 0\right)_{min} + b_2 \left(e^{-b_3\left((V_{S30},1130)_{min}-360\right)} - e^{-b_3(1130-360)}\right) ln(\frac{y_{ref}\ e^{\eta} + b_4}{b_4}) \quad (25)$$

In addition to the revised break-in-slope term in Equation 21, we also change the definition of R, from $(R_{RUP} + H)$ to $(\sqrt{R_{RUP}^2 + H^2})$, as shown in Equation 22. The new definition helps keep model extrapolation from reaching an unreasonably high value at very short distances.

The updated CY14 accounts for possible nonlinear soil response. We retain CY14's original nonlinear soil response model, even though we have not evaluated nor confirmed its applicability to data recorded on soft soil conditions. This issue is left for future study.

5.3.2 Data Selected for Updating CY14

The following record-selection criteria are utilized to select PIE data for use in updating CY14.

- Recordings were collected from the OK-Kan region.
- **M** is a genuine moment magnitude, not converted from other magnitude scale.
- Hypocentral depth $Z_{Hyp} \leq 10$ km.
- $R_{RUP} \leq 100$ km. This cutoff distance is selected to reflect the main application interest and to avoid modeling another break-in-slope observed in the distance range of 100 km to 200 km (see, for example, Figure 35).
- Recordings are noted by the record processor as having acceptable quality.
- Both horizontal components are available.

After record selection, following earthquake-selection criteria are then applied to select well-recorded earthquakes:

- Number of selected recordings in an earthquake is 5 or more, and at least one of the selected recordings is located within 20 km of earthquake source

- Distance ratio in an earthquake (defined as the ratio of the largest R_{RUP} to the smallest R_{RUP}) is greater than 5

The magnitude-distance distribution of the selected recordings is shown in Figure 37. The largest earthquake of the PIE dataset (the September 3, 2016, **M**5.8 earthquake) is excluded because its distance ratio is less than 5.

5.4 Regression Results

We use the mixed-effects model of Abrahamson and Youngs (1992) in the regression analysis of PIE data. As discussed previously, six coefficients ($c_1, c_2, c_3, c_4, c_{4x}, b_1$) have been adjusted. Their values are obtained from the mixed-effects regression analysis, except for c_2.

It is inappropriate to keep the original c_2 value because, if it were to be used together with the adjusted c_4, it would have lead to a massive over-saturation for $\mathbf{M} > 5.5$ and $R_{RUP} < 5$ km. Because all PIE data are from earthquakes of **M**5 or smaller, it is not possible to estimate c_2 based on the selected data. To remedy the over-saturation problem, we set the c_2 value by anchoring the updated CY14 to CY14's original magnitude scaling rate for $R_{RUP} = 0$; that is, imposing the constraint $(c_2 + c_4 c_6)_{\text{PIE}} = (c_2 + c_4 c_6)_{\text{CY14}}$.

Coefficient estimation for the other five coefficients is conducted in several iterations. In each iteration, estimates of one of the remaining unknown coefficients are smoothed over spectral periods and, in the subsequent iterations, the values of that coefficient are fixed to the smoothed values to estimate the next unknown coefficient. Iteration starts with the smoothing of c_4, then proceeds to c_{4x}, b_1, and c_3. Coefficient c_1, which is the last coefficient estimated, has not yet been smoothed.

5.4.1 Model Coefficients and Residual Standard Deviations of Updated CY14

The final model parameters (fixed-effect coefficients and the standard deviations of residuals) are shown in Figure 38, and listed in Table 4 and Table 5. Note that, although we have conducted regression for $PSA[T = 7.5s]$ and $PSA[T = 10s]$, we decide not to list their results in Table 4 and

Table 5. We have noted a relatively high percentage of outliers in the PSA of these two periods, and we judge that their regression results may be biased due to the outliers.

5.4.2 PIE Data and Fits by Updated CY14

Figure 39 to Figure 42 show the PIE data and fits by the updated CY14, for PSA of periods 0.01 (PGA), 0.1, 0.5, and 1 s. The decision to include a steep attenuation slope for short-period PSA at $R_{RUP} < 18$ km is clearly justified by Figure 39 and Figure 40. The outliers seen in the $(\mathbf{M}4 \pm 0.2)$ bin of Figure 42 are signs of outliers to come in longer periods.

5.4.3 Residuals of Updated CY14

Figure 43 shows the between-earthquake residuals (solid, reverse triangles) and within-earthquake residuals (open circles) as a function of $\mathbf{M}$ for PSA of $T = 0.01$ (PGA), 0.1, 0.5 and 1 s. Overall, the degree of fists to data are very reasonable; there isn't a noticeable trend with $\mathbf{M}$ in the data residuals.

Figure 44 shows the within-earthquake residuals as a function of R_{RUP} for PSA of $T = 0.01$ (PGA), 0.1, 0.5 and 1 s. Overall, updated CY14 fits the distance attenuation well, up to 100 km distance. A slight trend with R_{RUP} is seen in some of the residual plots. Such trend may be attributed to misfits to small-magnitude data (in the $\mathbf{M}3.2 \pm 0.2$ bin) and, in some other cases, to the smoothing of coefficient c_{4x}.

5.4.4 Scaling Relations of Updated CY14

The scaling relation between median PSA and magnitude is shown in Figure 45 for the original CY14 and the updated CY14. The median PSA is computed for $Z_{TOR} = 1$ km (used only by the original CY14), $V_{S30} = 500$ m/s, and $R_{RUP} = 1$, 10, and 20 km. The scaling relation between median PSA and source-to-site distance is shown in Figure 46.

Table 4: Coefficients of updated CY14.

Period (s)	c_1	c_2	c_3	c_n	c_M	c_4	c_{4x}	c_5	c_6	c_{HM}	b_1	b_2	b_3	b_4
0.0100	10.5071	1.9550	1.7821	16.0875	4.9993	-3.9236	-1.1500	6.4551	0.4908	3.0956	-0.1239	-0.1417	0.0070	0.1022
0.1000	10.7850	1.8634	1.8341	19.9992	5.0172	-3.6915	-1.1500	6.8305	0.5048	3.2381	-0.1326	-0.2943	0.0056	0.2307
0.1200	10.5125	1.8328	1.8596	18.7106	5.0315	-3.6310	-1.1500	7.1333	0.5048	3.3407	-0.1532	-0.3077	0.0057	0.2532
0.1500	9.9963	1.7867	1.9002	16.6246	5.0547	-3.5404	-1.1500	7.3621	0.5045	3.4300	-0.2026	-0.3113	0.0058	0.2665
0.1700	9.6694	1.7552	1.9290	15.3709	5.0704	-3.4804	-1.1500	7.4365	0.5036	3.4688	-0.2553	-0.3062	0.0060	0.2651
0.2000	9.1894	1.7078	1.9726	13.7012	5.0939	-3.3915	-1.1500	7.4972	0.5016	3.5146	-0.3479	-0.2927	0.0061	0.2553
0.2500	8.4363	1.6308	2.0415	11.2667	5.1315	-3.2483	-1.1500	7.5416	0.4971	3.5746	-0.4563	-0.2662	0.0064	0.2315
0.3000	7.7942	1.5590	2.1039	9.1908	5.1670	-3.1145	-1.1500	7.5600	0.4919	3.6232	-0.4569	-0.2405	0.0067	0.2073
0.4000	6.7848	1.4382	2.2094	6.5459	5.2317	-2.8867	-1.1500	7.5735	0.4807	3.6945	-0.3645	-0.1975	0.0071	0.1655
0.5000	6.0520	1.3513	2.2960	5.2305	5.2893	-2.7189	-1.1500	7.5778	0.4707	3.7401	-0.2749	-0.1633	0.0074	0.1338
0.7500	5.1736	1.2495	2.4679	3.7896	5.4109	-2.5142	-1.1500	7.5808	0.4575	3.7941	-0.1510	-0.1028	0.0081	0.0852
1.0000	4.9178	1.2261	2.5943	3.3024	5.5106	-2.4673	-1.1500	7.5814	0.4522	3.8144	-0.1054	-0.0699	0.0084	0.0586
1.5000	4.6394	1.2223	2.7405	2.8498	5.6705	-2.4606	-1.1500	7.5817	0.4501	3.8284	-0.0650	-0.0425	0.0077	0.0318
2.0000	4.4325	1.2184	2.8313	2.5417	5.7981	-2.4521	-1.1500	7.5818	0.4500	3.8330	-0.0413	-0.0302	0.0048	0.0197
3.0000	4.4018	1.2175	2.9360	2.1488	5.9983	-2.4500	-1.1500	7.5818	0.4500	3.8361	-0.0049	-0.0129	0.0018	0.0096
4.0000	4.3476	1.2175	3.0000	1.8957	6.1552	-2.4500	-1.1500	7.5818	0.4500	3.8369	0.0000	-0.0016	0.0015	0.0054
5.0000	4.2427	1.2175	3.0568	1.7228	6.2856	-2.4500	-1.1500	7.5818	0.4500	3.8376	0.0000	0.0000	0.0014	0.0032

$r_1 = 18$

Table 5: Residual standard deviations of updated CY14.

Period (s)	σ	ϕ
0.0100	0.4151	0.7418
0.1000	0.4365	0.7581
0.1200	0.4124	0.7483
0.1500	0.4074	0.7431
0.1700	0.3972	0.7576
0.2000	0.3742	0.7922
0.2500	0.3636	0.8210
0.3000	0.3487	0.8227
0.4000	0.3142	0.7931
0.5000	0.3132	0.7487
0.7500	0.3364	0.7004
1.0000	0.3630	0.6928
1.5000	0.3035	0.6436
2.0000	0.3323	0.6277
3.0000	0.4285	0.6881
4.0000	0.3849	0.7838
5.0000	0.4476	0.9242

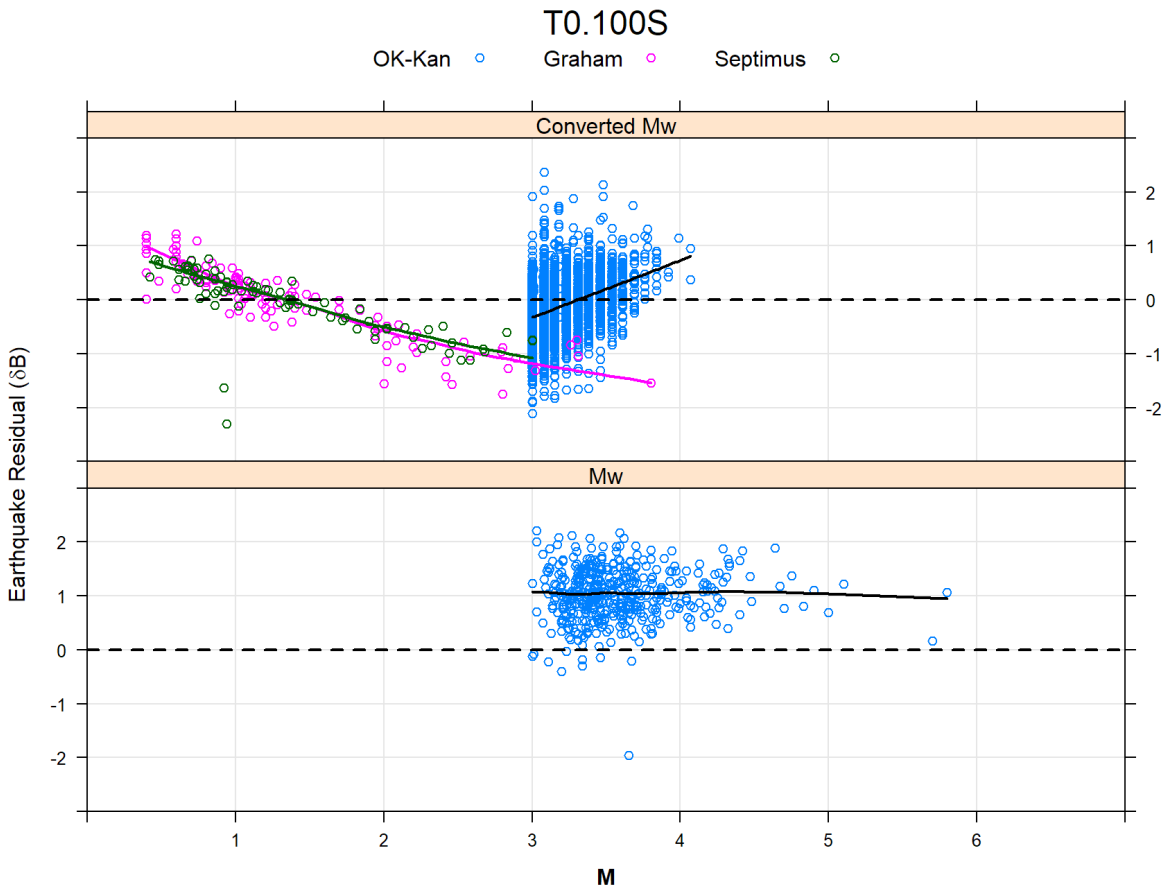


Figure 32: Earthquake residual of the original Chiou and Youngs (2014) versus **M**, $PSA[T = 0.1s]$. (Top) Earthquakes whose magnitude is converted from other magnitude scale. (Bottom) Earthquakes whose magnitude is a genuine moment magnitude.

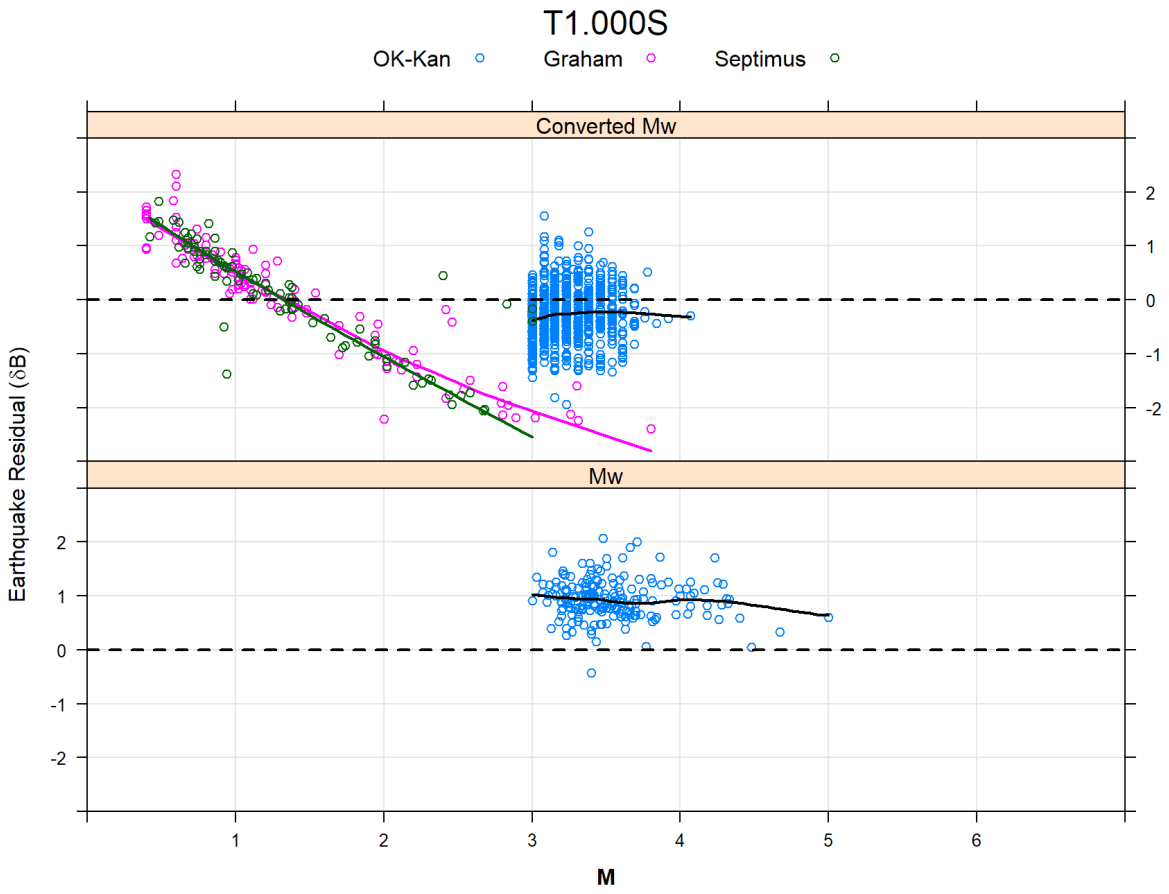


Figure 33: Earthquake residual of the original Chiou and Youngs (2014) versus **M**, $PSA[T = 1s]$. (Top) Earthquakes whose magnitude is converted from other magnitude scale. (Bottom) Earthquakes whose magnitude is a genuine moment magnitude.

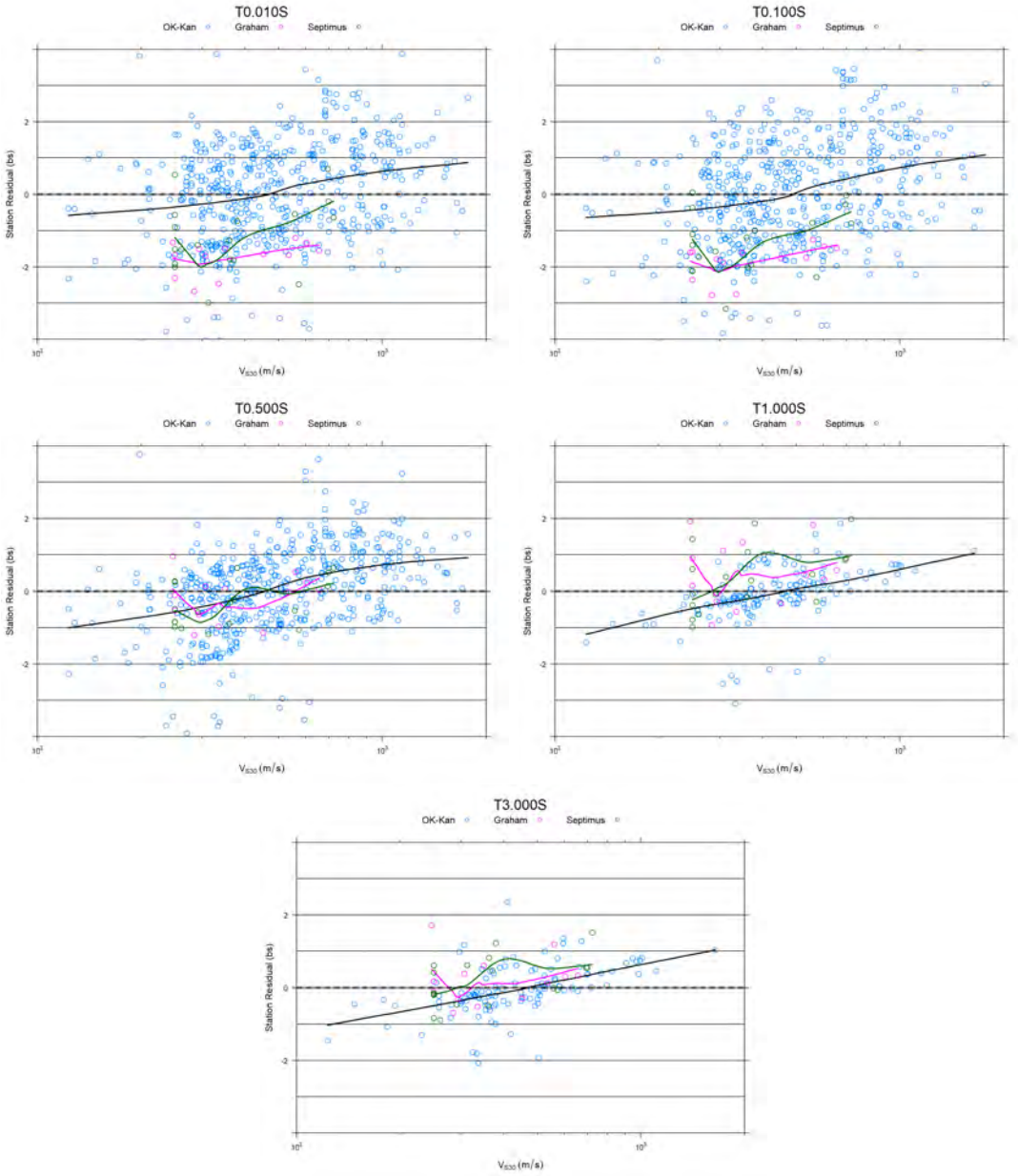


Figure 34: Station residuals (δS) of Chiou and Youngs (2014) versus V_{S30}.

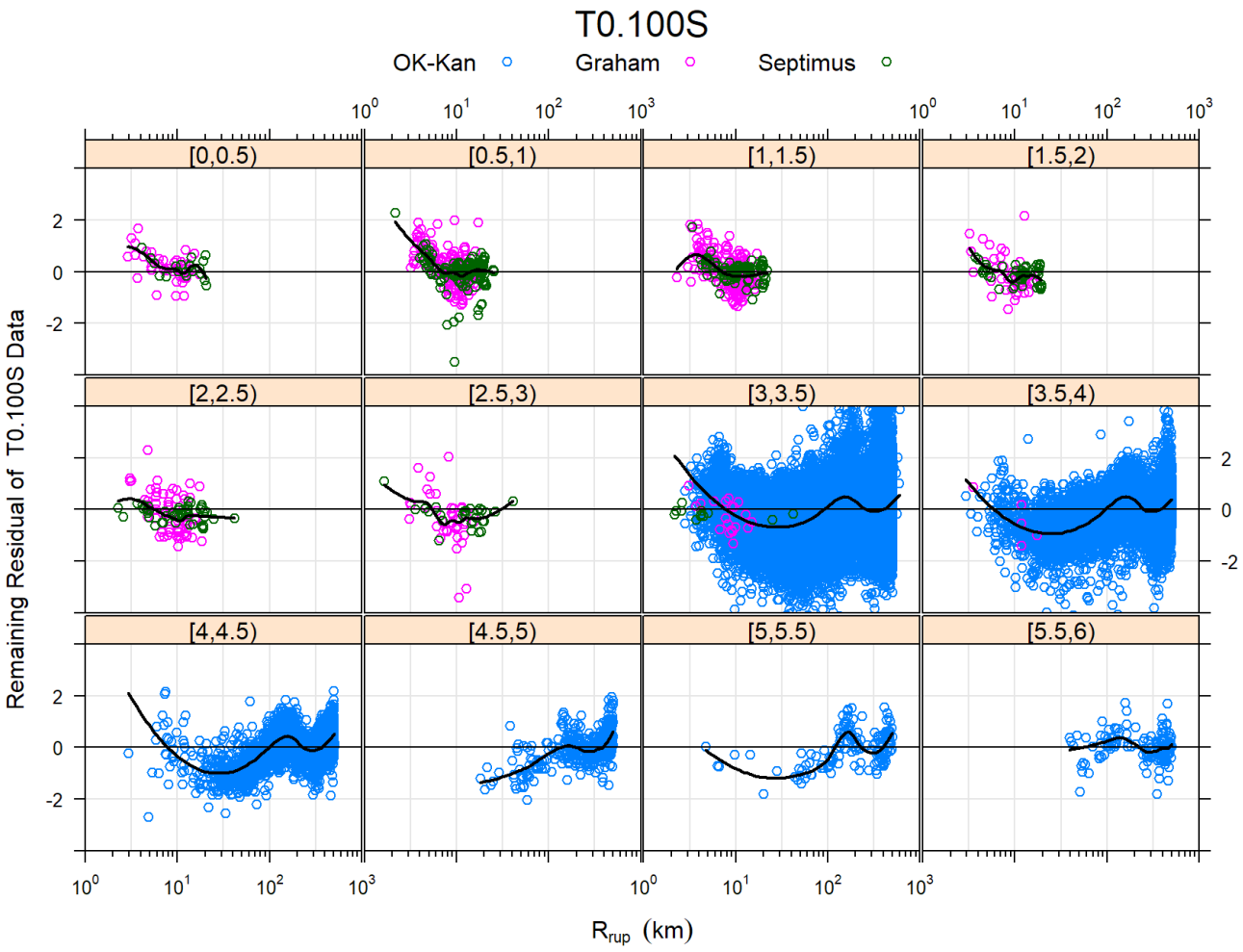


Figure 35: Record residual (δWS) of Chiou and Youngs (2014) versus R_{RUP}, $PSA[T = 0.1s]$.

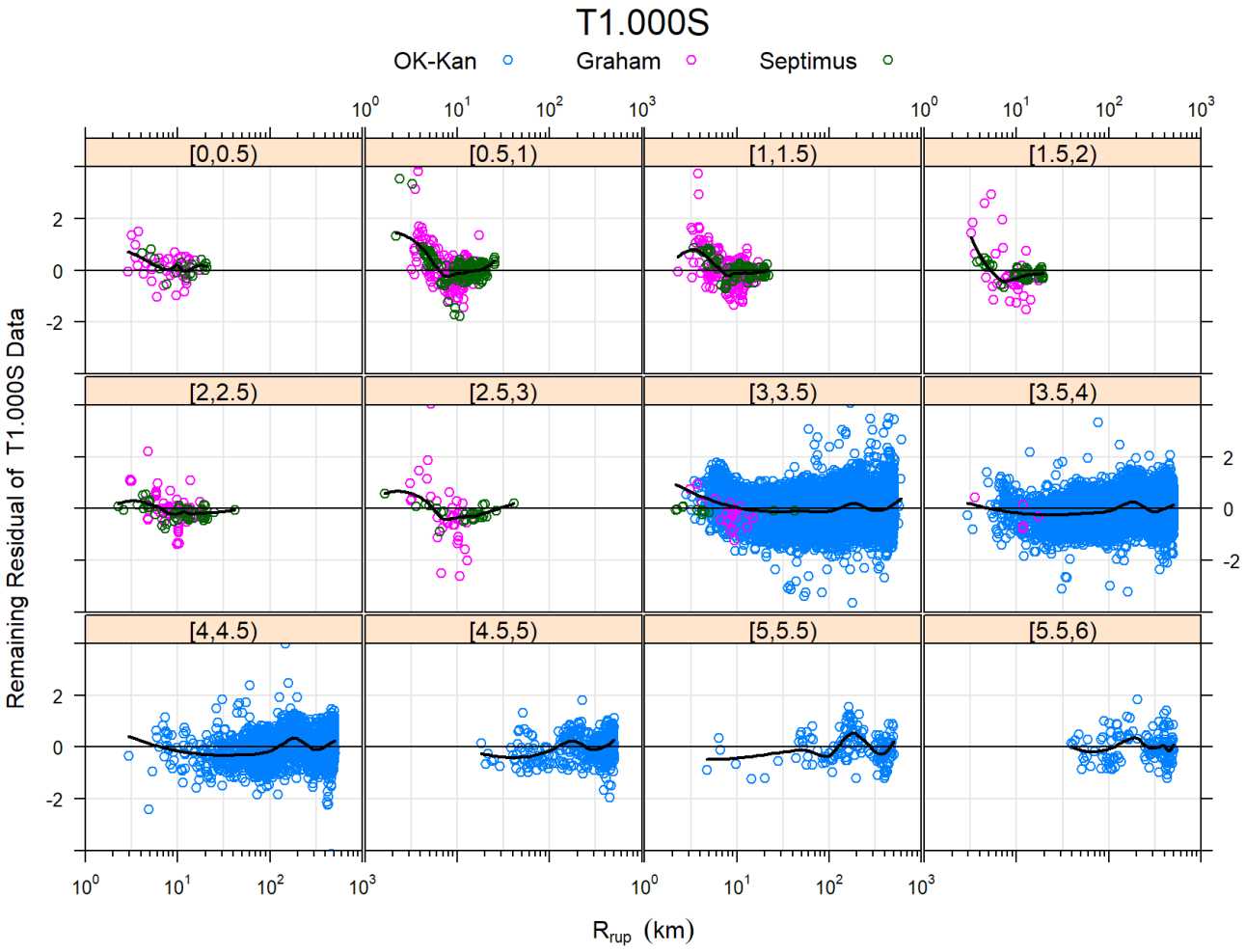


Figure 36: Record residual (δWS) of Chiou and Youngs (2014) versus R_{RUP}, $PSA[T = 1s]$.

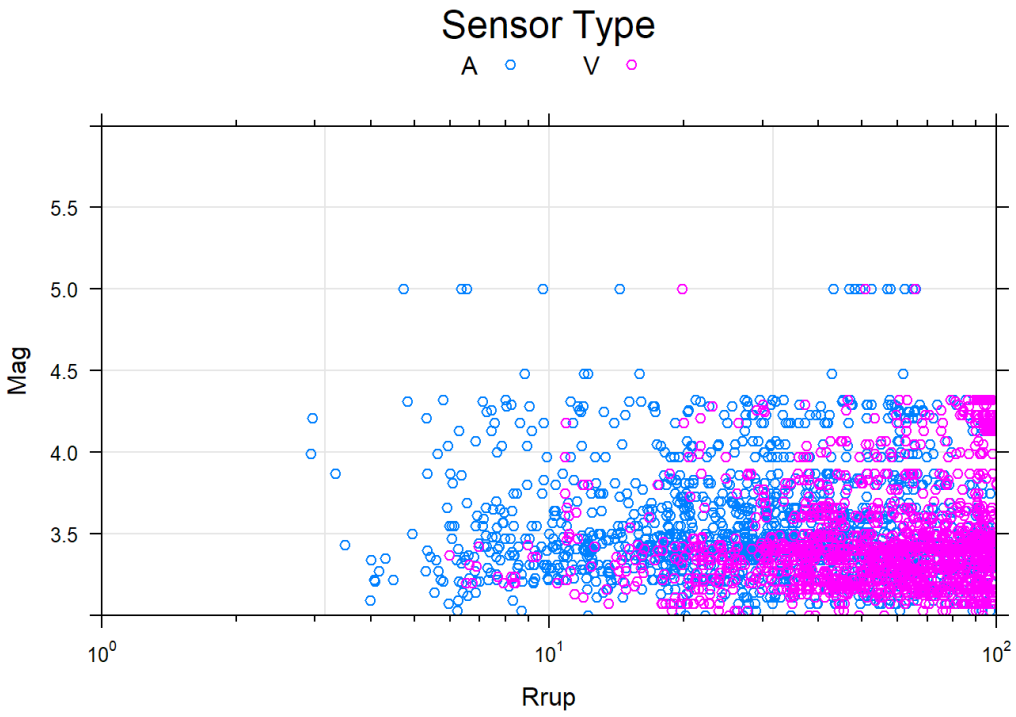


Figure 37: Magnitude-distance distribution of PIE data used in updating Chiou and Youngs (2014). Blue color is used for plotting data from the acceleration sensor, black for velocity sensor.

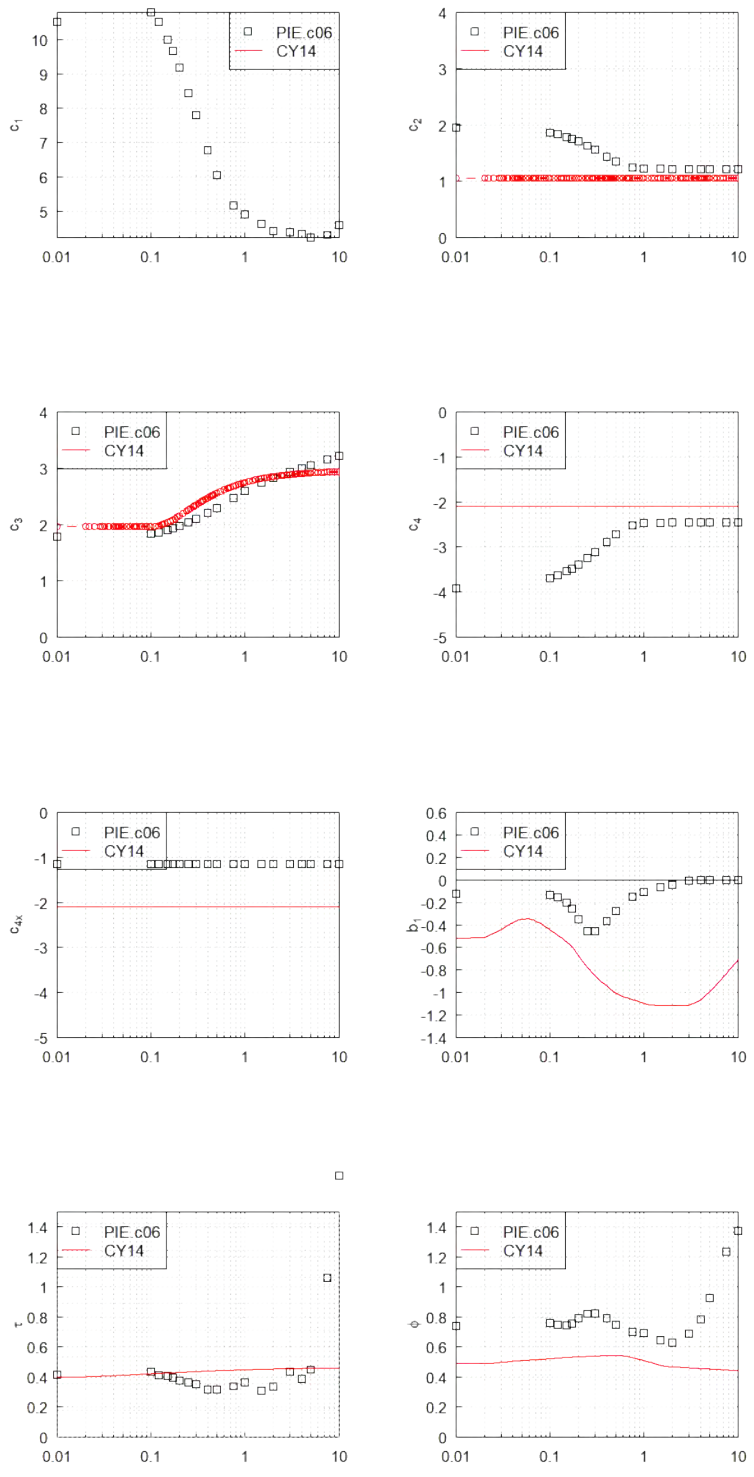


Figure 38: Fixed-effect coefficients and residual standard deviations of updated CY14.

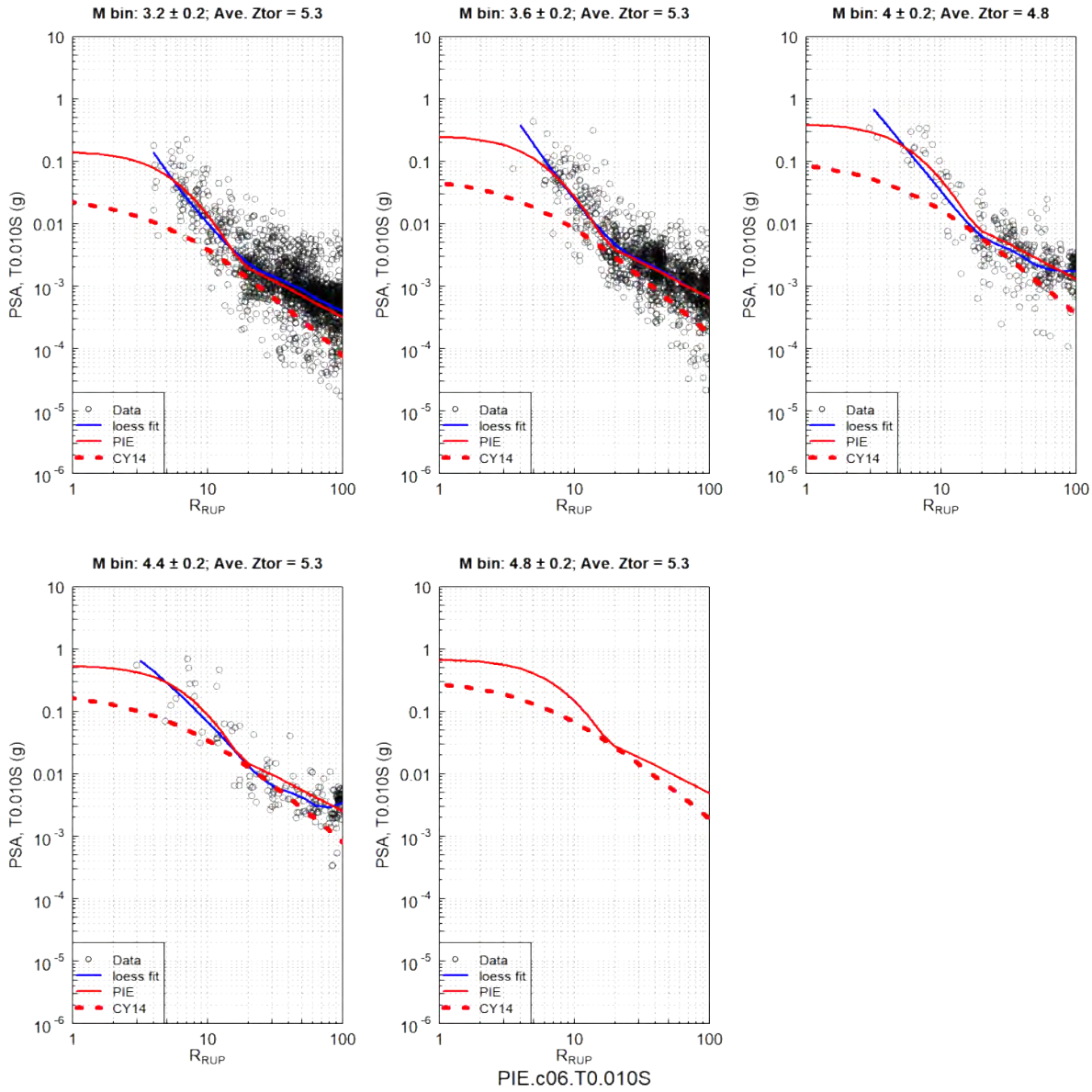


Figure 39: PGA ($PSA[T = 0.01s]$) data and fits by the original CY14 (dashed red) and the updated CY14 (solid red). Fits are computed for the average Z_{TOR} and the average V_{S30} of each specific magnitude bin.

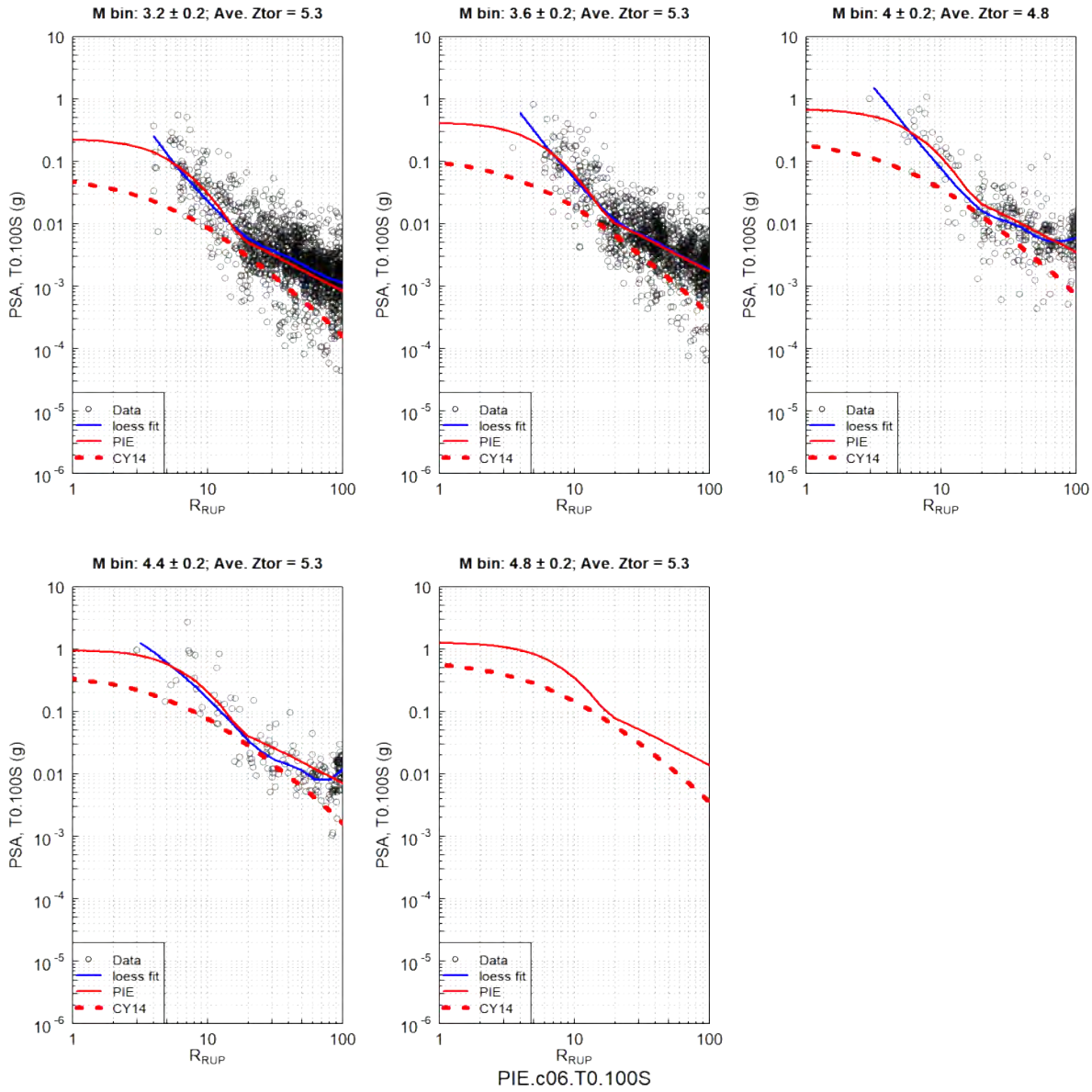


Figure 40: $PSA[T = 0.1s]$ data and fits by the original CY14 (dashed red) and the updated CY14 (solid red). Fits are computed for the average Z_{TOR} and the average V_{S30} of each specific magnitude bin.

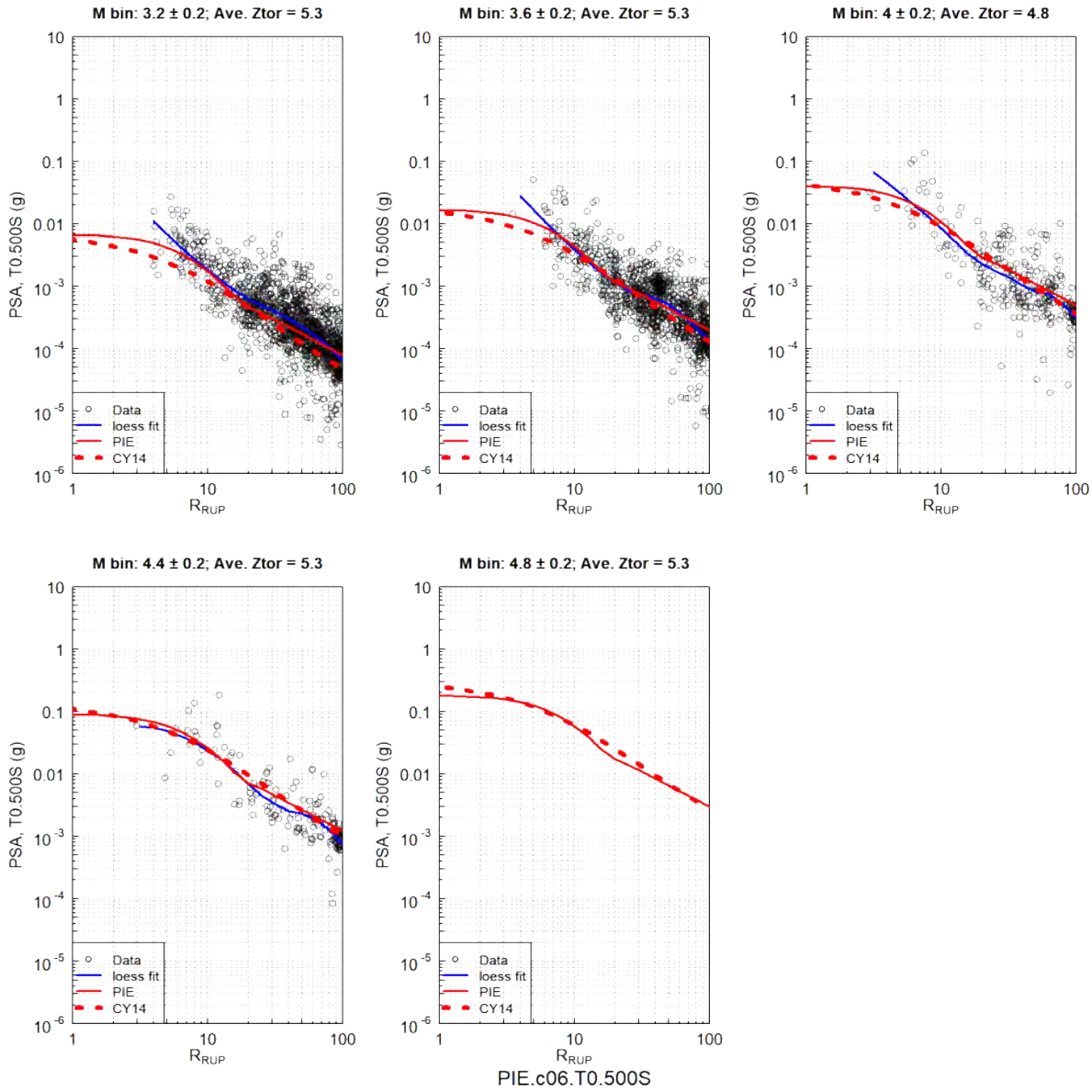


Figure 41: $PSA[T = 0.5s]$ data and fits by the original CY14 (dashed red) and the updated CY14 (solid red). Fits are computed for the average Z_{TOR} and the average V_{S30} of each specific magnitude bin.

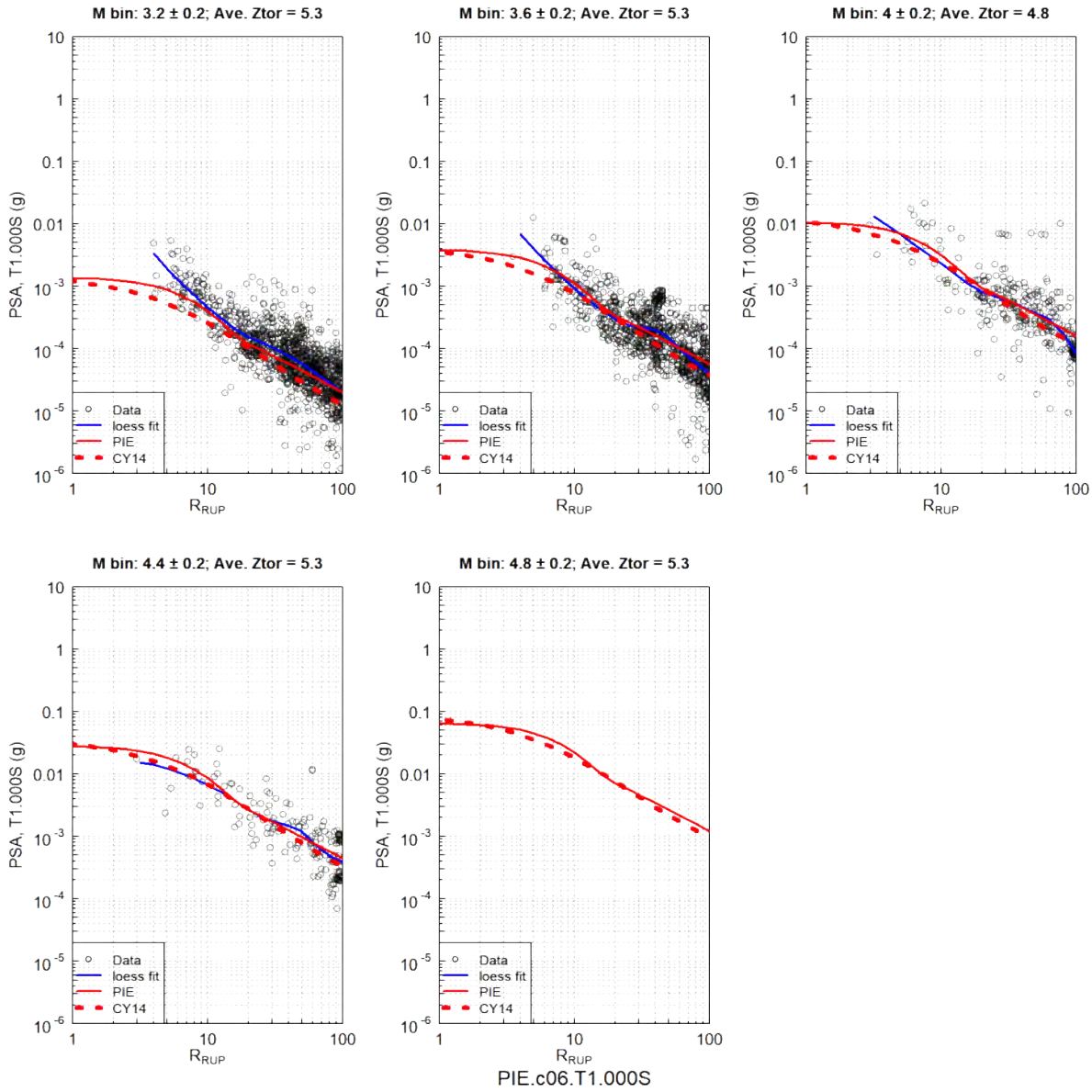


Figure 42: $PSA[T = 1s]$ data and fits by the original CY14 (dashed red) and the updated CY14 (solid red). Fits are computed for the average Z_{TOR} and the average V_{S30} of each specific magnitude bin.

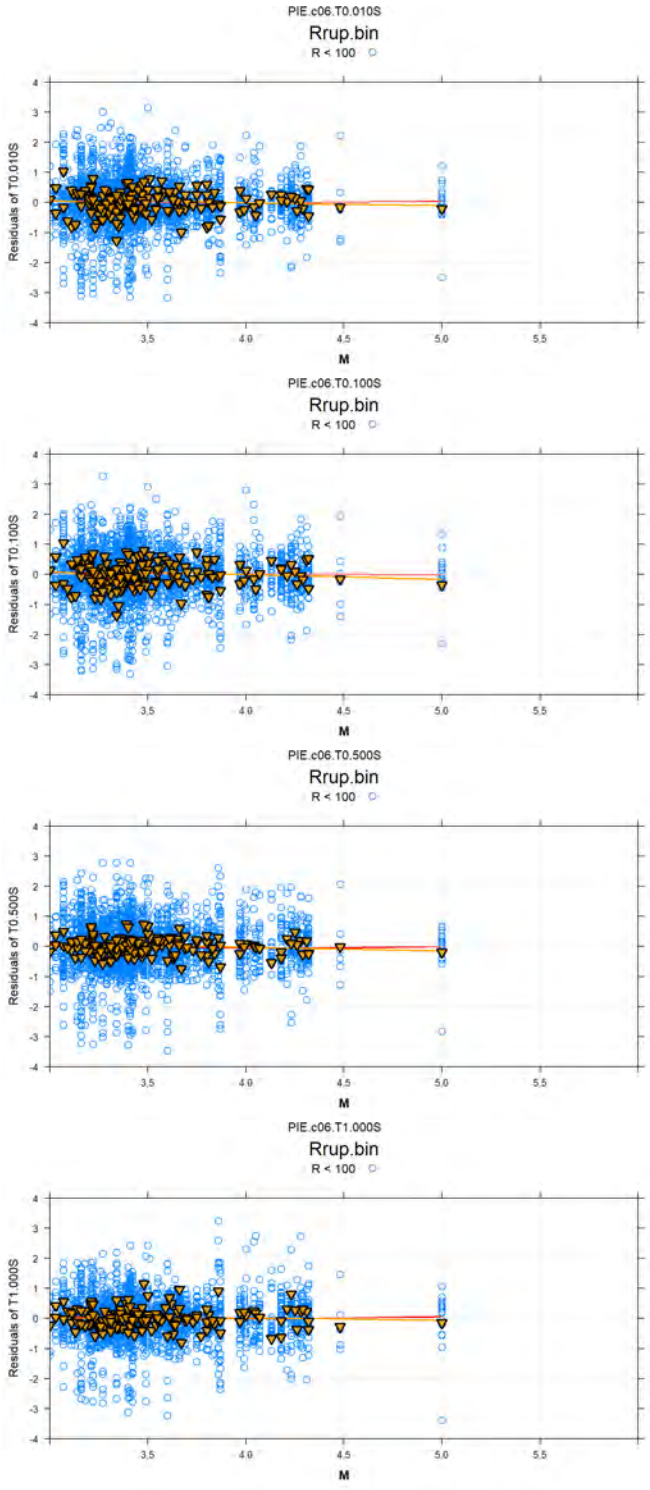


Figure 43: Residuals of updated CY14 vs. **M**. Between-earthquake residuals are marked by the solid reverse triangles, the within-earthquake residuals by open blue circles.

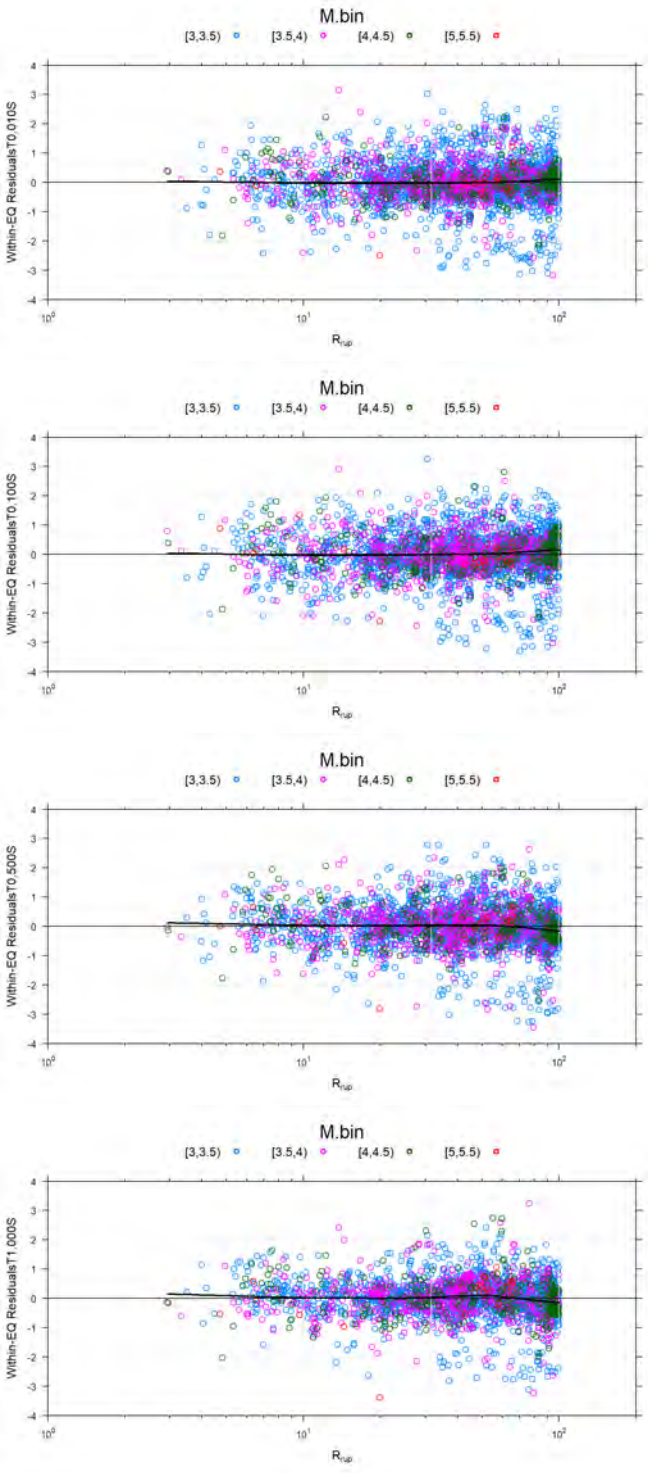


Figure 44: Within-earthquake residuals of updated CY14 vs. R_{RUP}.

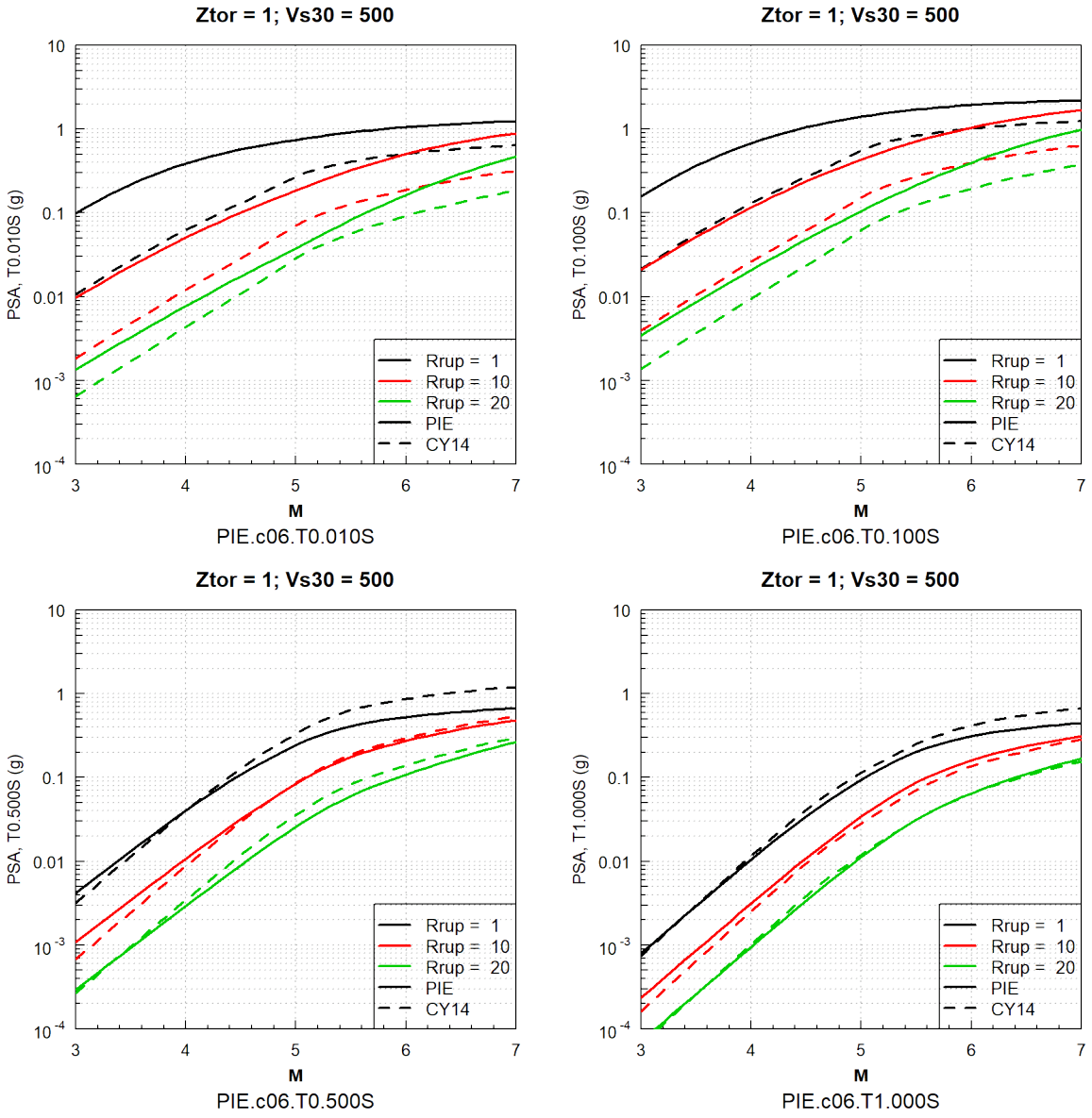


Figure 45: Scaling of PSA with magnitude, for the original CY14 and the adjusted CY14 for PIE.

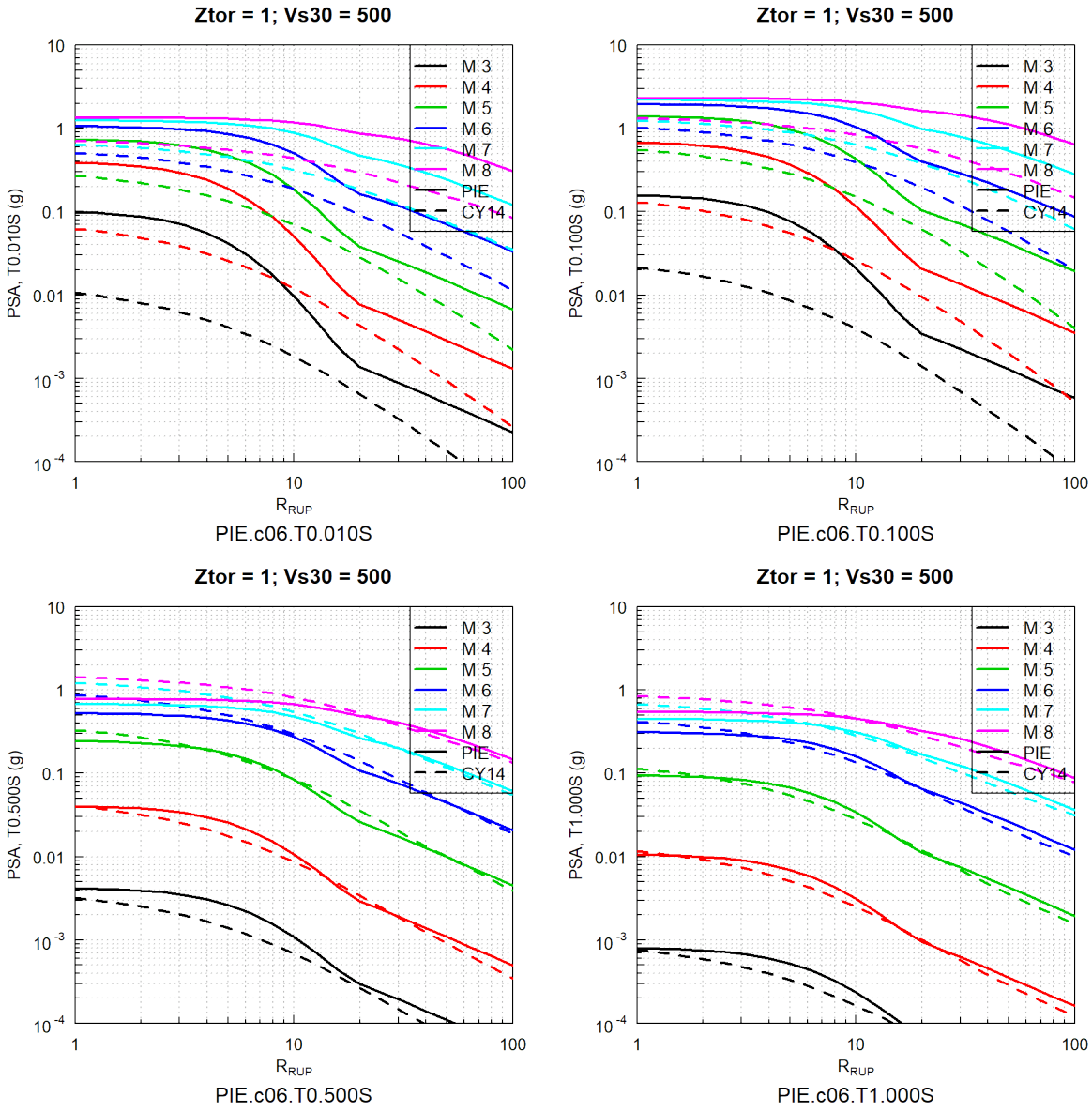


Figure 46: Scaling of PSA with distance, for the original CY14 and the adjusted CY14 for PIE.

6 Model Comparison

The distance scaling for the three modified GMMs are compared for M3 and M5 earthquakes for PGA and spectral periods of 0.1 sec, 0.2 sec, and 0.5 sec in Figures 47, 48, 49, and 50. Each figure alsp shows the distance scaling from the average of four unmodified NGA-W2 GMMs (REF).

At short spectral periods, the modified CY14 GMM tends to be on the high end of the range of the models and the modified CB14 GMM tends to be on the low end of the range. This difference reflects selection of the subset of data used. The modified CY14 model used only earthquakes with direct estimates of moment magnitude. This subset of earthquakes had higher ground motions than the subset for which the moment magnitude was estimated by a conversion. In contrast, the modified CB14 model used all of the data, which included the subset with converted magnitudes and their smaller ground motions for the given magnitude.

The modified ASK14 model tends to range between these two models. For example, as shown in Figure 47 for PGA, the modified ASK14 model for M3 is very similar to the modified CB14 model, but for M5, the modified ASK14 model becomes close to the modified CY14 model.

At T=0.5 sec (Figure (50), the three models are more similar for distances greater than 5 km.

The magnitude scaling of the modified GMMs for a rupture distance of 3 km are compared in Figures 51 and 52. For the short periods, shown in Figure 51, the modified CB14 and CY14 GMMs have similar magnitude slopes and mainly differ by a constant shift. In comparison to the unmodified NGA-W2 models, the magnitude slopes of these two models is steeper than the magnitude slope of the average NGA-W2 GMM.

Both the CB14 and CY14 models show monotonically increasing ground motion with increasing magnitude. In contrast, the slope of the magnitude scaling in the modified ASK14 model at this short distance range is not constant and there is over saturation with the largest ground motion from M5. The magnitude scaling of the modified ASK14 model ranges between the CB14 and CY14 models.

For the longer spectral periods (T=0.2 ad T=0.5), the magnitude scaling is again similar for the modified CB14 and CY14 models. The magnitude scaling is slightly flatter than the magnitude

scaling for the NGA-W2 models. For the modified ASK14 model, there is still some over saturation, but it is not as strong as for the short periods. At the longer periods, the three modified GMMs span a smaller range.

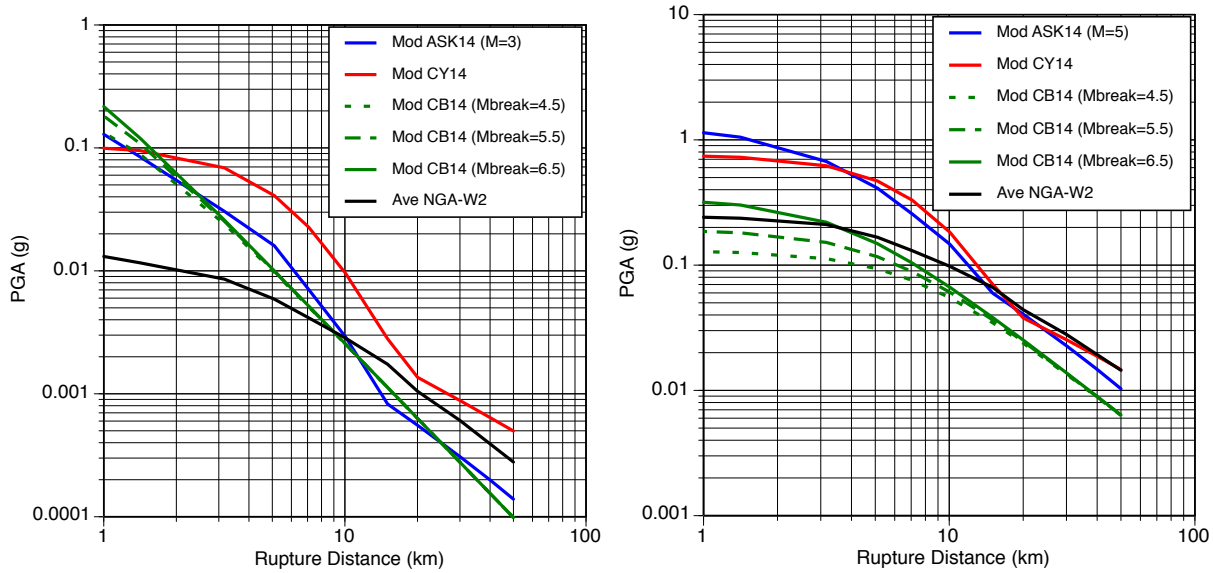


Figure 47: Comparison of the distance scaling of the modified GMMs for PGA for $Z_{TOR} = 1km$ and magnitudes M=3 (left frame) and M=5 (right frame).

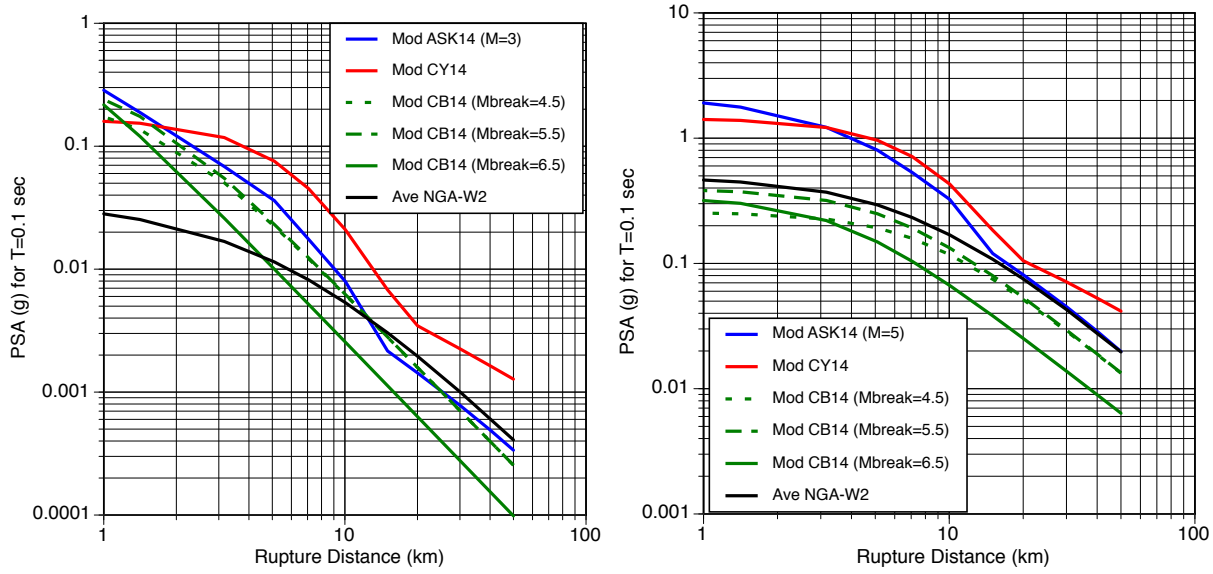


Figure 48: Comparison of the distance scaling of the modified GMMs for T=0.1 sec for $Z_{TOR} = 1km$ and magnitudes M=3 (left frame) and M=5 (right frame).

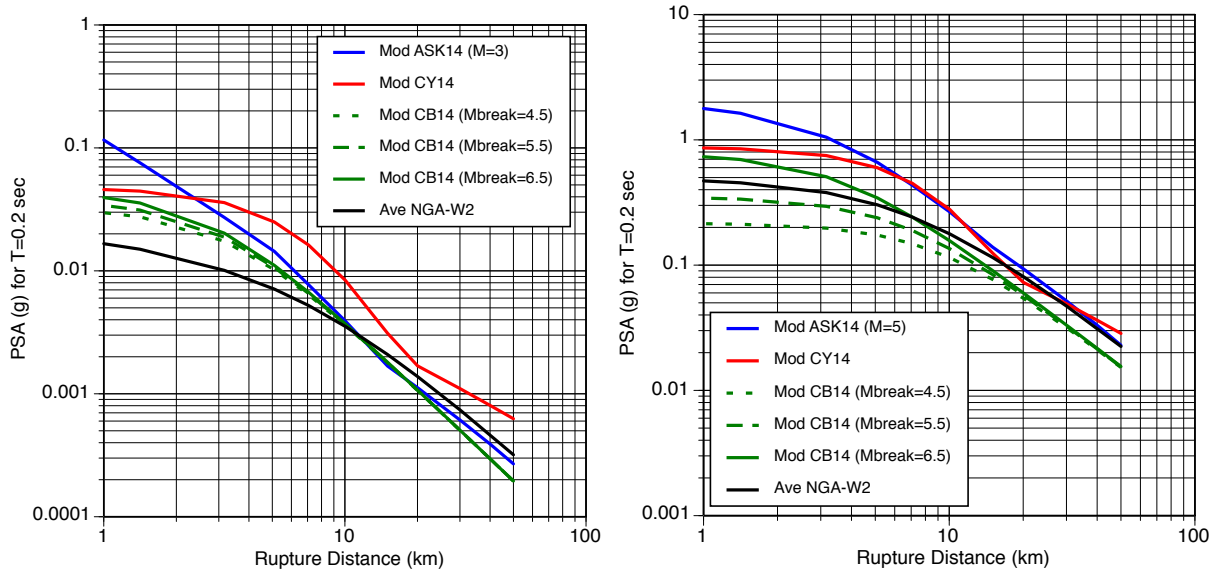


Figure 49: Comparison of the distance scaling of the modified GMMs for T=0.2 sec for $Z_{TOR} = 1km$ and magnitudes M=3 (left frame) and M=5 (right frame).

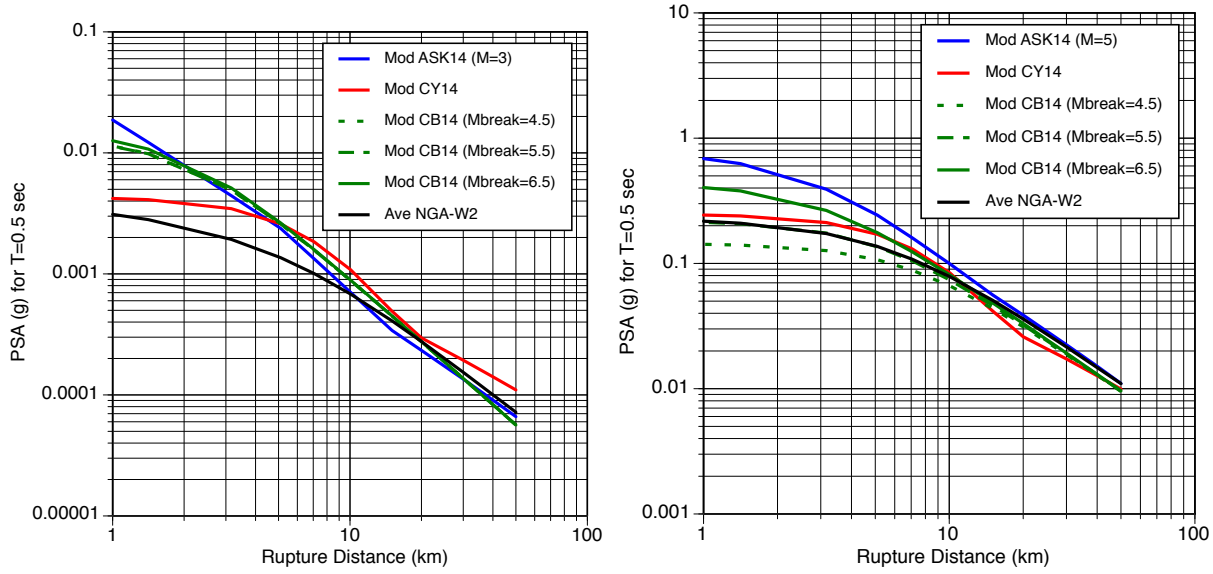


Figure 50: Comparison of the distance scaling of the modified GMMs for T=0.5 sec for $Z_{TOR} = 1km$ and magnitudes M=3 (left frame) and M=5 (right frame).

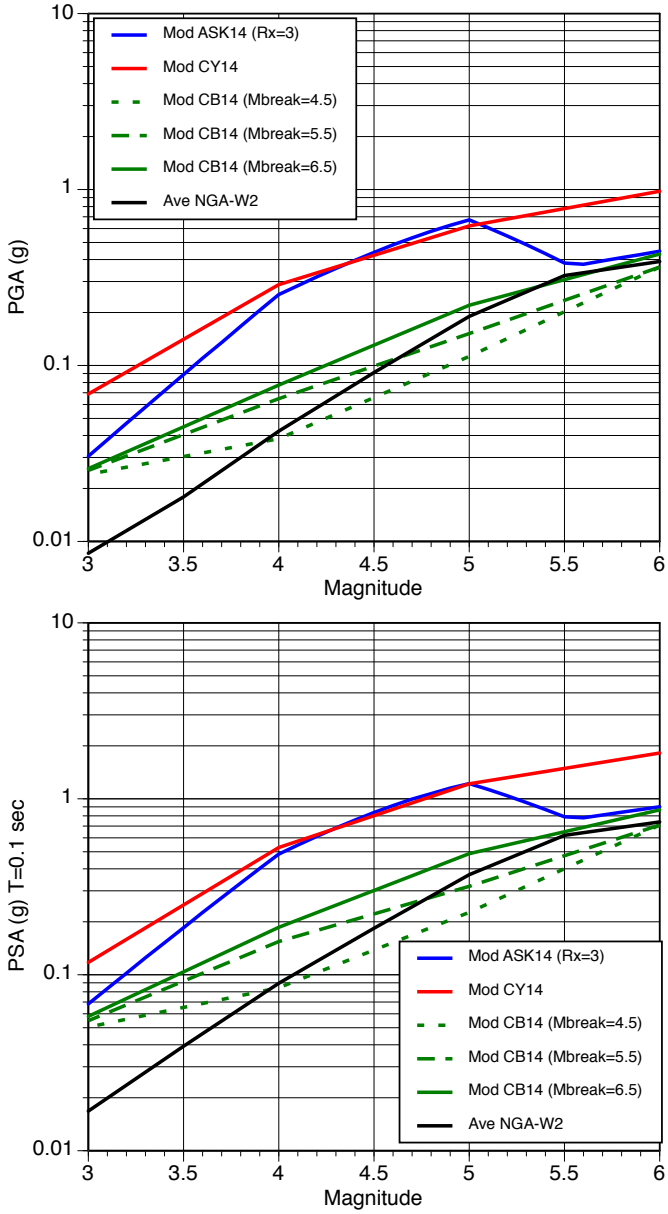


Figure 51: Comparison of the magnitude scaling of the modified GMMs for Rx = 3 km and $Z_{TOR} = 1km$. The left frame is for PGA and the right frame is for T=0.1 sec.

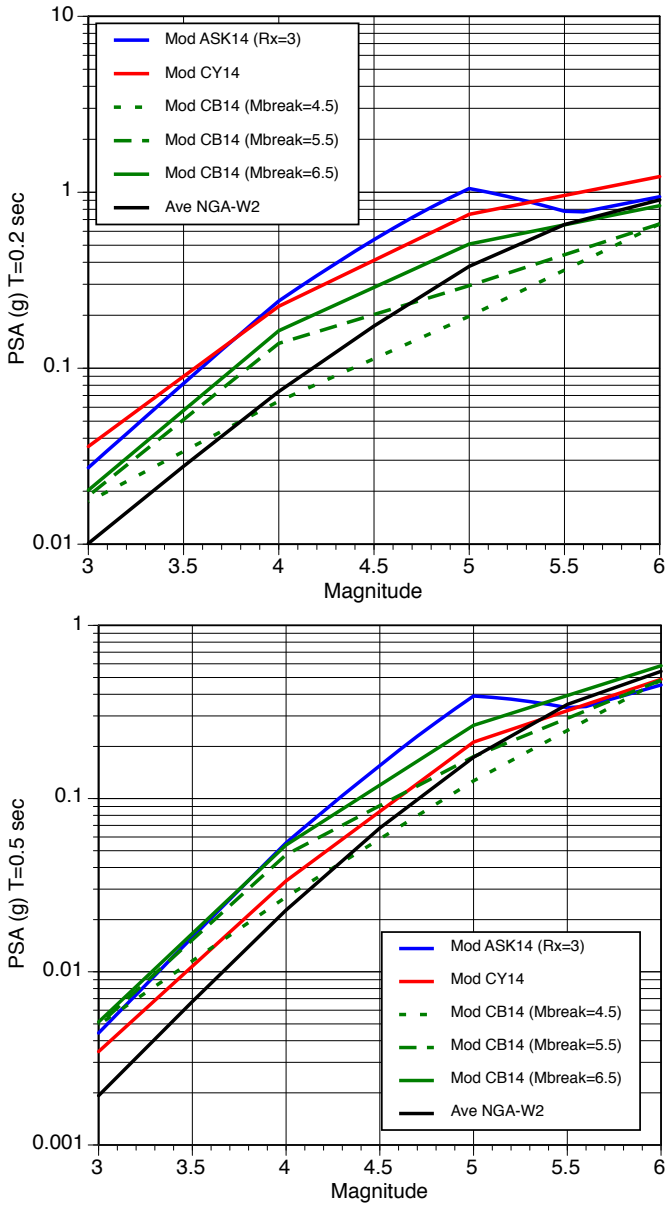


Figure 52: Comparison of the magnitude scaling of the modified GMMs for Rx = 3 km and $Z_{TOR} = 1km$. The left frame is for T=0.2 sec and the right frame is for T=0.5 sec.

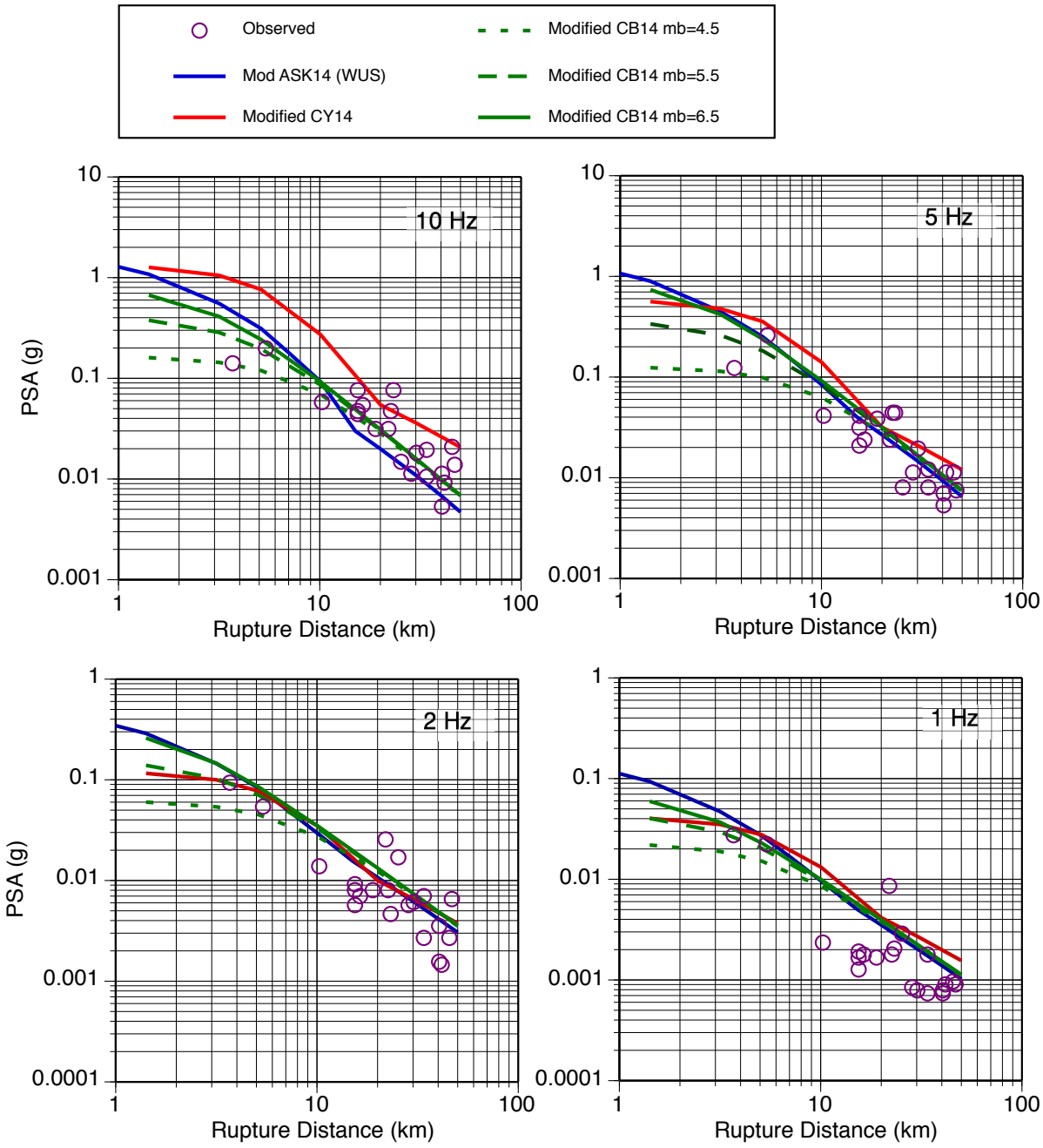


Figure 53: Comparison of the distance scaling for the 30 Nov 2018 M4.6 earthquake in BC with the adjusted GMMs for $V_{S30} = 760m/s$. The observed PSA values are the horizontal RotD50 values from Figure 5 of **?**.

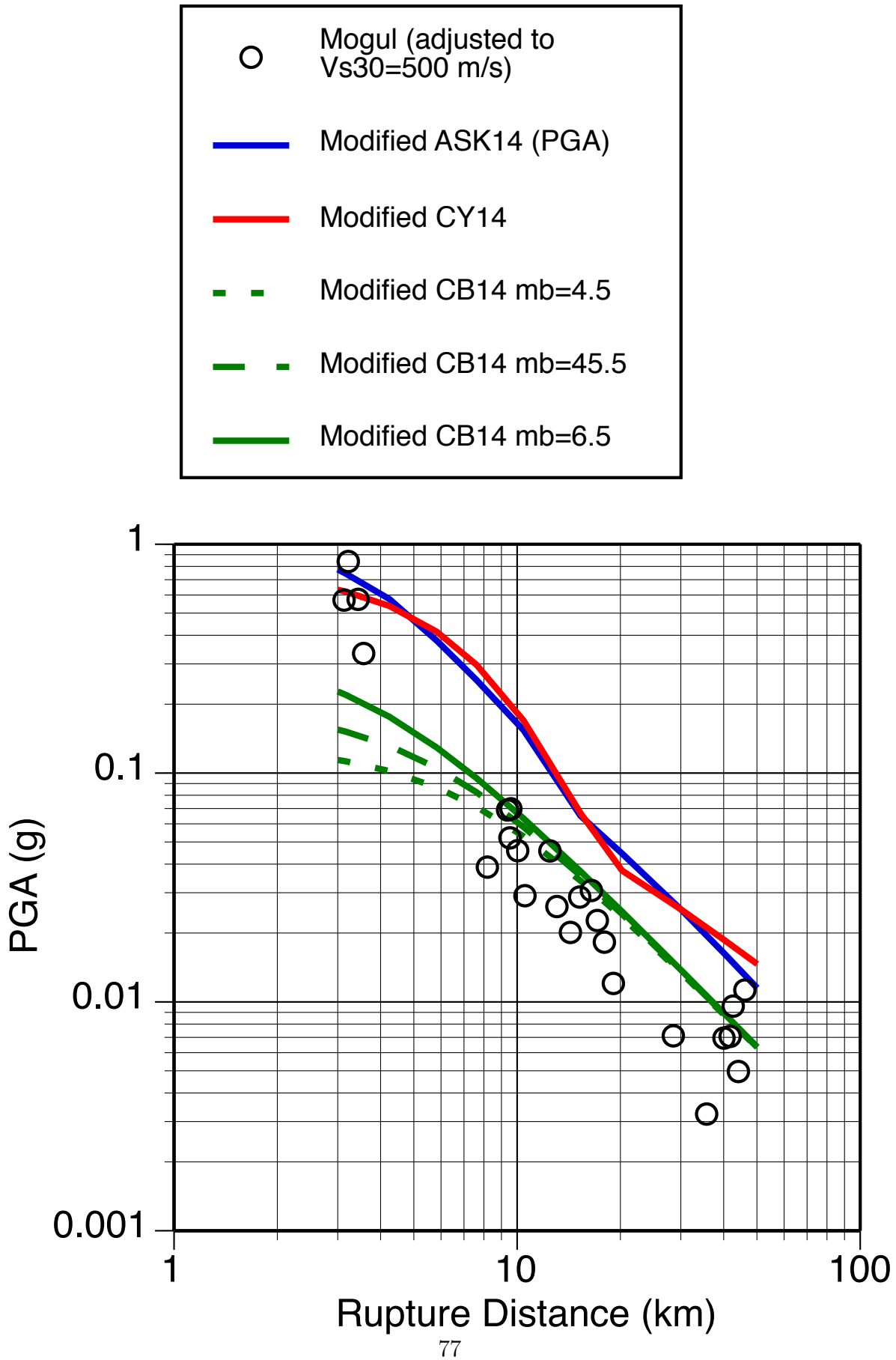


Figure 54: Comparison of the distance scaling for the 26 April 2008 Mogul earthquake in Nevada with the adjusted GMMs for $V_{S30} = 500m/s$. The observed PGA values are the geometric mean of the two as recorded components adjusted to $V_{S30} = 500m/s$

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